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Master Thesis

Energy-Guided Sampling for Controllable Text Generation: From Langevin Dynamics to Amortized Inference

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Stuttgart, July 22, 2026

(Sarthak Singh)

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Abstract

Controllable text generation asks a language model to produce text that satisfies a global property, such as a sentiment, a topic, or a factual constraint, that cannot be evaluated until a sequence is complete. A promising and increasingly popular family of methods approaches this problem without retraining the model: it treats the frozen likelihood of a pretrained autoregressive language model as an *energy function* over sequences and samples from that energy with gradient-guided Markov chain Monte Carlo, most commonly a discretization of Langevin dynamics. The appeal of the framework is that an arbitrary constraint can be added to the energy at inference time, so a single frozen model can, in principle, be steered toward any objective without a new fine-tuning run. This thesis sets out to realize that promise and, in the course of doing so, finds that its central premise does not hold.

We conduct a systematic study across five energy functions, spanning a supervised fine-tuned GPT-2 Large, three GFlowNet-tuned variants of the same base model, and a Llama-3 8B cross-architecture control, comprising one hundred and forty-five sampling configurations on a masked-token recovery benchmark constructed from ROCStories. We evaluate two samplers, a Discrete Langevin Sampler that remains in token space and a Continuous Langevin Sampler that operates in the embedding space, under a factorial combination of Metropolis–Hastings correction, gradient normalization, and annealing schedule length. The headline result is negative and it is consistent: the gradient of a frozen autoregressive likelihood is not a usable search direction on the discrete text manifold. In a controlled ablation that isolates the direction of the gradient from its magnitude, following the exact gradient is statistically indistinguishable from following a random direction of the same norm, on every model tested.

Rather than stopping at this observation, the thesis explains it. We establish three interlocking mechanisms. First, a *linearization failure*: the first-order Taylor surrogate that the discrete sampler uses to rank candidate tokens is essentially uncorrelated with the true change in sequence energy, because the smallest admissible discrete move in the embedding space is far larger than the radius over which the local linear model is valid. Second, a *likelihood trap*: on our own models, the text of highest likelihood is the most degenerate, so any procedure that concentrates probability mass on low-energy regions is optimizing toward poor text by construction; we measure a monotone relationship between per-token likelihood and repetition rate. Third, a *Metropolis–Hastings breakdown* that is specific to the continuous sampler:

because the energy is defined through a nearest-neighbour projection into the vocabulary, its drift is discontinuous at every cell boundary, and we measure that the acceptance ratio is destroyed by the proposal term rather than the target term, so the correction rejects precisely the moves that would change a token.

We then ask whether amortizing the search escapes these mechanisms. Using a GFlowNet, which learns a sampling policy from a scalar reward and never differentiates the energy, we observe three distinct failure modes and show, in a unifying experiment, that Langevin sampling on the GFlowNet-tuned energy is indistinguishable from Langevin sampling on the untuned base model. The pathology survives amortization because it is a property of the training objective of autoregressive language models, which shapes local conditional distributions and never a global energy over sequences.

The thesis states its conclusions as three claims of deliberately different strength. First, at full strength and established both by measurement and by proof: the gradient of the frozen autoregressive likelihood carries no usable search direction on discrete text, statistically indistinguishable from a random direction of equal norm on every model tested, and provably identically zero at the final token, the one position where the energy difference between candidate tokens is maximally informative. Second, stated more precisely than a blanket null: an additive constraint gradient does carry a directional signal, unlike the fluency gradient, but it cannot rescue generation once the sample leaves the fluent manifold, and even when fluency is held fixed by a trust region its transfer to an independent judge is bounded by how far two separately trained classifiers happen to agree rather than by any fixed reachable direction. Third, the training-free premise on which the whole framework rests is tested and refuted: a score-trained diffusion model on the same tokenizer supplies the very local direction the autoregressive gradient lacks and natively performs the recovery the gradient machinery could not, so the capability the promise assumed for free is available only at the price of an objective trained to provide it. The contribution of this thesis is therefore a mechanism, established by independent routes and confirmed by a positive control, for why energy-guided sampling on frozen autoregressive language models fails and for what would repair it.

Contents

List of Figures

List of Tables

1 Introduction

Text generation with large language models has, in the space of a few years, moved from a research curiosity to a technology that underpins a wide range of applications, from writing assistants and conversational agents to summarization, translation, and code synthesis. The dominant paradigm behind these systems is **autoregressive generation**: a model factorizes the probability of a sequence into a product of next-token conditionals and produces text one token at a time, from left to right, sampling or maximizing each token given the tokens already committed (Radford et al., 2019; Brown et al., 2020). This factorization is what makes modern language models trainable at scale and fast at inference, and it is the reason they work as well as they do. It is also, as this thesis will argue at length, the reason a particular and popular approach to controlling them does not.

The same factorization that makes autoregressive models practical is the source of a persistent and practically important limitation. Because generation proceeds strictly from left to right, and because each token is fixed the moment it is emitted, an autoregressive model has no mechanism for revising an earlier decision in light of a later one. If the model, having committed to the first half of a sentence, discovers as it generates the second half that the sentence has become logically inconsistent or has drifted into a tone the user did not want, it cannot go back. Its only recourse is to continue, or to discard the entire generation and start again. This is not a problem when the objective is simply to produce fluent continuations, which is what these models are trained to do. It becomes a problem the moment we want to *control* what the model produces.

The reason control is hard has a precise structure, and it is worth stating carefully because it motivates everything that follows. Many of the properties we care about most, such as the overall sentiment of a passage, its topical focus, its adherence to a factual constraint, or the absence of toxic content, are **global properties** of a completed sequence rather than local properties of any single token. Whether a paragraph is positive in sentiment cannot be read off its first word; it is a property of the whole. A greedy, left-to-right decision about the next token cannot see such a property, for the simple reason that the object which would be evaluated against it, the finished text, does not yet exist at the moment the decision is made. Enforcing a global constraint autoregressively therefore requires something like generating many complete sequences and keeping the ones that happen to satisfy the constraint, which is wasteful when the constraint is easy to satisfy and effectively impossible when it is not. The problem of steering a language model toward such global objectives, while keeping its output fluent, is the problem of **controllable text generation**, and it is the subject of this thesis.

1.1 Two Established Routes to Control, and Their Costs

There are, broadly, two established ways of imposing control on a language model, and each carries a cost that motivates the search for a third.

The first is to bake the desired behaviour into the parameters of the model itself, through supervised fine-tuning on curated data or, more commonly today, through reinforcement learning from human feedback (Ouyang et al., 2022). This approach is effective and now standard; the aligned assistants in wide use are its product. But the control it provides is *static* in a specific sense: a model tuned to be helpful and non-toxic embodies exactly those constraints and no others, and introducing a genuinely new constraint at inference time, a new attribute the user cares about today but did not anticipate at training time, requires collecting new preference data and running a new and expensive training procedure. The control is fixed at training time, and adapting it is not cheap.

The second is to leave the model untouched and instead filter or re-rank its outputs, generating many independent candidates and discarding those that violate the constraint. This is conceptually simple and it preserves the model, but it inherits precisely the intractability described above: for a constraint that a fluent generation satisfies only rarely by chance, the number of candidates one must draw to find a satisfying one grows without practical bound.

What both approaches lack, and what would be genuinely valuable, is the ability to introduce a fresh constraint at the moment of generation, cheaply, without touching the underlying model and without generating a haystack to find a needle. This is the capability that the energy-based approach promises, and it is the promise that this thesis sets out to examine.

1.2 Energy-Based Control and Its Central Assumption

An elegant alternative reframes decoding itself. Rather than generating a sequence token by token, one can define a probability distribution over the space of all possible sequences and *sample* from it directly. Following the standard formulation of **energy-based models** (LeCun et al., 2006; Hinton, 2002), the distribution is written in the Boltzmann form

$$p(\boldsymbol{x}) = \frac{1}{Z}\exp\bigl(-E(\boldsymbol{x})\bigr), \tag{1.1}$$

where $E(\boldsymbol{x})$ is the **energy** of a sequence $\boldsymbol{x}$, a scalar that is low for desirable sequences and high for undesirable ones, and Z is a normalizing constant. The constant Z is intractable to compute, because it sums over every possible sequence, but the great advantage of the sampling framework is that it need never be computed: the methods used to draw samples from (1.1) depend only on *ratios* of probabilities or on gradients of the log-probability, in both of which the constant cancels.

The attraction of this view for controllable generation is that the energy can be assembled additively from independent pieces. One writes

$$E(\boldsymbol{x}) = -\log p_{\mathrm{LM}}(\boldsymbol{x}) + \lambda\, C(\boldsymbol{x}), \tag{1.2}$$

where the first term is the negative log-likelihood of a frozen, pretrained language model and supplies *fluency*, pulling samples toward text the model considers natural, and the second term $C(\boldsymbol{x})$ is an arbitrary differentiable *constraint*, weighted by a coefficient λ that trades off fluency against constraint satisfaction. Because the constraint is simply added on, it can be specified at the moment of generation and swapped freely. A practitioner who wants positive sentiment today writes C as the negative log-probability of the positive class under a sentiment classifier; one who wants factual consistency tomorrow writes a different C; and in both cases the language model is untouched. This is the promise of plug-and-play, inference-time controllable generation, and it is an appealing one. A substantial and growing body of recent work pursues it, through gradient-guided decoders such as PPLM (Dathathri et al., 2020), FUDGE (Yang and Klein, 2021), and GeDi (Krause et al., 2021), and, most directly related to this thesis, through sampling-based methods such as COLD (Qin et al., 2022) and MuCoLa (Kumar et al., 2022), which draw samples from the energy in (1.2) using continuous relaxations and dynamics derived from Langevin sampling.

Every method in this family rests on a single assumption, and because that assumption is the pivot of the entire thesis it is worth isolating with care. To sample from the energy in (1.2) using a gradient-guided method, two things must be true. First, one must be able to compute a gradient of $-\log p_{\mathrm{LM}}(\boldsymbol{x})$ that points, at least on average, in the direction of better sequences. Second, one must be able to *follow* that gradient across the space of possible texts, moving from a worse sequence to a better one by steps the gradient recommends. Taken together, these amount to the presupposition that the frozen likelihood of an autoregressive language model, viewed as a function over whole sequences, defines a *landscape that a sampler can actually navigate*: a surface with meaningful slopes, whose downhill direction can be computed locally and trusted to lead somewhere good. This presupposition is intuitive. It is also, remarkably, almost never tested directly in the literature that relies on it. Testing it, and explaining what is found, is the central business of this thesis, and the central finding is that for standard autoregressive language models on discrete text, the presupposition does not hold.

1.3 From Sampling to Amortization

The natural response to a gradient that will not cooperate is to stop using the gradient. There is a version of the energy-based idea that does not differentiate the energy at all. If a per-sequence energy can still be *evaluated*, that is, if one can take a finished sequence and compute its score, even when one cannot usefully *differentiate* that score to decide how to improve a sequence, then one can treat the language model as a black box that ranks finished sequences and learn a separate policy that generates in proportion to that ranking. This is the idea of **amortized inference**: rather than paying the cost of a search procedure afresh for every sample, one pays a one-time training cost to learn a generator that produces good samples directly.

The most principled recent realization of this idea for sequence generation is the **Generative Flow Network**, or GFlowNet (Bengio et al., 2021; 2023), which trains a generative policy so that the probability of producing a complete sequence is proportional to its reward. Applied to language models, GFlowNets have been proposed as a way to sample diverse, high-reward text without the mode-collapse that afflicts reward-maximizing reinforcement learning (Hu et al., 2024). The second half of this thesis asks a specific and critical question about this approach: does amortizing the search, and thereby never touching the gradient, escape the difficulties that defeat the gradient-guided samplers? The answer, established in Chapter 5, is that it does not, and the reason it does not is illuminating, because it shows that the difficulty was never in the search procedure at all but in the energy that both the sampler and the amortized policy inherit from the frozen model.

1.4 The Shape of the Investigation

It will help the reader to know in advance that this thesis is structured as an investigation rather than as the presentation of a working method, and that this structure is a consequence of what the experiments found rather than a stylistic choice. The work began with the goal of building controllable generation on faithful Langevin dynamics, and it proceeded by first establishing whether the foundation, energy-guided sampling on a frozen model, was sound. It was not. Rather than build control on a foundation shown to be unsound, which would have produced numbers without interpretable meaning, the work turned to the more valuable question of *why* the foundation fails, and it pursued that question until it had an answer supported by direct measurement and reached by two independent routes. The narrative of the results chapter follows this investigative path: each experiment is motivated by a question left open by the one before it, each states what outcome was expected, and each ends by naming the new question its result raises. This is, we will argue, the honest way to present a diagnostic result, because the reasoning chain is itself the contribution.

1.5 Research Questions and Contributions

The work is organized around four research questions, which the results chapter answers in turn and the discussion chapter revisits explicitly.

RQ1. Can the frozen likelihood of an autoregressive language model be sampled effectively with theoretically faithful Langevin dynamics on the discrete text manifold, and if not, why not?

RQ2. What is the role of the Metropolis–Hastings correction in each of the discrete and continuous samplers, and does making the samplers theoretically correct also make them work?

RQ3. Does amortizing the energy with a GFlowNet, which never differentiates the reward, escape the failures observed for the gradient-guided samplers?

RQ4. Can an additive constraint term steer generation toward a target attribute on this energy landscape, as the plug-and-play framework requires?

In answering these questions the thesis makes the following contributions. It establishes, through a controlled ablation that separates the direction of the gradient from its magnitude, that the frozen autoregressive gradient carries no usable information for discrete sampling, and it replicates this finding across five distinct energy functions. It identifies and measures the *linearization failure* that is the geometric cause of this result, showing that the region over which the sampler's first-order approximation is valid is smaller than the smallest possible discrete step, so that the gradient is not merely imprecise but inapplicable. It measures the *likelihood trap* on the study's own models, demonstrating that the lowest-energy text is the most degenerate, so that the very objective the methods pursue is undesirable. It provides a precise, measured account of the *Metropolis–Hastings breakdown* in continuous space, attributing the collapse of the acceptance ratio to the proposal term rather than the target term and connecting it to the non-Lipschitz drift induced by nearest-neighbour projection. It documents three mechanistically distinct failure modes of GFlowNet fine-tuning on a frozen-likelihood reward, and shows in a unifying experiment that Langevin sampling on the amortized energy is indistinguishable from sampling on the untuned base, thereby locating the cause in the energy rather than the search. It reports a constrained-generation extension showing that, unlike the fluency gradient, the constraint gradient does carry a directional signal, but that this signal cannot rescue generation off the fluent manifold and, even with fluency held fixed by a trust region, transfers to an independent judge only as far as two separately trained classifiers agree. And it runs, rather than merely proposes, the positive control: a score-trained diffusion language model on the same tokenizer supplies the local direction the autoregressive gradient lacks and performs the recovery the gradient machinery could not, isolating the training objective as the causal variable and confirming the thesis's central explanation.

1.6 Structure of the Thesis

Chapter 2 introduces the background needed to follow the technical development: the energy-based view of generation, Langevin dynamics and the Metropolis–Hastings correction, the discrete and continuous samplers studied here, and GFlowNets, with an emphasis throughout on the specific assumption each tool makes. Chapter 3 situates the work within the literature on controllable and energy-based text generation, traces how the field arrived at energy-based sampling, and identifies the unexamined assumption on which it rests. Chapter 4 describes the experimental platform, the diagnostic task and why it was chosen, the five energy functions, the sampler configurations, the evaluation metrics and the reasoning behind each, and the four diagnostic experiments designed to isolate the mechanisms. Chapter 5 presents the results as a connected investigation, from the calibration of the samplers through the annealing dynamics, the Metropolis–Hastings breakdown, and the central gradient-fallacy result, to the GFlowNet experiments and the constrained-generation extension. Chapter 6 draws the findings together into a single account, answers each research question explicitly, connects the results to the related work, and reports the diffusion positive control that tests the central explanation. Chapter 7 concludes.

2 Background Work

This chapter develops the technical background that the remainder of the thesis relies on. It begins with the energy-based view of sequence generation and the precise way in which a frozen language model is repurposed as an energy function, dwelling on a distinction, between a well-trained local conditional and a well-behaved global energy, that turns out to be the seed of the entire investigation. It then introduces Langevin dynamics as a gradient-guided sampling method, explaining not only the update rule but the intuition for why it samples rather than merely optimizes, and the regularity condition it quietly assumes. It presents the Metropolis–Hastings correction that makes such samplers asymptotically exact, and it is careful about a subtlety in the correction that becomes decisive later. It then describes the two concrete samplers, discrete and continuous, that are studied in the experiments, and the geometry each one must confront. It closes with an account of GFlowNets as an amortized alternative that avoids the gradient altogether. Throughout, the aim is not to survey these tools exhaustively but to expose the specific assumptions each one makes, because it is precisely those assumptions that the experiments in Chapter 5 put under strain.

2.1 Autoregressive Language Models as Energy Functions

An autoregressive language model defines a probability distribution over a sequence of tokens $\boldsymbol{x} = (x_1, \ldots, x_T)$ by the chain rule of probability,

$$\log p_{\mathrm{LM}}(\boldsymbol{x}) = \sum_{t=1}^{T} \log p_{\mathrm{LM}}(x_t \mid x_{<t}), \tag{2.1}$$

where each conditional $p_{\mathrm{LM}}(x_t \mid x_{<t})$ is produced by a neural network, in the models studied here a Transformer (Vaswani et al., 2017), applied to the prefix $x_{<t}$. The network reads the prefix, produces a vector of logits over the vocabulary, and a softmax turns those logits into the conditional distribution for the next token. The model is trained by **teacher forcing**: on a large corpus, with the true prefix supplied at every position, it is asked to maximize the log-likelihood in (2.1), which decomposes

into a sum of independent next-token prediction problems, one per position.

It is essential for everything that follows to be precise about what this training objective does and does not shape, because the entire diagnosis of this thesis turns on the difference. The objective shapes each conditional $p_{\mathrm{LM}}(x_t \mid x_{<t})$ to be an accurate predictor of the next token given a *correct* history drawn from the data distribution. That is what it optimizes, and modern models optimize it extraordinarily well. What the objective never does is ask the joint quantity $\log p_{\mathrm{LM}}(\boldsymbol{x})$ to behave in any particular way as a function of $\boldsymbol{x}$ when $\boldsymbol{x}$ is varied away from the data, for instance by substituting one token for another in the middle of an otherwise fixed sequence. The model is never trained on such counterfactual sequences, never asked to assign them calibrated scores, and never penalized for the shape of the surface it induces over them. There is, in other words, no reason internal to the training objective for the joint log-likelihood to be a smooth function of the sequence, or a well-calibrated measure of sequence quality, or a function whose local gradient points toward better sequences. This distinction, between a well-trained local conditional and a well-behaved global energy, is easy to state and easy to overlook, and it is the origin of the central result of this thesis. We return to it, having accumulated the evidence, in Chapter 6.

With that caveat noted rather than resolved, the energy-based reformulation is immediate. Identifying the Boltzmann form of (1.1) with the language model, one simply sets

$$E_{\mathrm{LM}}(\boldsymbol{x}) = -\log p_{\mathrm{LM}}(\boldsymbol{x}), \tag{2.2}$$

so that sequences of high probability are sequences of low energy and vice versa. Sampling text from the language model is then, formally, sampling from the Boltzmann distribution of this energy, and the two descriptions are interchangeable. For controllable generation, an additive constraint term $\lambda\, C(\boldsymbol{x})$ is appended as in (1.2), where C is typically the negative log-probability that an attribute classifier assigns to the desired label, so that low energy now means both fluent and on-attribute. The task of generation becomes the task of drawing low-energy samples from the combined energy, and the central computational question, to which the next two sections are devoted, is how one draws such samples at all.

2.2 Langevin Dynamics

Suppose, for a moment, that the state $\boldsymbol{s}$ lived in a continuous space $\mathbb{R}^D$ and that the energy were differentiable everywhere. Then one could sample from the distribution $p(\boldsymbol{s}) \propto \exp(-E(\boldsymbol{s}))$ using **Langevin dynamics**, a gradient-guided Markov chain Monte Carlo method with a long history in statistics and physics (Roberts and Tweedie, 1996; Welling and Teh, 2011). The discretized update, which is what one actually runs, is

$$\boldsymbol{s}_{t+1} = \boldsymbol{s}_t + \frac{\epsilon_t}{2} \nabla_{\boldsymbol{s}} \log p(\boldsymbol{s}_t) + \sqrt{\epsilon_t}\, \boldsymbol{\xi}_t, \qquad \boldsymbol{\xi}_t \sim \mathcal{N}(\mathbf{0}, \boldsymbol{I}), \tag{2.3}$$

where ϵ_t is a step size at iteration t. The update has two parts, and the interplay between them is exactly what distinguishes sampling from optimization, so it is worth dwelling on.

The first part, the **drift**, is a step of size $\epsilon_t/2$ in the direction of the gradient of the log-density. Taken on its own, iterating the drift alone is gradient ascent on $\log p$, and it drives the state monotonically toward the nearest mode of the distribution, where it comes to rest. If one wanted to *find* the single most probable point, the drift alone would suffice. But finding the mode is not the goal; the goal is to draw a *representative sample*, and a distribution is more than its mode. This is where the second part enters. The **diffusion** term injects fresh Gaussian noise at every step, scaled by $\sqrt{\epsilon_t}$, and this noise is what prevents the chain from simply collapsing onto the mode. It knocks the state off the deterministic ascent path, allowing it to climb away from a mode, cross into a neighbouring region, and in the limit visit every part of the distribution with the correct frequency.

A physical intuition makes the balance vivid. Imagine a marble rolling on an uneven surface, where the height of the surface at each point is the negative log-density, so that valleys are regions of high probability. The drift is gravity, pulling the marble downhill into valleys. The diffusion is a vibration applied to the whole surface, shaking the marble so that it does not simply settle into the first valley it finds but continues to explore, occasionally hopping over ridges into other valleys. If the vibration is held constant, the marble wanders forever and never settles. But if the vibration is gradually reduced to nothing, the marble's final resting place is a fair draw from the landscape: it settles preferentially in the deep, wide valleys, which is exactly where the probability mass is. This gradual reduction is **annealing** the step size, and Welling and Teh (2011) showed that annealing ϵ_t toward zero on a suitable schedule makes the distribution of $\boldsymbol{s}_t$ converge to the target $p(\boldsymbol{s})$ rather than to its mode. The annealing schedule, and the artifacts it can produce when the landscape is not what the theory assumes, will be a recurring theme in the results.

The convergence guarantee, however, rests on a regularity condition that is easy to pass over and that will matter a great deal later. For the Metropolis-adjusted variant of the method, discussed in the next section, to be valid, the drift term must be **Lipschitz continuous** (Roberts and Tweedie, 1996): loosely, the gradient must not change too abruptly from one point to a nearby one, so that a small step lands where the gradient at the starting point predicted it would. When the energy landscape is smooth, as the theory assumes, this condition holds and nothing needs to be said about it. When, as in the continuous sampler of Section 2.4, the energy is defined through a discontinuous projection into a discrete vocabulary, the condition fails, and the failure is not a technicality. It is, as Chapter 5 will show by direct measurement, the precise reason the theoretically correct version of that sampler cannot move.

2.3 The Metropolis–Hastings Correction

The update in (2.3) is a discretization of a continuous-time process, and any finite step size introduces a bias: the chain does not sample exactly from the intended distribution $p(\boldsymbol{s})$ but from a perturbed distribution that depends on the step size. For many purposes this bias is tolerated. If one wants asymptotic correctness, however, it can be removed exactly by the **Metropolis–Hastings** (MH) correction (Metropolis et al., 1953; Hastings, 1970), which turns a biased proposal into an exact sampler by accepting or rejecting each proposed move. Having proposed a move from the current state $\boldsymbol{s}$ to a new state $\boldsymbol{s}'$ according to a proposal density $q(\boldsymbol{s}' \mid \boldsymbol{s})$, one accepts the move with probability

$$\alpha(\boldsymbol{s}' \mid \boldsymbol{s}) = \min\left\{1,\ \frac{p(\boldsymbol{s}')\, q(\boldsymbol{s} \mid \boldsymbol{s}')}{p(\boldsymbol{s})\, q(\boldsymbol{s}' \mid \boldsymbol{s})}\right\}, \tag{2.4}$$

and otherwise rejects it and remains at $\boldsymbol{s}$, counting the current state again. When Langevin dynamics is equipped with this correction the method is known as the Metropolis-adjusted Langevin algorithm, or MALA.

The acceptance ratio factors into two meaningful parts, and understanding each is necessary to follow the central diagnostic of Chapter 5. The **target ratio** $p(\boldsymbol{s}')/p(\boldsymbol{s})$ compares the probability of the proposed state to that of the current state under the distribution being sampled; it is greater than one when the move is toward higher probability, and it is the part of the ratio that expresses the sampler's preference for good states. The **proposal ratio** $q(\boldsymbol{s} \mid \boldsymbol{s}')/q(\boldsymbol{s}' \mid \boldsymbol{s})$ compares how likely the proposal mechanism was to suggest the reverse move against the forward move; it is what enforces **detailed balance**, the condition that guarantees the chain leaves p invariant and therefore samples from it. Intuitively, the proposal ratio penalizes moves that are easy to make but hard to undo, because such moves, if accepted freely, would let the chain drift in a preferred direction and distort the distribution.

Two facts about the correction deserve emphasis, because both recur in the results. First, a great many energy-based decoders in the literature omit the correction entirely. Without it, the noise-augmented gradient update of (2.3) is not a sampler at all but a biased optimizer with a particular kind of stochasticity, and calling its output a sample from the energy is, strictly, incorrect. The distinction is not merely pedantic: a biased optimizer stopped early and a correct sampler run to convergence can produce very different distributions of outputs, and much of the apparent success of correction-free methods is, on inspection, the success of optimization under early stopping rather than of sampling. Second, and more subtly, the proposal ratio is delicate to compute correctly for a Langevin proposal. The reverse proposal density $q(\boldsymbol{s} \mid \boldsymbol{s}')$ must be evaluated using the drift computed *at the proposed point* $\boldsymbol{s}'$, not at the starting point, because the reverse move would start from $\boldsymbol{s}'$. If the drift at $\boldsymbol{s}'$ differs wildly from the drift at $\boldsymbol{s}$, which is exactly what happens when the landscape is discontinuous and the two points fall on opposite sides of a discontinuity, then the reverse proposal can place the original state $\boldsymbol{s}$ deep in the tail of a Gaussian centred far away, driving $q(\boldsymbol{s} \mid \boldsymbol{s}')$ toward zero and with it the entire acceptance probability. This is the mechanism that paralyzes the continuous sampler, and Chapter 5 measures it directly, decomposing the acceptance ratio into its two terms and showing that the proposal term alone accounts for the collapse.

2.4 Sampling in a Discrete Space: DLS and CLS

The difficulty that motivates the entire thesis is that text is not continuous. A sequence is a list of discrete tokens drawn from a finite vocabulary V, and there is no state *between* two tokens: one cannot be halfway between the word *cat* and the word *dog*. Langevin dynamics, which assumes a continuous differentiable space in which infinitesimal steps are meaningful, cannot be applied to such a state without modification. Two strategies for adapting it are studied in this thesis, and they differ precisely in where they confront the discreteness, which is why studying both, rather than either alone, is what allows the results to be attributed to the energy rather than to one sampler's idiosyncrasies.

The **Discrete Langevin Sampler** (DLS), following the discrete gradient-based samplers of Grathwohl et al. (2021) and Zhang et al. (2022), never leaves token space at all. Its state is always a genuine sequence of tokens. It uses the gradient of the log-likelihood, taken with respect to the continuous input embedding of a masked position, only as a *score* for constructing a categorical proposal distribution over which token to place at that position next. Concretely, let $\boldsymbol{e}(x_i)$ denote the embedding of the current token at a masked position i, and let $\boldsymbol{g} = \nabla_{\boldsymbol{e}_i} \log p_{\mathrm{LM}}(\boldsymbol{x})$ denote the gradient of the sequence log-likelihood with respect to that embedding. For a candidate token v with embedding $\boldsymbol{e}(v)$, the proposal assigns a logit that combines a term penalizing the embedding displacement $\|\boldsymbol{e}(v) - \boldsymbol{e}(x_i)\|$, which keeps proposals local, with a term proportional to $\frac{1}{2}\boldsymbol{g}^\top(\boldsymbol{e}(v) - \boldsymbol{e}(x_i))$, which biases the proposal toward tokens the gradient suggests will lower the energy. This second term is a **first-order Taylor surrogate**: it approximates the true change in log-likelihood that would result from substituting v for x_i by the inner product of the gradient with the embedding displacement, which is exactly the first-order term in the Taylor expansion of the log-likelihood around the current embedding. Whether this surrogate actually predicts the true change, and therefore whether the gradient is a useful ranking signal, is the single most important question the diagnostic experiments answer, and the reader should hold the definition of this surrogate in mind, because Section 5.6 tests it directly.

To make the role of the surrogate precise, because it is the object the central diagnostic tests, it is worth writing the proposal out in full. The discrete sampler adapts the continuous Langevin proposal of (2.3) to a categorical distribution over the vocabulary by the construction of Zhang et al. (2022). In the continuous case, a Gaussian proposal centred at $\boldsymbol{s} + \frac{\alpha}{2}\nabla \log p(\boldsymbol{s})$ assigns to a candidate $\boldsymbol{s}'$ a log-density proportional to $-\frac{1}{2\alpha}\left\|\boldsymbol{s}' - \boldsymbol{s} - \frac{\alpha}{2}\nabla \log p(\boldsymbol{s})\right\|^2$. Expanding this and discarding terms constant in $\boldsymbol{s}'$ leaves a term linear in the displacement, $\frac{1}{2}\nabla \log p(\boldsymbol{s})^\top(\boldsymbol{s}' - \boldsymbol{s})$, plus a quadratic penalty on the displacement itself. Transplanting this to the discrete setting, where the candidate states are token embeddings $\boldsymbol{e}(v)$ and the current state is $\boldsymbol{e}(x_i)$, gives a categorical proposal whose logit for token v is

$$\log q(v \mid x_i) \propto \frac{1}{2}\boldsymbol{g}^\top\big(\boldsymbol{e}(v) - \boldsymbol{e}(x_i)\big) - \frac{1}{2\alpha}\left\|\boldsymbol{e}(v) - \boldsymbol{e}(x_i)\right\|^2, \tag{2.5}$$

where $\boldsymbol{g} = \nabla_{\boldsymbol{e}_i} \log p_{\mathrm{LM}}(\boldsymbol{x})$. The first term is the surrogate: it rewards candidates whose displacement from the current embedding aligns with the gradient, and it is precisely the first-order Taylor estimate of how much the log-likelihood would change if the substitution were made, since to first order $\log p(\boldsymbol{x}_{i\to v}) - \log p(\boldsymbol{x}) \approx \boldsymbol{g}^\top(\boldsymbol{e}(v) - \boldsymbol{e}(x_i))$. The second term keeps the proposal local, penalizing distant candidates. Everything the sampler knows about which token is good, as opposed to merely near, is contained in that first term, and if it is uninformative the proposal reduces to a distance-penalized random choice. This is exactly why Section 5.6 tests the correlation between that first term and the true change: the surrogate is the sampler's entire claim to be gradient-guided rather than random, and the diagnostic measures whether the claim holds.

The **Continuous Langevin Sampler** (CLS), following the continuous-relaxation approach of methods such as MuCoLa (Kumar et al., 2022), takes the opposite route. It relaxes the state out of token space and into the continuous embedding space $\mathbb{R}^D$, where it runs genuine Langevin dynamics according to (2.3), and it projects the continuous state back onto the nearest token embedding whenever a discrete sequence is required, for instance to evaluate the energy. The energy it navigates is therefore defined *through* a **nearest-neighbour projection** into the vocabulary, and this construction has an immediate and consequential geometric structure. The embedding space is partitioned into **Voronoi cells**, one per token, where a cell is the region of space closer to that token's embedding than to any other. Within a cell the projected token is constant, so the energy, which depends on the sequence only through its projected tokens, is also constant: the cell is a flat plateau. Across a cell boundary the projected token changes abruptly, so the energy jumps discontinuously. The landscape the continuous sampler must traverse is thus not the smooth surface the Langevin theory assumes but a collection of flat plateaus separated by cliffs, and it is exactly on such a landscape that the Lipschitz condition of Section 2.3 fails, because the drift, which depends on the energy, is discontinuous at every boundary. The consequences of this geometry are measured in Section 5.3.

Making the energy explicit clarifies where the discontinuity enters. Writing $\mathrm{proj}_V(\boldsymbol{s})$ for the nearest-neighbour map that sends the continuous state at each masked position to the token whose embedding is closest in Euclidean distance, the energy the continuous sampler navigates is

$$E_{\mathrm{CLS}}(\boldsymbol{s}) = -\log p_{\mathrm{LM}}\big(\mathrm{proj}_V(\boldsymbol{s})\big), \tag{2.6}$$

the negative log-likelihood of the discrete sequence obtained by projecting the state back to tokens, to which the additive constraint λC of (1.2) is appended when a constraint is imposed. The projected token, and hence the energy, changes only when the state crosses a cell boundary, which is the sense in which the surface is piecewise constant, while the gradient the sampler differentiates is taken with respect to the continuous state as it enters the network on the input side, so it is well defined within a cell and yet blind to the boundary that actually matters. It is this split, a smooth interior and a jump at the boundary, that breaks the correction.

The split has a direct consequence for the acceptance probability that is worth deriving once, because the results measure exactly this quantity. Consider a proposed move that stays within a single Voronoi cell, so that $\mathrm{proj}_V(\boldsymbol{s}') = \mathrm{proj}_V(\boldsymbol{s})$. The projected sequence, and therefore the energy of (2.6), is essentially unchanged, so the target ratio $p(\boldsymbol{s}')/p(\boldsymbol{s})$ is at or near unity and the acceptance probability of (2.4) is governed almost entirely by the proposal ratio $q(\boldsymbol{s} \mid \boldsymbol{s}')/q(\boldsymbol{s}' \mid \boldsymbol{s})$. For the discrete sampler a within-cell move is the proposal of the same token, which is its own reverse and for which the forward and backward proposal probabilities coincide, so the proposal ratio is one and the move is accepted with certainty. For the continuous sampler a within-cell move is still a continuous displacement in embedding space, and its reverse proposal density is evaluated from the Langevin drift at the proposed point, which even inside the same cell need not make the return move likely, so the proposal ratio is not one and within-cell acceptance is not protected. The measured split in Section 5.3 bears this out precisely: the discrete sampler accepts within-cell moves with probability one, while the continuous sampler accepts them at well under a percent, and it is boundary-crossing moves, the only ones that change a token, on which the correction collapses.

2.5 Score Matching and the Score Function

The diagnosis that this thesis arrives at, and the positive-control experiment it proposes, both turn on a notion that is worth introducing in advance: the **score function**. For a differentiable density $p(\boldsymbol{s})$, the score is the gradient of its log-density, $\nabla_{\boldsymbol{s}} \log p(\boldsymbol{s})$, and it is exactly the quantity the Langevin drift of (2.3) requires. A model that provides an accurate score at every point can be sampled by Langevin dynamics; a model that does not cannot, no matter how the sampling is arranged. The question that runs through this thesis can therefore be restated compactly: does a frozen autoregressive language model provide a usable score over sequences? The results answer no.

What makes this framing useful is that there exist models trained specifically to provide a score, and the theory connecting them to Langevin sampling is well established. **Score matching** (Vincent, 2011) is a training objective that fits a model directly to the score of the data distribution, and the foundational observation behind modern score-based and diffusion generative models (Song and Ermon, 2019; Ho et al., 2020) is that learning to denoise a corrupted input is equivalent to learning the score. A diffusion model, trained to reverse a gradual noising process, therefore learns by construction the very gradient field that the autoregressive likelihood was found not to provide. This equivalence is the reason the diffusion language model is the natural positive control for the thesis's central claim, as developed in Chapter 6: it is a model whose training objective shapes a score, where the autoregressive objective does not, and it is the cleanest available way to test whether the training objective is indeed the causal variable.

2.6 Amortized Inference with GFlowNets

Both samplers above require the gradient of the energy: the discrete sampler to score its proposals, the continuous sampler to compute its drift. An entirely different strategy is to give up on the gradient and instead *learn* to sample. A **Generative Flow Network** (Bengio et al., 2021; 2023) treats generation as a sequential process of constructing an object one piece at a time, and it trains a policy so that the probability of terminating at a complete object is proportional to a non-negative reward $R(\boldsymbol{x})$ assigned to that object. The name carries a useful metaphor. Imagine a fixed quantity of flow, like water, entering the process at an initial empty state and being divided among the available branches at each step according to the policy. The amount of flow that arrives at each complete object is then determined by the policy's choices along the way, and the training objective adjusts the policy so that this arriving flow matches the object's reward. Generation amounts to releasing a unit of flow and following where it goes. Because the policy distributes flow in proportion to reward rather than concentrating it all on the single highest-reward object, the objects it produces are *diverse*, and this is the property that distinguishes a GFlowNet from a reward-maximizing reinforcement-learning agent, which would learn to produce only the one best object and would therefore be useless as a sampler.

For text, the object under construction is a sequence and the construction proceeds from left to right, one token at a time, so the space of partial constructions is a tree: there is exactly one path from the empty state to any given complete sequence. This tree structure simplifies the training objectives, which enforce a consistency condition on the flow along trajectories; the objectives used in practice include trajectory balance (Malkin et al., 2022) and its sub-trajectory variants (Madan et al., 2023), which improve the assignment of credit along long generative trajectories and are what the reference implementation this thesis builds on employs. The single feature of the whole approach that matters most for the present investigation is this: the reward $R(\boldsymbol{x})$ is only ever *evaluated*, never *differentiated*. The policy learns from scalar scores of complete sequences, and the discontinuity and non-smoothness of the underlying energy landscape, which cripple the two gradient-guided samplers, are simply invisible to it. This makes the GFlowNet the ideal instrument for a specific question: if the difficulty with energy-guided sampling were purely a property of the gradient, then a method that never uses the gradient ought to escape it. Whether it does is the subject of Section 5.10. Following Hu et al. (2024), the GFlowNet is realized here as a parameter-efficient low-rank adapter (Hu et al., 2022) on the very same frozen base model that supplies the energy for the Langevin samplers, so that the amortized and gradient-guided halves of the study operate on one shared model and the comparison between them is not confounded by any difference in the underlying network.

3 Related Work

The work in this thesis sits at the intersection of three lines of research: controllable text generation, energy-based and sampling-based decoding, and the study of the geometry and behaviour of language-model representations. This chapter reviews each in turn. Rather than cataloguing methods, it tells the story of how the field arrived at energy-based sampling as an approach to control, because understanding that trajectory is what makes the unexamined assumption at its heart visible, and it is that assumption which the thesis tests. The chapter closes by stating precisely the gap in the literature that the work addresses.

3.1 Controllable Text Generation

Approaches to controllable generation can be organized by how much they change the underlying model, and reviewing them in that order reveals a progression toward ever lighter-touch intervention, of which energy-based sampling is the logical endpoint.

At one extreme are the **fine-tuning methods**, which adjust the model's parameters so that the desired behaviour is internalized. The earliest of these simply continued training on attribute-specific data, and the modern and dominant form is reinforcement learning from human feedback (Ouyang et al., 2022), which optimizes the model against a learned reward that encodes human preferences. These methods work, and the widely used aligned assistants are their fruit, but the control they produce is fixed once training is complete. A new attribute requires new data and a new training run, and the cost of that adaptation is precisely what motivates the search for methods that impose control at inference time. Conditioning methods such as CTRL (Keskar et al., 2019) occupy a middle ground, prepending control codes to the input so that a single model can be steered among a fixed set of attributes anticipated during training, but they too cannot accommodate a genuinely novel constraint after the fact.

At the other extreme are the **decoding-time methods**, which leave the model's parameters entirely untouched and intervene only during generation. These are the methods most relevant to this thesis, because they are the ones that promise the inference-time flexibility that fine-tuning cannot offer. Plug and Play Language Models (Dathathri et al., 2020) perturb the hidden states of a frozen model at each step in the direction indicated by an attribute classifier's gradient, nudging the generation toward the desired attribute without altering any weight. FUDGE (Yang and Klein, 2021) and GeDi (Krause et al., 2021) take a complementary approach, reweighting the next-token distribution using a discriminator that predicts, from a prefix, whether the eventual completion will bear the desired attribute. These methods are ingenious and effective within their scope, but they share a structural feature that limits them: they remain autoregressive. They modify the local, per-token decision, but they retain the strictly left-to-right generation order and therefore inherit its fundamental inability to revise an earlier token in light of a later one. A method that could edit any part of the sequence in light of the whole would be strictly more powerful, and it is the pursuit of that capability that leads to the sampling-based methods.

3.2 Energy-Based and Sampling-Based Decoding

The decisive conceptual move, and the one that defines the family of methods this thesis studies, is to abandon left-to-right generation entirely and to cast decoding as sampling from an energy defined over whole sequences. Once a sequence is treated as a single object drawn from a distribution, rather than as a stream of tokens produced in order, a global constraint can be imposed on the completed object directly, and any position in the sequence can be revised in light of any other. This is the promise that makes the approach attractive, and several influential systems realize it.

COLD decoding (Qin et al., 2022) performs energy-based constrained generation by running Langevin dynamics on a continuous relaxation of the sequence, combining a fluency energy derived from a language model with task-specific constraint energies, and it demonstrated the approach on tasks such as constrained generation and abductive reasoning. MuCoLa (Kumar et al., 2022) samples in the continuous embedding space and projects back to tokens, and it is the closest published antecedent of the continuous sampler studied in this thesis; its formulation of the sampling problem, and its use of a projection to return to discrete text, are exactly the ingredients whose geometric consequences Chapter 5 examines. Related efforts include gradient-based inference for lexically constrained generation and infilling (Sha, 2020), and a body of work on discrete samplers adapted from the energy-based-model literature, including the gradient-informed discrete proposals of

Grathwohl et al. (2021) and the discrete Langevin proposals of Zhang et al. (2022) on which the discrete sampler of this thesis is directly based.

Reading across this literature, a feature recurs that is rarely brought to the foreground, and it is the feature this thesis makes central. The Metropolis–Hastings correction, which as Section 2.3 explained is what separates genuine sampling from biased optimization, is frequently omitted. Methods are described in the language of sampling from an energy while in practice running a noise-augmented gradient descent, and where the correction is included, the difficulty of computing its reverse-proposal term on a landscape defined by projection is seldom examined. This omission is understandable, because including the correction faithfully often makes these methods stop working, but the reason it makes them stop working is itself informative, and recovering that reason is one of the contributions of this thesis. Taking the correction seriously, implementing it faithfully for both a discrete and a continuous sampler, and measuring exactly what its inclusion reveals is a lens through which the failure of the whole approach comes into focus.

A distinct branch of the same literature deserves to be isolated here, because it anticipates the diagnosis this thesis reaches. Not every method that samples from a frozen-language-model energy uses the *gradient* of that energy. Mix-and-Match (Mireshghallah et al., 2022) performs controllable generation by Metropolis–Hastings over a product-of-experts energy, proposing token substitutions from an auxiliary masked language model and scoring them with the frozen energy, so that the energy is only ever evaluated, never differentiated. The twisted sequential Monte Carlo methods of Lew et al. (2023) and Zhao et al. (2024) likewise sample from an intractable posterior defined by a frozen model using importance weights and resampling rather than energy gradients. That these gradient-free samplers operate successfully on essentially the same frozen-LM energies that defeat the gradient-guided methods is, in retrospect, a strong hint about where the difficulty lies, and this thesis makes the hint concrete: a gradient-free Gibbs sampler run on exactly the energy the Langevin samplers navigate, reported in Section 5.5.2, recovers coherent text on the same benchmark. That in-platform result is the clean separation between evaluating the energy, which works, and differentiating it, which does not, and it is what licenses narrowing the thesis's claim to the gradient rather than the energy.

3.3 Amortized Sampling and GFlowNets

Generative Flow Networks were introduced by Bengio et al. (2021) as a method for sampling discrete, compositional objects in proportion to a reward, with the explicit motivation of producing diverse high-reward samples rather than the single mode that reward-maximizing reinforcement learning tends to collapse onto. The framework was subsequently formalized and given a theoretical foundation (Bengio et al., 2023), and a family of training objectives was developed to address the practical difficulty of assigning credit along long generative trajectories, including trajectory balance (Malkin et al., 2022) and its partial-episode sub-trajectory refinement (Madan et al., 2023). The application most directly relevant here is that of Hu et al. (2024), who framed a range of language tasks, including infilling and reasoning, as sampling from an intractable posterior defined by a frozen language model, and trained a GFlowNet to perform that sampling, reporting that the resulting policies yield diverse samples that transfer better to downstream use than those obtained by reward maximization.

The GFlowNet component of this thesis builds directly on that formulation and on the accompanying codebase, using the same modified sub-trajectory objective and the same parameter-efficient adaptation of a frozen base model (Hu et al., 2022). The question posed here, however, is narrower and more critical than the one those works set out to answer. Their concern is whether amortized sampling is useful, and they show that it is, on tasks where the reward defines a well-behaved target. The concern here is whether amortized sampling *escapes a specific pathology*, namely that of the frozen autoregressive likelihood used as an energy over free-form sequences, which the first half of this thesis shows defeats the gradient-guided samplers. These questions are compatible, and their answers turn out to be compatible too: the difference lies entirely in whether the reward defines a landscape with a coherent-text optimum, and the contribution of Chapter 5 is to show, by direct experiment, that the frozen likelihood does not, so that amortization inherits rather than escapes the problem.

3.4 The Geometry and Behaviour of Language-Model Representations

A final and complementary line of work concerns the representations in which these samplers operate and the behaviour of the likelihood they optimize, and it supplies part of the explanation for the results. Two phenomena from this literature are directly relevant.

The first is **anisotropy**. A well-documented property of the embeddings produced by Transformer language models, both contextual and static, is that they do not spread uniformly through the representation space but instead occupy a narrow cone, so that two randomly chosen token embeddings have high average cosine similarity rather than being roughly orthogonal as one might naively expect (Gao et al., 2019; Ethayarajh, 2019). This has a direct consequence for any method that treats Euclidean or cosine distance in embedding space as a meaningful measure of similarity, because anisotropy compresses the dynamic range of those distances: the difference between a good and a bad token, measured in the embedding metric, is smaller than the geometry would suggest. This thesis measures the anisotropy of both models it studies and shows that it renders Euclidean distance an unreliable evaluation metric, which is one reason the experiments adopt a divergence-based measure of contextual fit instead, and it shows in addition that the anisotropy scale differs enough between models to make a single step-size schedule non-transferable, a consequence the prior work did not draw.

The second is the failure of high-probability text to be high-quality text. Holtzman et al. (2020) demonstrated that the most probable sequences under a neural language model are frequently degenerate, repetitive, and unnatural, and that maximization-based decoding such as beam search produces markedly worse text than sampling from the same model. This observation is central to the interpretation of the present results, because it means that any procedure which succeeds in concentrating probability mass on the lowest-energy regions of the frozen-likelihood energy is thereby concentrating it on degenerate text. This thesis reproduces the phenomenon quantitatively on its own models, terms it the *likelihood trap* in the context of energy-based sampling, and uses it to argue that the objective the samplers pursue is itself undesirable, quite apart from whether they can pursue it effectively.

3.5 Diffusion Models for Text and the Score Function

A final body of work is directly relevant not to the methods this thesis evaluates but to the positive control it proposes, and so it is reviewed here to prepare the discussion of Chapter 6. Diffusion models, which generate by gradually reversing a noising process, have become the dominant approach to image generation and have been adapted to text in two broad families. The first embeds tokens into a continuous space and runs a continuous diffusion there, as in Diffusion-LM (Li et al., 2022), which demonstrated controllable generation by applying classifier guidance to the continuous latents at each denoising step. The second operates directly on discrete tokens, as in the score-entropy discrete diffusion of Lou et al. (2024), which learns the ratios of the data distribution over a discrete state space and has narrowed much of the quality gap between diffusion and autoregressive language models.

What makes these models pertinent to this thesis is not their generation quality but their training objective. The foundational result connecting them to the samplers studied here is that of Vincent (2011): learning to denoise a corrupted input is mathematically equivalent to learning the score of the data distribution, the gradient of its log-density, and it is this equivalence that gives score-based generative modeling its name and its mechanism (Song and Ermon, 2019; Ho et al., 2020). A diffusion model, unlike an autoregressive one, is therefore trained precisely to provide the gradient field that Langevin dynamics requires, at every noise level and across the whole space rather than only on the data manifold. This is exactly the property that the results of this thesis find the autoregressive likelihood to lack, which is why, as Chapter 6 develops, a diffusion language model is the natural instrument for a positive control: it holds the sampling method fixed and changes only the training objective that shapes the energy, and so it can isolate that objective as the cause of the failure this thesis diagnoses.

3.6 The Gap Addressed by This Thesis

The literature reviewed above establishes energy-based sampling as an attractive and actively pursued route to controllable generation, and it supplies both the samplers this thesis studies and the amortized alternative it examines. What the literature does not supply is a direct, controlled test of the assumption on which the entire enterprise rests, namely that the gradient of a frozen autoregressive likelihood is a usable search direction on the discrete text manifold, together with a mechanistic account of what happens when that assumption fails. Negative observations, where they appear, tend to be reported in passing, as brittleness, as sensitivity to hyperparameters, or as a need for careful tuning, rather than isolated as a phenomenon and explained. This thesis addresses that gap directly. It designs an ablation that cleanly separates the direction of the gradient from its magnitude, and thereby tests the assumption rather than a proxy for it; it constructs diagnostic experiments that identify the geometric, statistical, and analytic mechanisms behind the result; and it shows that the same mechanism survives amortization, which locates the cause in the training objective of the language model rather than in any particular sampling algorithm. The result is not merely the observation that these methods fail, which is implicit in the difficulty the field has had with them, but an explanation of why, supported at every step by measurement.

4 Methodology

This chapter describes the experimental platform in the detail needed to reproduce the study, and it does so in a particular order: it first establishes the task and the models, then the samplers and the metrics, and only then the diagnostic experiments, because the diagnostics were designed after the main experiments and in response to what those experiments revealed. Where a choice was made, the reasoning behind it is given, because in a diagnostic study the design of each measurement is part of the argument. The chapter sets out the masked-recovery task and why it was chosen, the five energy functions, the two samplers and the factorial space of configurations, the evaluation metrics and the justification for each, the GFlowNet training setup, the constrained-generation extension, and finally the four diagnostic experiments, each introduced together with the question it was built to answer.

4.1 Task: Masked Token Recovery

A sampler can only be studied if there is a way to tell whether it is being guided toward good sequences or is merely wandering. An open-ended generation task provides no such handle, because there is no reference against which to judge the output: a fluent but arbitrary continuation is neither clearly a success nor clearly a failure of the sampler, and one cannot separate the sampler's contribution from the difficulty of the underlying generation. The study therefore adopts **masked token recovery**, also called infilling, as its diagnostic bench. A sequence is drawn from the corpus, one or more of its tokens are replaced with random tokens, and the sampler is asked to restore a coherent sequence by resampling only the corrupted positions while the rest of the sequence is held fixed. Because the original sequence is known, the recovery can be scored; because the corruption and the surrounding context are controlled, the behaviour of the sampler can be attributed to the method rather than to the open-endedness of the task. This is the standard reason infilling is used as a diagnostic for controllable generation, and it is the reason it is used here.

An important subtlety shapes the choice of evaluation metric and is worth stating at the outset, because it recurs when the metrics are defined. The token a sampler recovers need not be the original token in order to be correct. If the corrupted word *roommate* is recovered as *dorm*, the resulting sequence may be perfectly coherent even though it does not match the reference. An exact-match criterion would wrongly score this as a failure, and would thereby confound the quality of the sampler with the arbitrariness of the particular reference sentence. For this reason the primary evaluation is based not on token identity but on how well the recovered sequence fits its context under the model's own predictive distribution, a point developed in Section 4.4.

The corpus is **ROCStories** (Mostafazadeh et al., 2016), a collection of short five-sentence everyday narratives. It is chosen for three reasons that matter for a controlled diagnostic. Its sentences are short and syntactically simple, so a single corrupted token has a reasonably well-defined and locally checkable correct recovery. Its domain is narrow and consistent, so the model's uncertainty at a masked position reflects the behaviour of the sampler rather than the boundless open-endedness of unrestricted text. And it is the corpus used by the amortized-inference reference implementation of Hu et al. (2024), so the GFlowNet half of the study operates on the same data distribution as the work it builds on, which removes a possible confound from the comparison between the gradient-guided and amortized halves of the study.

4.2 Energy Functions

The study evaluates five energy functions, chosen to let the central findings be tested for robustness across training regimes and across architectures.

The reference energy is a **GPT-2 Large** model (Radford et al., 2019), of 774 million parameters and embedding dimension $D = 1280$, supervised fine-tuned on ROCStories. This same fine-tuned model serves as the base for the amortized experiments, so that the Langevin and GFlowNet halves of the study share weights and are not confounded by any difference in the underlying network, an identity that is essential to the unifying experiment of Section 5.10. Three **GFlowNet-tuned variants** of this base model are included, produced as described in Section 4.5: two trained with the unnormalized-length reward setting, at 500 and 2000 optimization steps respectively, so that the effect of longer training can be observed, and one trained with the length-normalized setting at 500 steps. Finally, a **Llama-3 8B** model (Dubey et al., 2024), with embedding dimension $D = 4096$, serves as a cross-architecture control. Because Llama-3 uses a different tokenizer and a markedly different embedding geometry, and therefore a different step-size scale, it is never placed on a shared numerical axis with the GPT-2 family; it is used to establish that the qualitative findings, such as the gradient fallacy and the Metropolis–Hastings breakdown, hold across architectures rather than being artifacts of one model. The four GPT-2-based models, by contrast, share the tokenizer, the corruption seeds, and the step-size schedule, and their numbers are therefore directly comparable without qualification.

4.3 Samplers and Configurations

The two samplers of Section 2.4, the Discrete Langevin Sampler (DLS) and the Continuous Langevin Sampler (CLS), are each run under a factorial combination of design choices. The two are studied together, rather than either alone, because they represent the two opposite strategies for confronting the discreteness of text and therefore isolate different failure modes: the DLS, following Grathwohl et al. (2021) and Zhang et al. (2022), keeps the state in token space and uses the gradient only to build a proposal, which tests whether the gradient is an informative ranking signal; the CLS, following Kumar et al. (2022), relaxes into continuous space and runs genuine Langevin dynamics, which tests whether continuous Markov chain Monte Carlo is compatible with a projected discrete energy. A finding that holds for both is a finding about the energy rather than about one sampler's idiosyncrasies, which is why both are needed.

The design axes are as follows. The **proposal method** is one of three. The *policy* method uses the true gradient of the energy. The *grad-norm-preserved random* method replaces the gradient with a random vector rescaled to the exact norm of the true gradient, so that it differs from the policy only in direction. The *random* method uses a random vector of random magnitude. The comparison among these three is the central ablation of the thesis, isolating respectively the full gradient, the effect of its direction alone, and a pure baseline, and the reasoning behind this three-way design is developed where it is used, in Section 5.5. The **Metropolis–Hastings correction** is either enabled or disabled, so that the effect of theoretical correctness can be observed directly. **Gradient normalization**, which rescales the gradient to unit norm for numerical stability, is either enabled or disabled; this choice interacts with the proposal-method ablation in a way that turns out to be critical and is explained in Section 5.5. The **annealing schedule** runs for either 50 or 100 steps, with the step size decreasing linearly; for the GPT-2 family the schedule runs from $\epsilon_{\text{start}} = 10.5$ to $\epsilon_{\text{end}} = 0.1$, a scale established by the calibration described in Section 5.1. An *oracle* variant, in which the ideal step size at each iteration is selected using knowledge of the ground-truth token, is included as an upper-bound reference; its purpose is to rule out the possibility that the samplers fail merely for want of a well-tuned schedule, since if even an oracle-tuned schedule does not let the gradient outperform a random direction, the failure cannot be one of calibration. The oracle sweep is run independently for each proposal method, with a separate per-method oracle run rather than a single schedule shared across methods, so the schedule selection cannot advantage the policy gradient over the random comparators. In total, across the five energy functions, the study comprises 145 sampling configurations, each evaluated on 200 sequences; the arithmetic of that count is given in Appendix A.2.

4.4 Evaluation Metrics

Three quantities are recorded, and the choice among them is itself informative.

The first is the **Euclidean distance** in embedding space between the recovered token and the ground-truth token. This is the most obvious measure of recovery, and one might expect it to be the natural criterion. It proves to be unreliable, for a reason established quantitatively in Section 5.8: because language-model embeddings are anisotropic, occupying a narrow cone in the representation space (Ethayarajh, 2019), Euclidean distance compresses the distinction between good and bad tokens, so that a token can be geometrically close to the target while being grammatically inadmissible, and conversely a perfectly good synonym can lie far away. It is reported for completeness, and studied as an object in its own right, but it is not relied upon as the primary criterion.

The primary criterion is the **KL divergence** (Kullback and Leibler, 1951) between the model's predictive distribution at the recovered position and a reference distribution, which measures how well the recovered token fits its context, that is, its *contextual coherence*. Concretely, let $\mathcal{M}$ be the set of masked positions in a sequence of length L, and let $\mathcal{M}' = \{\, m \in \mathcal{M} : m < L - 1 \,\}$ be those masked positions that have a right neighbour, the only ones at which the metric is defined. For a position $m \in \mathcal{M}'$, write $p_{\text{ref}}^{(m)}$ for the model's next-token distribution at m when every masked position holds its *ground-truth* token, and $p_{\text{pred}}^{(m)}$ for the same distribution when the masked positions instead hold the *recovered* fill. The metric is the mean divergence over these positions,

$$\overline{\mathrm{KL}} = \frac{1}{|\mathcal{M}'|} \sum_{m \in \mathcal{M}'} \mathrm{KL}\Big(p_{\mathrm{ref}}^{(m)} \,\Big\|\, p_{\mathrm{pred}}^{(m)}\Big)\,, \tag{4.1}$$

which is exactly the quantity logged by the sampler in `core/base_sampler.py` and recomputed in the revision diagnostics. By construction the ground-truth fill gives $\overline{\mathrm{KL}} = 0$, since then $p_{\mathrm{pred}}^{(m)} = p_{\mathrm{ref}}^{(m)}$ at every position, so the metric has a hard floor of zero that a perfect recovery attains. This metric is chosen over exact-match accuracy for the reason given in Section 4.1: a recovered token that differs from the reference may still be perfectly coherent, and a metric that penalized it would confound the sampler's quality with the arbitrariness of the reference. A divergence between predictive distributions measures contextual fit directly and is insensitive to which particular admissible token was chosen. One limitation of the divergence measure must be acknowledged, because it shapes how the results are argued: its magnitude depends on the position of the corrupted token within the sequence, since corrupting an early token perturbs the conditional probability of a long suffix and yields a large divergence, whereas corrupting a final token yields almost none. For this reason no model is ranked on small differences in KL divergence; the central results are argued from the *absence* of a gap between conditions rather than from a small measured gap, and a null gap is robust to this positional dependence in a way that a narrow measured win would not be. To make "absence of a gap" a decision rather than an impression, the analysis fixes an equivalence margin in advance, at five percent of the policy method's mean KL for each configuration, and treats two methods as practically equivalent when their paired mean difference falls inside that margin. The margin is calibrated to the pipeline's own noise rather than chosen to fit the result: as Section 4.8 reports, the across-seed standard deviation of the final KL for the flagship configuration is 0.183, comfortably inside the corresponding margin of about 0.327, so the threshold below which a difference is deemed negligible is smaller than the study's stated ceiling on it yet still larger than a single standard deviation of seed noise.

Finally, for the generation-quality analyses, the study records **repetition rate**, the fraction of repeated n-grams, and **distinct-n**. These are the standard instruments for detecting the degeneration studied by Holtzman et al. (2020), and they are therefore the appropriate measures for the likelihood trap. For the constrained-generation extension the study records classifier accuracy and **BERTScore** (Zhang et al., 2020), the latter chosen because it measures semantic similarity through contextual embeddings rather than surface overlap, and so does not penalize a valid paraphrase of the reference.

4.5 GFlowNet Training

The GFlowNet-tuned variants are produced following the amortized-inference formulation of Hu et al. (2024), using their modified sub-trajectory-balance objective (Malkin et al., 2022; Madan et al., 2023). The GFlowNet is chosen as the amortized alternative, over reward-maximizing reinforcement learning, for a reason specific to the question of this thesis: a reward-maximizing agent would converge on the single highest-reward sequence and so could not distinguish a sampler that explores the energy from one that merely finds its mode, whereas a GFlowNet is trained to sample in proportion to reward and therefore tests amortized *sampling* rather than amortized optimization. Crucially for the argument, it only ever evaluates the reward and never differentiates it, so it is blind to exactly the gradient pathology that afflicts the Langevin samplers, which makes it the ideal instrument for asking whether that pathology can be sidestepped.

The policy is a low-rank adapter (Hu et al., 2022) on the frozen GPT-2 Large base, and the reward is the log-likelihood assigned by that frozen base to the completed sequence, so that the GFlowNet is trained to sample in proportion to the model's own likelihood, exactly the energy the Langevin samplers navigate. The task is a story-infilling formulation in which, given a preceding context and a known ending, the policy generates a connecting span, and the reward scores the concatenation of context, generated span, and ending under the frozen model. A single hyperparameter, denoted `len_beta`, controls the length normalization of the reward: at `len_beta` $= 0$ the reward is the raw, unnormalized log-likelihood, and at `len_beta` $= 1$ it is fully normalized by sequence length. The consequences of this single knob, which turn out to be dramatic and to produce two opposite degeneracies, are one of the central findings of the GFlowNet study and are developed in Section 5.10.

4.6 Constrained Generation Extension

To test the plug-and-play claim of (1.2) directly, which is the object of RQ4, a constrained-generation extension augments the energy with a sentiment constraint, in the manner of the gradient-based constrained samplers of Kumar et al. (2022) and Qin et al. (2022). A sentiment classification head is trained on the representations of the GPT-2 Large base and used to construct the constraint term $C(\boldsymbol{x})$. Generation is then run under five constraint modes designed to isolate the contribution of the constraint gradient. The *lm_only* mode uses only the fluency energy, providing a baseline with no steering. The *cons_only* mode uses only the constraint, showing what the constraint can do in isolation. The *full* mode combines them with weights $\beta_{\text{LM}} = 0.8$ and $\beta_{\text{C}} = 0.2$, which is the intended plug-and-play configuration. The *random* mode replaces the combined gradient with noise, and the *cons_random* mode replaces only the constraint gradient with noise while retaining the fluency gradient, so that the specific contribution of the constraint direction can be separated from that of the fluency direction. The steering effect is measured as the change in the fraction of generations classified as the target sentiment relative to the unconstrained baseline. Both the discrete sampler in the formulation of this thesis and a MuCoLa-style continuous baseline are evaluated, on both infilling and continuation, so that the finding does not depend on a single sampler or task.

4.7 Diagnostic Experiments

The configurations above establish *that* the methods fail. They do not, on their own, establish *why*, and the why is the contribution. Four additional diagnostic experiments were therefore designed, after the main results were in hand and in direct response to them, each targeting one candidate mechanism. Unlike the configurations above, these do not sample; they measure properties of the energy landscape and of the gradient directly, and each is introduced here together with the question that motivated its construction.

The **linearization experiment** was built to answer the question raised by the gradient-fallacy result: if the gradient carries no usable information, is that because the first-order surrogate the discrete proposal relies on fails to predict the true change in energy? For each of 200 sequences and a masked position, the exact gradient at that position is computed, and for a stratified set of 2000 candidate tokens the surrogate predicted change $\hat{\Delta}(v) = \boldsymbol{g}^\top(\boldsymbol{e}(v) - \boldsymbol{e}(x_i))$ is compared against the true change $\Delta(v) = \log p(\boldsymbol{x}_{i\to v}) - \log p(\boldsymbol{x})$ obtained by an exact forward pass, with the embedding distance of each candidate recorded so that the correlation can be examined as a function of distance. The true change is further decomposed into a *self* term, the change in the log-probability of the candidate given its own left context, and a *future* term, the change in the log-probability of the suffix, because the gradient with respect to the input embedding is structurally able to register only the latter, and measuring both quantifies how much signal the gradient is blind to.

The **acceptance experiment** was built to answer the question raised by the Metropolis–Hastings breakdown: when the correction rejects nearly every useful move on the continuous sampler, is the rejection driven by the target term, meaning the moves are genuinely bad, or by the proposal term, meaning the moves are good but the correction cannot verify their reversibility? For every proposal it logs whether the proposal crossed a Voronoi cell boundary, whether it was accepted, and the decomposition of the acceptance ratio into its target and proposal terms, which lets the cause be located unambiguously.

The **trajectory experiment** logs the raw state of the sampler at every step for a small number of sequences, so that the path can be projected into two dimensions and inspected, and it records the distance from the continuous state to the nearest token embedding, which quantifies how far the sampler drifts from the token manifold over the course of a run. The **likelihood-trap experiment** was built to answer a more basic question that the failure of the samplers raised: even if the samplers could reach low energy, would low energy be a good place to be? It decodes with several strategies, from greedy and beam search to ancestral sampling, and measures the relationship between the per-token likelihood of the generated text and its repetition rate, establishing on the study's own models whether the lowest-energy text is in fact the worst text. A final **anisotropy analysis** measures the distribution of inter-token distances and related geometric quantities in the two embedding spaces, underpinning both the step-size result and the unreliability of Euclidean distance.

The scale of the configuration grid, one hundred and forty-five sampling configurations each evaluated on two hundred sequences, together with the separate diagnostic runs across five models, required an experimental infrastructure that could distribute work reliably across several machines and recover cleanly from interruptions. The runs are organized as a manifest of jobs, each annotated with its memory requirement, and dispatched by a resumable file-queue system that assigns jobs to available devices according to that requirement, so that a large model and a small one are not placed on the same device when their combined memory would exceed it. Completed jobs are marked so that an interrupted sweep resumes rather than restarts, which matters when a single sweep runs for many hours. All experiments use this system, and all sampling code was verified against a reference implementation with a bitwise-equivalence test suite before the runs reported here were launched, so that the diagnostic instrumentation is known not to have altered the behaviour of the samplers it measures.

4.8 Reproducibility, Seeds, and Hardware

Because the central claims of this thesis are claims about the *absence* of an effect, the run-to-run stability of the pipeline is itself part of the evidence and is quantified directly rather than assumed. Rerunning the flagship configuration, the discrete sampler on GPT-2 Large with the Metropolis–Hastings correction and gradient normalization enabled over a fifty-step schedule, under four independent corruption seeds, 1000 through 1003, yields final KL divergences of 6.348, 6.589, 6.150, and 6.291, with a mean of 6.345, a standard deviation of 0.183, and a range of 0.438. This across-seed standard deviation fixes the scale below which a difference in KL cannot be told apart from noise, and it is the empirical basis both for the equivalence margin defined in Section 4.4 and for the bounded-effect reading of the gradient-fallacy comparisons in Section 5.5. The corruption procedure is deterministic per sample index, with the seed set to the data seed plus the running index, so the corrupted inputs, and therefore the paired comparisons, are identical across the three method runs of a configuration and can be reproduced bit for bit; this is what allows the statistical analysis of Section 5.5 to pair the policy method against its comparators on the same sample without any rerun.

All reported experiments were run on a university GPU server equipped with 49 GB A6000-class accelerators, under PyTorch 2.12 with CUDA 13.0. For code and data availability, the sampler and diagnostic implementations, together with the bitwise-equivalence suite (`verify_equivalence_suite.py`) that certifies the instrumented code against the reference, and the manifests that define every configuration in the grid, are retained with the thesis so that the full set of runs can be regenerated from the seeds above. The accompanying `README.md` documents the environment, the exact rerun command for every experiment family, and an artifact map that links each table and figure in this thesis to the script that produces it and the result file it draws from, so that every quantitative claim is traceable from the number in the text to the file on disk.

5 Results

This chapter presents the experimental findings as a connected investigation. It proceeds in the order in which the study actually developed, because that order is also the logical one: each experiment is motivated by a question that the previous result left open, and each ends by naming the question that motivates the next. The reader who follows the chapter in order will therefore see not only what was found but why each step was taken, which for a diagnostic result is the substance of the contribution. The investigation begins with a practical obstacle, the calibration of the samplers, which unexpectedly reveals the influence of embedding geometry. It proceeds to the annealing dynamics of the discrete sampler, where an apparent convergence turns out to be an artifact. It then confronts the Metropolis–Hastings correction, and finds that the same correction which rescues the discrete sampler destroys the continuous one, for a reason it measures precisely. This leads to the central result, the gradient fallacy, that following the true gradient is indistinguishable from following a random direction, on all five energy functions. Three diagnostic sections then supply the mechanism, through the linearization radius, the likelihood trap, and the anisotropy of the embedding space. The chapter closes by asking whether amortization escapes the failure, and finding that it does not, and by testing the plug-and-play constraint directly, and finding that it does not steer. Throughout, each reported number is annotated in the source of this document with a comment naming the results file from which it was drawn, so that every quantitative claim is traceable to the experiment that produced it, without cluttering the text the reader sees.

5.1 A First Obstacle: Step-Size Calibration and Embedding Geometry

The investigation began with a simple intention: to run the two samplers on GPT-2 Large using a step-size schedule that had already been made to work on Llama-3, and to compare their behaviour. It is worth being exact about what "work" means in this sentence, because the whole chapter turns on the distinction. Here a schedule works if it produces *calibrated motion*: the sampler's projected tokens change over the run and its metrics evolve rather than freezing, so the chain is at least exploring the space. It says nothing yet about whether that motion is *guided* toward better sequences, which is the separate and harder property the later sections test and find absent. The expectation was that the schedule would transfer, perhaps with minor adjustment, since both are Transformer language models and the samplers are the same. That expectation was wrong, and the way in which it was wrong is the first substantive finding.

On GPT-2 Large, with the Llama-tuned schedule, the samplers did not merely perform poorly; they did not move at all. The entropy of the discrete proposal collapsed to zero within a few steps and no token was ever changed. The sampler was not broken but trapped, and diagnosing why led directly to the geometry of the embedding space. The anisotropy analysis, reported in full in Section 5.8, measures the mean distance between neighbouring token embeddings as approximately 1.82 in GPT-2 Large against approximately 0.59 in Llama-3, and the mean pairwise distance as 2.77 against 0.84, a difference of roughly a factor of three in each case. The reason for the trapped sampler follows immediately. A step size calibrated for the tighter geometry of Llama-3 is simply too small to carry the continuous state across a Voronoi cell boundary in the more dispersed geometry of GPT-2, so the projected token never changes and the sampler sits frozen in its initial cell. Recovering the correct scale required an oracle sweep over candidate step sizes, which yielded a schedule running from $\epsilon_{\text{start}} = 10.5$ to $\epsilon_{\text{end}} = 0.1$ for the GPT-2 family, and this schedule is used throughout the rest of the study.

This obstacle is worth reporting rather than quietly fixing, for two reasons. The immediate practical lesson is that there is no universal step-size schedule for these methods; the correct scale is a property of the specific model's embedding geometry and must be rediscovered for each model. That alone qualifies the plug-and-play framing, since a method that must be recalibrated to each model's geometry before it will move at all is not, in the strong sense, model-agnostic. But the obstacle also raises a sharper question, and it is this question that motivates the ablation at the heart of the thesis. Once the step size is calibrated so that the sampler does move, does it move *toward better sequences*, guided by the gradient, or does it merely move? Calibration ensures motion; it does not ensure that the motion is informed. The oracle configuration, which selects the ideal step size at each iteration using knowledge of the ground-truth token, was introduced precisely to separate these possibilities, because it establishes the best any schedule could do and thereby distinguishes a failure of calibration from a failure of guidance. This question is taken up in earnest in Section 5.5, and answering it consumes the rest of the chapter. This section, meanwhile, has answered the first half of RQ1: theoretically faithful

Langevin dynamics cannot be applied off the shelf, because the step size that makes the sampler move at all is model-specific.

5.2 An Apparent Convergence: Annealing Dynamics and the Quenching Effect

With the step size calibrated, the discrete sampler moves, and the next question is how it behaves over the course of a run. Run without the Metropolis–Hastings correction, it exhibits a striking and suggestive pattern: its metrics remain noisy and unimproving for the majority of the annealing schedule and then, near the end, drop abruptly to much better values. The natural reading of this pattern, and the one we initially entertained, is that the sampler has finally converged: that after a long period of exploration it has located a good sequence and settled into it. If that reading were correct, it would be modest good news for the method.

A controlled test refutes the reading, and the design of the test is the point. If the late drop marked genuine convergence, it would occur when the model had found a good sequence, and its timing would be governed by the difficulty of the problem, not by the schedule. If instead the drop were an artifact of the annealing schedule, its timing would track the schedule: lengthen the schedule, and the drop would move. We therefore extended the schedule from 50 steps to 100 and observed where the drop occurred. It moved, from around step 40 to around step 85, tracking the schedule rather than remaining fixed at the point where a good sequence might have been found. The drop is therefore not convergence but what we term a **quenching effect**, and Figures 5.1 and 5.2 show it at the two schedule lengths.

The mechanism is a direct consequence of the annealing schedule interacting with the absence of the correction. Early in the schedule the step size is large and the injected Gaussian noise dominates the update, so the sampler, lacking any mechanism to reject a bad proposal, bounces stochastically across the vocabulary, exhibiting high entropy and poor contextual fit. As the schedule anneals the step size toward zero, the noise vanishes, and the update degenerates into deterministic gradient descent, which collapses greedily into whatever local optimum happens to be nearest. The abrupt late-schedule drop is the signature of this collapse. The name is borrowed from metallurgy, where quenching denotes a rapid cooling that freezes a material into whatever configuration it happened to occupy at the moment the cooling began, and the analogy is exact: removing the noise freezes the sampler into its current local optimum rather than allowing it to continue exploring toward

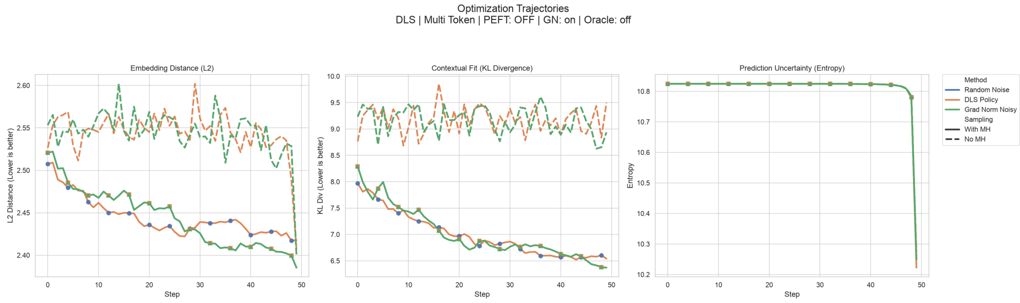

Figure 5.1: Discrete Langevin Sampler trajectories on GPT-2 Large over a 50-step schedule, averaged across sequences. The metrics remain unimproving for most of the run and drop abruptly near the end as the noise schedule approaches zero, the signature of quenching rather than convergence.

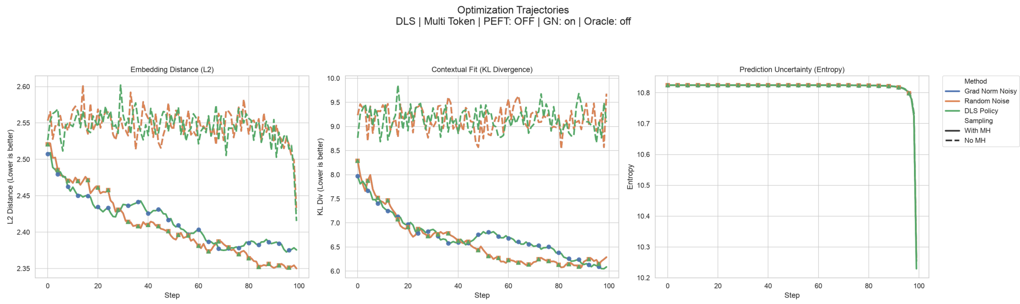

Figure 5.2: The same sampler over a 100-step schedule. The abrupt drop has moved from around step 40 to around step 85, tracking the annealing schedule rather than remaining fixed. This confirms that the drop is an artifact of cooling and not evidence of model convergence.

a better one. The drop looks like convergence because the metrics improve, but it is the improvement of a system that has stopped moving, not one that has found the right place.

There is a revealing contrast when the Metropolis–Hastings correction is enabled: the quenching pattern does not occur. The chain converges smoothly and early, within the first twenty steps, and remains stable thereafter, maintaining a low divergence for the rest of the schedule. In the discrete sampler, then, the theoretically motivated correction is also the empirically better choice, and this is worth marking clearly precisely because it is the exception. It is the one place in the entire study where making the sampler correct also makes it work. That observation naturally raises the next question, and it is a question the correction itself poses: if the correction rescues the discrete sampler, what does it do to the continuous one, whose landscape, as Section 2.4 warned, violates the smoothness the correction assumes? The next section takes up that question, and the answer is the opposite of what the discrete case might lead one to expect.

5.3 A Correction That Breaks the Sampler: The Metropolis–Hastings Breakdown in Continuous Space

The result of the previous section, that the Metropolis–Hastings correction improves the discrete sampler, sets up a natural expectation: that the correction, being theoretically motivated, ought to improve the continuous sampler too, or at least not harm it. Section 2.3 gave a reason to doubt this expectation, namely that the continuous sampler's energy is defined through a projection whose drift is discontinuous, violating the Lipschitz condition the correction relies on. The experiment resolves the tension decisively and in favour of the doubt, and it produces what is, in our assessment, the single most striking result of the sampler study.

Applied to the continuous sampler, the correction does not rescue it, and the trace experiment shows why in a way the discrete case makes vivid. Decomposing the logged proposals by whether they crossed a Voronoi cell boundary, and therefore changed the projected token, the continuous policy sampler with gradient normalization disabled and *without* the correction accepts within-cell proposals, which change no token, 0.03% of the time and boundary-crossing proposals 3.7% of the time; enabling the correction raises these only to 0.63% within a cell and 8.56% across a boundary. The discrete sampler under the same correction, by contrast, accepts within-cell moves with probability 1.0, because such moves are exactly reversible, and accepts boundary-crossing moves at 9.3%. That single difference, a within-cell acceptance of essentially one for the discrete sampler against essentially zero for the continuous one, is what the correction turns on: it rescues the discrete sampler, whose token can then move freely inside a cell and cross to a neighbour, and leaves the continuous sampler, which cannot even accept a move within its own cell, barely leaving its initial cell across an entire run, as the next summary shows.

The question this raises is the one the acceptance experiment was built to answer: is the correction rejecting these boundary-crossing moves because the moves are bad, or because the correction cannot verify their reversibility? The two possibilities correspond to the two terms of the acceptance ratio, and the decomposition separates them cleanly. For proposals that crossed a boundary, the mean *target* log-ratio is $+4.60$. That is positive, which means these moves, on average, *improve* the sequence probability: the distribution being sampled would welcome them. The moves are good. Yet the mean *proposal* log-ratio for those same moves is -1325.

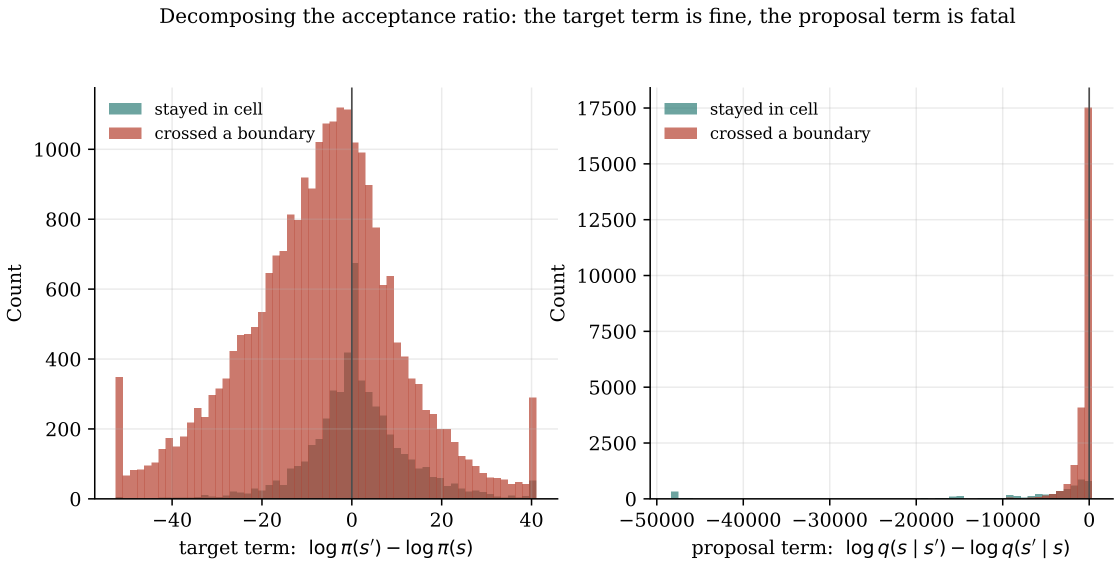

Figure 5.3: Decomposition of the Metropolis–Hastings acceptance ratio in the continuous sampler, for proposals that stay within a Voronoi cell against those that cross a boundary. The target term (left) is well-behaved and often favourable for boundary-crossing moves; the proposal term (right) collapses to large negative values for exactly those moves. The rejection is driven by the proposal term, not the target term.

The reverse proposal assigns effectively zero density to the return move, and this term alone, dwarfing the favourable target term by three orders of magnitude, drives the acceptance probability to zero. Figure 5.3 shows the two distributions side by side: the target term is centred near zero and frequently positive for boundary-crossing moves, while the proposal term for exactly those moves is enormous and negative. The rejection is driven by the proposal term, not the target term.

This is the measured form of the Lipschitz argument of Section 2.3, and it confirms the theory of Roberts and Tweedie (1996) in the breach: the Metropolis-adjusted Langevin algorithm requires a Lipschitz-continuous drift, and the continuous sampler's energy does not have one. When a proposal crosses a cell boundary, the reverse proposal mean is computed from the drift on the far side of the boundary, which, because the drift is discontinuous there, points somewhere unrelated to the direction back toward the original state. The original state therefore falls deep in the tail of the reverse Gaussian, the reverse density vanishes, and the correction, doing exactly what it is designed to do, rejects the move as irreversible. The correction is not malfunctioning; the landscape is violating the assumption on which the correction depends. The consequence is both precise and damaging: the continuous sampler only appears to function when the correction is omitted, that is, when it is run as a biased optimizer that ignores reversibility. Making it a correct sampler paralyzes it entirely. The same behaviour is observed on both GPT-2 Large and

Llama-3, so it is a property of the problem class rather than of a single model, and it is best summarized as a fundamental incompatibility between continuous Markov chain Monte Carlo and a discrete text representation accessed by projection. This answers RQ2 for the continuous sampler: making it theoretically correct does not make it work, it does the opposite, and the reason is now measured rather than conjectured.

The two sampler results so far, the quenching of the discrete sampler and the breakdown of the continuous one, concern how the samplers *move*. Neither has yet asked the more fundamental question that underlies both: whether the gradient they follow, when they do move, carries any useful information at all. That question is the central one, and the next section confronts it directly.

5.4 A Spatial View of the Breakdown: Trajectory Analysis

The acceptance statistics of the previous section describe the breakdown in terms of proposals accepted and rejected. A complementary and more concrete view comes from watching where the sampler actually goes, which is what the trajectory experiment records: for a handful of sequences it logs the full continuous state at every step, along with the distance from that state to the nearest token embedding and the identity of the Voronoi cell the state occupies. Reducing these trajectories to two simple summaries makes the breakdown visible as a matter of geometry rather than of acceptance ratios, and a reduced-dimensionality projection of the same paths into embedding space is given in Appendix A.4.

The first summary is the number of distinct Voronoi cells a run visits over its fifty steps, which is a direct count of how many times the projected token actually changed. The continuous sampler with the Metropolis–Hastings correction enabled visits, on average, about two cells across the entire run, and in half the traced sequences exactly one, so it is essentially frozen at its starting cell. This is the same paralysis reported in Section 5.3, now seen from the side: a sampler that accepts almost no move which changes a token is, spatially, a sampler that barely leaves its initial cell, and the two descriptions are the same fact. By contrast, the continuous sampler with the correction disabled visits an average of about 45 of a possible 50 cells, changing its token at almost every step. It moves freely precisely because, without the correction, nothing checks whether its moves are reversible.

The second summary is the distance from the continuous state to the nearest token embedding, which measures how far the sampler has drifted from the manifold of actual tokens. Here the contrast between the two samplers is at its sharpest and it exposes a difficulty that the discrete sampler avoids by construction. With the correction *on*, the continuous state is quenched between 118 and 151 units from the nearest token across the whole run; with the correction *off*, in the course of forcing its way across cell boundaries, it drifts as far as roughly 980 units from the nearest token embedding, ending its run an average of about 17 units away. Given that the mean distance between neighbouring tokens is only 1.82 and the mean pairwise distance 2.77, an off-manifold distance of 118 to 151 is a factor of roughly 65 to 83 times the nearest-neighbour spacing, so the continuous state spends much of its run in essentially empty space between tokens, where the energy is defined only by an increasingly distant projection. The discrete sampler, by contrast, sits exactly on the token manifold at every step, at distance zero, because its state is always a genuine token and it never enters the continuous interstice at all. It visits an average of about five cells over the run, moving deliberately from token to token rather than either freezing or drifting.

These three trajectories, the frozen continuous sampler under the correction, the freely drifting continuous sampler without it, and the on-manifold discrete sampler, are a compact picture of the whole continuous-sampler predicament. The continuous relaxation buys the ability to run genuine Langevin dynamics, but it pays for that ability by leaving the token manifold, and once off the manifold it faces an impossible choice: either respect the correction and freeze, or ignore the correction and wander through the empty space between tokens where the energy is a poor guide. Neither option samples the intended distribution. The discrete sampler escapes this dilemma by never leaving the manifold, which is why, of the two, it is the one that at least moves sensibly, even though, as the next section shows, the signal it moves by is no better than noise.

5.5 The Central Result: The Gradient Fallacy

Everything to this point has concerned the mechanics of the samplers. The result in this section concerns the signal that drives them, and it is the central finding of the thesis. The samplers use the gradient of the frozen likelihood to decide which way to move; the question is whether that gradient points anywhere useful. The ablation of Section 4.3 was designed to answer exactly this, by comparing three proposal methods: the true gradient (*policy*), a random direction rescaled to the gradient's norm (*grad-norm-preserved*), and a fully random vector (*random*). The three-way design is deliberate and is the methodological core of the thesis, because it separates the two things a gradient can contribute. If the gradient's *direction* is informative, the policy method should outperform the norm-preserved random method, which differs from it only in direction. If the gradient's *magnitude* is informative, the norm-preserved method should outperform the fully random method, which differs from it only in magnitude. The expectation, going in, was that the direction at least would matter: that a gradient computed from the model would point toward better tokens more often than a random vector would, even if the magnitude were noisy.

Before the comparison can be trusted, a confound must be removed, and identifying it is itself part of the contribution, because it is the reason a naive version of this experiment would have given an uninterpretable answer. Gradient normalization, which is applied by default for numerical stability, rescales every proposal direction to unit norm. Under normalization, the norm-preserved random method and the fully random method become the same object, a random unit vector, and so an observed equality between the policy method and the random method admits two incompatible explanations: either the direction of the gradient is genuinely uninformative, or the normalization has thrown away a real signal that lived in the magnitude. The normalized ablation cannot tell these apart. The extended ablation therefore repeats the entire comparison *with gradient normalization disabled*, so that the policy and norm-preserved methods both carry the gradient's full unnormalized magnitude and differ from each other only in direction. Any gap between them is then attributable to direction alone, cleanly. This single design decision, doubling the ablation to disentangle direction from magnitude, is what turns a suggestive observation into a claim that survives scrutiny, and it is the reason the study comprises 145 configurations rather than half as many.

With the confound removed, the result is unambiguous, and it contradicts the expectation. Table 5.1 reports the final KL divergence for the policy and norm-preserved methods with normalization disabled and the correction enabled, across four energy functions, together with their difference. The differences are small and, tellingly, inconsistent in sign. On GPT-2 Large the policy method is worse than the random direction by 0.044; on one GFlowNet variant it is worse by 0.008; on another it is better by 0.146; and on the length-normalized variant the random direction is better by a substantial 0.670. This last gap is the one exception that deserves separate comment, because it exceeds the across-seed noise scale of about 0.18 nats measured in Section 4.8 and so is not attributable to run-to-run variation:

Energy function	Policy	Grad-norm-preserved	Difference
GPT-2 Large (SFT)	6.474	6.430	+0.044
GFlowNet, `len_beta`= 0, 500	6.009	6.001	+0.008
GFlowNet, `len_beta`= 0, 2000	6.698	6.552	+0.146
GFlowNet, `len_beta`= 1, 500	6.008	5.338	+0.670

Table 5.1: Final KL divergence (lower is better) for the true gradient against a norm-preserved random direction, with gradient normalization disabled and the correction enabled, on the discrete sampler at 50 steps. The differences are small and inconsistent in sign, indicating that the direction of the gradient contributes no reliable improvement over a random direction of equal magnitude.

on that variant the true gradient is not merely uninformative but reliably *worse* than a random direction, which points the wrong way for gradient guidance and, far from softening the result, sharpens it. In no case does the true gradient reliably outperform a random direction of the same magnitude. Following the exact gradient is statistically indistinguishable from following a random direction. The direction of the frozen autoregressive gradient carries no usable information for discrete sampling. And because, in the separate comparison, the norm-preserved and fully random methods are themselves indistinguishable, the magnitude carries none either. The gradient, in both its direction and its magnitude, is no better than noise.

To bound this equivalence rigorously rather than rest it on the eye, the flagship configuration was analyzed as a paired comparison over its two hundred sequences, pairing the policy method against the norm-preserved random direction on each sample. For the discrete sampler on GPT-2 Large with the correction and gradient normalization enabled over fifty steps, the paired mean difference in final KL is +0.171 nats, with a 95% bootstrap confidence interval of $[-0.285, +0.619]$ that straddles zero, a Wilcoxon signed-rank p of 0.40, and a difference detectable at eighty percent power only if it were as large as 0.652. These three figures say the same thing three ways: the observed difference is small, it is not statistically distinguishable from zero, and the experiment was powered to detect only differences several times larger than the one it found. The formal equivalence test of Section 4.4 stops just short of certifying equivalence here, but for an instructive reason: the confidence interval, at this sample size, is wider than the equivalence margin of 0.327, so the data cannot yet close the gap in either direction. That is not evidence of an effect; it is a statement that any real effect must be smaller than two hundred sequences can resolve, which is precisely the bounded-effect reading the thesis relies on. The result is therefore reported throughout as “no reliable difference” rather than as an exact numerical equality.

There is a single statistically clear contrast in the entire matrix, and it points the wrong way for gradient guidance rather than the right way. On Llama-3, with the correction disabled, the true gradient produces a *higher* final KL divergence than a random direction: there the gradient is not merely uninformative but actively misleading. We defer the mechanistic explanation of this to the next section, but note the shape of it here, because it is instructive: a gradient that points toward the minimizer of a *local linearization* of a globally discontinuous landscape may point somewhere systematically unhelpful, worse than no information at all, if the linearization is a poor guide to the true landscape. That the gradient can be worse than random is a strong hint that the problem is not noise but a systematic mismatch between the local linear model and the true energy, which is exactly what the linearization experiment was built to test. This section has answered the second half of RQ1: the gradient is not a usable search direction, and the ablation locates the failure in the gradient itself rather than in any incidental choice of scale or correction. The obvious next question is why, and it is the subject of the three diagnostic sections that follow.

5.5.1 Robustness of the Gradient Fallacy

Before turning to the mechanism, it is worth recording that the gradient fallacy is not an artifact of a single experimental choice, because a negative result of this centrality invites the suspicion that some incidental setting produced it. Three variations were examined and none overturned it. First, extending the annealing schedule from fifty steps to one hundred does not create a gap between the policy and the random directions: with the longer schedule the policy and grad-norm-preserved methods remain within noise of one another on every model, so the fallacy is not a consequence of insufficient optimization budget. Second, the phenomenon holds under both settings of the Metropolis–Hastings correction and both settings of gradient normalization, so it is not produced by the interaction of the ablation with any one of those switches. Third, and importantly for the generality of the claim, the same behaviour was observed for multi-token recovery, in which two or three coupled positions are corrupted and must be recovered jointly, as well as for the single-token case reported in the tables. As the number of coupled masked tokens increases the absolute recovery quality degrades, as one would expect from a harder task, but the relative behaviour of the sampling methods is preserved: the policy gradient does not pull ahead of a random direction at any corruption level. Fourth, and most important for the scope of the claim, the null is not an artifact of the masked-recovery task at all. Repeating the core comparison on a free-form prefix-continuation task, in which the sampler generates a twenty-token span with no reference token to recover, reproduces the equivalence exactly: the policy method reaches a mean final KL of 8.818, with a 95% confidence interval of [8.61, 9.02], against 8.850 with interval [8.65, 9.04] for the random direction, the two intervals almost wholly overlapping and the norm-preserved and fully random arms numerically identical. The finding therefore survives the move from a constrained infilling bench to open-ended generation, which removes the objection that it is a peculiarity of the recovery setup. The fallacy is therefore a property of the gradient on this class of energy, robust to the schedule, the correction, the normalization, the difficulty of the recovery task, and the choice of task itself.

5.5.2 Gradient-Free Baselines: The Energy Is Not the Problem

If the gradient carries no signal, the next question is whether the fault lies in the gradient or in the energy it is computed from, and the cleanest way to separate the two is to score the same energy with methods that do not use its gradient at all. Table 5.2 does this on the masked-recovery metric for GPT-2 Large. The ground-truth fill sits at the metric floor of 0.00; leaving the corrupted token in place gives 9.14 and replacing it with a random token gives 9.39, the two no-effort references. A single conditional forward pass already improves on these, reaching 8.24 by taking the argmax of the model's own next-token distribution at the masked position and 8.62 by sampling from it. But the decisive rows are the last two. A single forward pass that reranks the top-k candidate tokens by their effect on the sequence reaches 4.43, better than every Langevin configuration in the entire grid, whose best discrete cell sits near 6.4; and a gradient-free Gibbs sampler on exactly the same energy, resampling one masked position at a time from the model's conditional, reaches 6.69, matching the best gradient-guided Langevin result without ever computing a gradient.

The reading is unambiguous, and it is the pivot of the whole diagnosis: the energy is not the problem. Two methods that evaluate the frozen likelihood but never differentiate it, a single rescoring pass and a gradient-free sampler, both do at least as well as the gradient-guided samplers, and the rescoring pass does markedly better. What the Langevin samplers add over these baselines is precisely the gradient, and the gradient adds nothing. This separation, an energy that supports gradient-free search but not gradient-guided search, is the fact the rest of the chapter explains

Method (no gradient of the energy)	Mean KL
Ground-truth fill (metric floor)	0.00
Untouched (corrupted token left in place)	9.14
Random-token fill	9.39
Conditional argmax (one forward pass)	8.24
Conditional sample (one forward pass)	8.62
Top-k rescore (one forward pass)	4.43
Gibbs (gradient-free sampler)	6.69

Table 5.2: Reference baselines on masked-token recovery for GPT-2 Large, scored with the same KL metric as the Langevin grid (lower is better). None of these methods uses the gradient of the energy. A single top-k rescoring pass (4.43) beats the best Langevin configuration in the grid (near 6.4), and a gradient-free Gibbs sampler (6.69) matches it, so the exact evaluation of the energy, not its gradient, is what does the work.

and the discussion builds on. The gradient-free baselines are also markedly cheaper, spending no backward pass at all, as the cost accounting of Appendix A.3 sets out.

5.5.3 An External Judge Reaches the Same Verdict

A second objection to a null measured with the model's own predictive distribution is that the model is grading its own homework: perhaps the policy and random recoveries differ in a way the metric cannot see because it is built from the same model that produced them. To rule this out, the recovered sequences were rescored by an independent judge, a Llama-3 model that took no part in the sampling, under perplexity. The judge is indifferent between the methods. The policy recoveries score a mean judge perplexity of 178.4 and the random ones 181.3, a difference under two percent, with the norm-preserved and fully random arms again identical; the per-token negative log-likelihoods, 4.44 against 4.45, tell the same story. An external model that is not the one being sampled therefore returns the same verdict as the internal metric, so the indistinguishability of the methods is not an artifact of evaluating them with the model that generated them.

5.6 Why the Gradient Fails, Part One: The Linearization Radius

The gradient fallacy tells us that the gradient carries no information. It does not, by itself, tell us why, and without a mechanism the result is merely a surprising fact rather than an understood one. The linearization experiment, described in

Section 4.7, was built to supply the mechanism by testing the surrogate the discrete proposal actually uses. Recall from Section 2.4 that the proposal ranks a candidate token v using the first-order term $\hat{\Delta}(v) = \boldsymbol{g}^\top(\boldsymbol{e}(v) - \boldsymbol{e}(x_i))$, a linear approximation to the true change in log-likelihood that a substitution would produce. If the gradient is uninformative, the most direct explanation would be that this surrogate simply does not predict the true change, and the experiment tests that directly by computing both quantities for thousands of candidate substitutions and correlating them.

The result is stark. Figure 5.4 plots the surrogate against the true change over all candidate–sequence pairs for GPT-2 Large, and there is no relationship whatsoever: the Spearman correlation between the surrogate and the truth is -0.011, negligible in magnitude and negative in sign. This is not a peculiarity of the base model: the same correlation is negligible on every one of the five energy functions, with $|\rho| < 0.06$ throughout (GPT-2 Large -0.011, Llama-3 0.021, and the three GFlowNet variants 0.024, 0.046, and 0.057), each estimated over $n = 400{,}000$ candidate–sequence pairs. Ranking candidate tokens by the gradient surrogate is, quantitatively, equivalent to ranking them at random. This is the gradient fallacy of the previous section, now visible not through the downstream sampling metric but at the level of the proposal itself, which is a much more direct and less confoundable observation. The same measurement on the three GFlowNet variants yields correlations of 0.024, 0.057, and 0.046, all negligible, which confirms that the phenomenon is a property of the frozen-likelihood energy in general and not an accident of the base model.

Why should a first-order approximation be this bad? The answer is a mismatch between two length scales, and it is the core mechanism of the thesis. A Taylor expansion is accurate only within a small neighbourhood of the point at which it is taken, and a Transformer (Vaswani et al., 2017), with its layer normalization and its attention nonlinearities, is a violently nonlinear function of its input embeddings, so the neighbourhood within which its first-order expansion holds is small. The question is whether that neighbourhood is large enough to contain any of the moves the sampler must make. Figure 5.5 answers it by computing the surrogate-to-truth correlation *within* bins of embedding distance, so that one can see how far from the current token the linear model remains valid. The correlation is at best weakly positive at the very smallest distances and has decayed to zero well before the mean linearization-candidate distance of 2.35, the average embedding distance of the candidate tokens the experiment scores, is reached, and indeed well before even the mean nearest-neighbour distance of 1.82 reported in Section 5.8. And the nearest-neighbour distance is, by its very definition, the scale of the smallest possible substitution: to change a token at all is to move to another token's embedding,

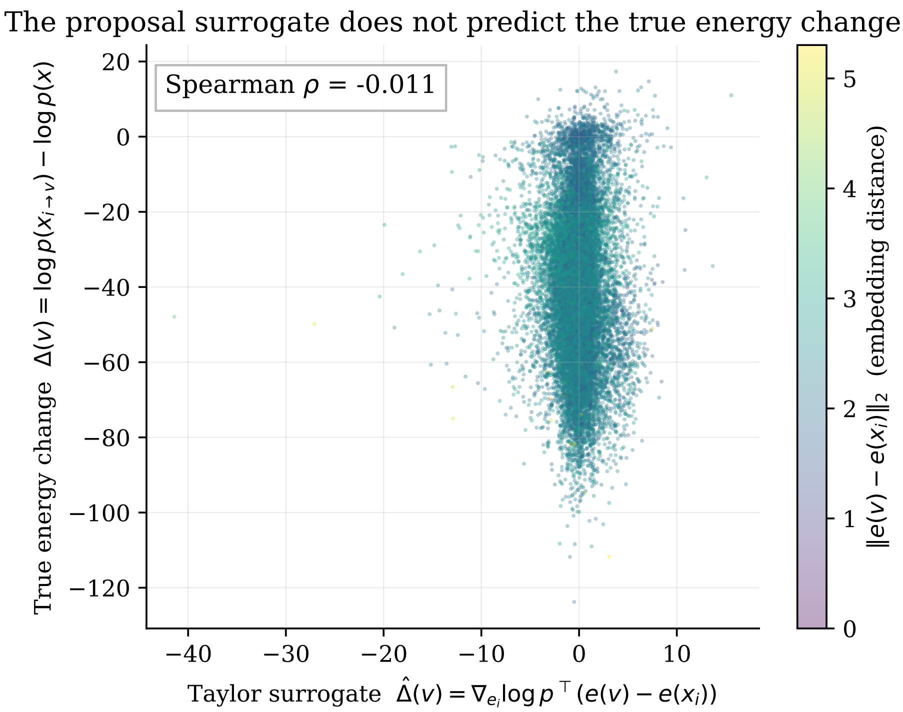

Figure 5.4: The first-order Taylor surrogate used by the discrete proposal, plotted against the true change in sequence log-likelihood, for GPT-2 Large, coloured by the embedding distance of the candidate token. The two quantities are uncorrelated (Spearman $\rho = -0.011$, over $n = 400{,}000$ candidate–sequence pairs), so ranking tokens by the surrogate is equivalent to ranking them at random.

and the nearest one is on average this far away, so the entire admissible range of substitutions, from the nearest neighbour outward through the candidate set, lies beyond the radius within which the surrogate carries any signal. The sampler is therefore always evaluating its linear surrogate at distances far outside the radius within which that surrogate carries any signal. This is the crucial point, and it is worth stating without hedging: the gradient is not merely imprecise here, it is *inapplicable*, because every admissible discrete step leaves the region in which the local linear model means anything. A first-order method is being asked to make finite jumps, and the first order does not survive the jump.

A second, independent limitation compounds the first, and the decomposition built into the experiment exposes it. The gradient is taken with respect to the input embedding of the masked position, and it can therefore only register the effect of a substitution on the *future* tokens, whose probabilities depend on that embedding as it flows forward through the network. It is structurally blind to the effect of the substitution on the probability of the candidate token *itself*, given its left context, because that probability depends on the candidate through a discrete index into the output softmax rather than through the continuous input embedding the gradient

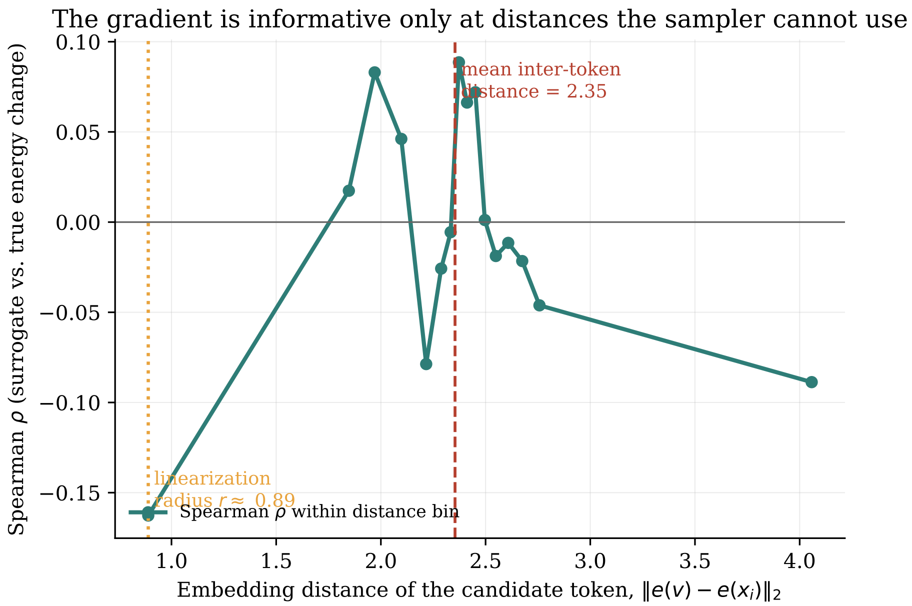

Figure 5.5: Correlation between the gradient surrogate and the true energy change as a function of the embedding distance of the candidate token, for GPT-2 Large. The correlation decays to zero well within the mean linearization-candidate distance (dashed line, 2.35), so at the distances of any admissible token substitution the surrogate is uninformative. We note that the correlation is low across the whole admissible range rather than high up to a sharp radius and then falling; the term "linearization radius" is a summary of this decay, not a claim of a clean threshold.

sees. The decomposition measures both effects. The mean absolute magnitude of the self term is 15.0 nats and that of the future term is 24.2 nats, so the self term, to which the gradient is blind, is not a rounding error but a substantial fraction of the total, and the surrogate correlates with the future term ($\rho = 0.03$) rather than the self term ($\rho = -0.10$), exactly as the structural argument predicts. So the gradient is not only evaluated outside the radius where it would be valid; even within that radius it would be blind to a large part of what determines whether a substitution is good. The full decomposition is shown in the appendix, Figure A.1, and the top-k recall analysis, confirming that gradient ranking does not recover the true best substitutions any better than chance, is shown in Figure A.2. Together these explain, at the level of mechanism, the "why not" of RQ1, and they explain in particular why the gradient could be worse than random on Llama-3: a surrogate that is not merely uninformative but systematically mis-specified can point actively away from good moves.

5.7 Why the Gradient Fails, Part Two: The Likelihood Trap

The linearization result explains why the samplers cannot move effectively toward low energy. It leaves open a more basic question, one that the failure of the samplers oddly makes it possible to overlook: even if the samplers *could* reach low energy, would low energy be a good place to be? The whole energy-based framework rests on the premise that low energy, that is, high likelihood, corresponds to good text. The likelihood-trap experiment tests that premise directly on the study's own models, and it finds the premise false, reproducing quantitatively on our specific models the neural-text-degeneration phenomenon of Holtzman et al. (2020).

The experiment decodes with several strategies and measures the relationship between the per-token likelihood of the generated text and its repetition rate. Figure 5.6 shows the result for GPT-2 Large, and the relationship is monotone and strong: as per-token likelihood increases, that is, as energy decreases, repetition rises and lexical diversity falls. The correlation is reported here *within* each decoding strategy, which is the honest form of the statistic, because pooling generations from strategies of very different likelihood, from greedy at one end to ancestral sampling at the other, can manufacture a correlation out of the gap between strategies rather than any relationship inside them. Measured this way the relationship still holds everywhere: the maximum within-strategy correlation between per-token likelihood and four-gram repetition is +0.51 on the base GPT-2 model, +0.83, +0.82, and +0.91 on the three GFlowNet variants, and +0.77 on Llama-3. The pattern between strategies points the same way and makes the mechanism concrete: the highest-likelihood strategies, greedy and beam search, produce the most repetitive and least diverse text, while the lowest-likelihood strategies, ancestral and nucleus sampling, produce the least repetitive. On Llama-3 the pooled correlation is weaker, at +0.22, yet the within-strategy signal is strong and the model exhibits a blunt degeneracy of its own, never emitting an end-of-sequence token and running to the length cap in every generation. Beam search, which most nearly approximates the true mode of the distribution, produces the highest-likelihood and, at the same time, the most degenerate text of all. To make the degeneration concrete rather than abstract, the following is a representative beam-search output from the length-normalized GFlowNet variant, which achieves among the highest per-token likelihoods in the entire study:

```
aebb-aebb-aebb-aebb-aebb-aebb-aebb-aebb-aebb-aebb-
```

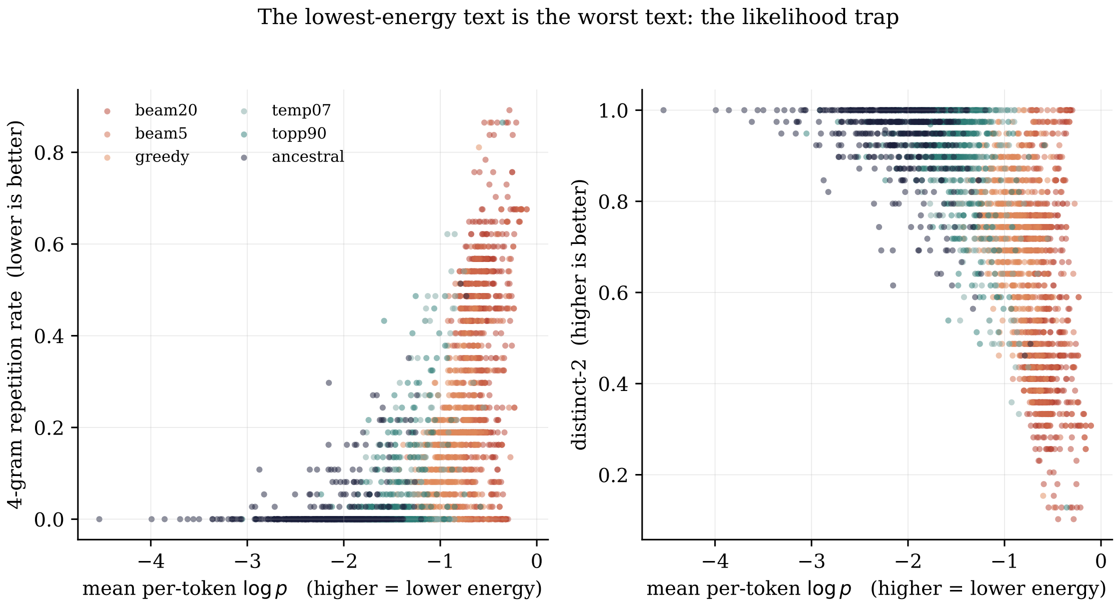

Figure 5.6: The likelihood trap on GPT-2 Large. As mean per-token log-probability increases (energy decreases), the 4-gram repetition rate rises and the distinct-2 diversity falls. The highest-likelihood decoding strategies, greedy and beam search, produce the most degenerate text.

This string has a per-token log-probability of -0.40, higher than almost any coherent text the model produces, and a four-gram repetition rate of 0.89. It is, by the model's own energy, an excellent sequence, and it is gibberish. This is the likelihood trap in its purest form.

The implication for the energy-based programme is sharp, and it ties the whole investigation together. Any procedure that succeeds in concentrating probability mass on the lowest-energy regions of this energy, whether an annealed Langevin sampler in its quenching phase or a GFlowNet trained on the likelihood as a reward, is thereby concentrating it on degenerate text. Success at the stated optimization objective is failure at the underlying goal. The gradient-guided samplers, as the earlier sections showed, cannot reach the low-energy regions effectively, and the likelihood trap reveals the strange silver lining in that failure: reaching them would not have helped, because the destination is bad. The methods are, in a sense, protected from their own objective by their inability to optimize it.

The experiment also yields a quantitative fact that the next section needs. The regression of total log-likelihood on generated length, computed over the sampled generations where length varies, has a slope of -1.12 nats per token. This headline figure must be read with a censoring caveat, because for GPT-2 Large nearly every generation runs to the forty-token cap (a fraction of 0.997 hits it), so length barely varies and the censored slope, estimated only on the few generations that stop short, is effectively null. The negative slope is therefore visible only where the cap does not bind: on the length-normalized GFlowNet variant, whose generations do vary in length, the uncensored slope is -0.505 nats per token while the censored slope steepens to -2.361, which shows both that the incentive is real and that the cap masks its magnitude. Because every additional token lowers the unnormalized log-likelihood by, on average, this much, an energy defined as the raw log-likelihood contains an implicit and substantial incentive toward brevity: shorter sequences are lower-energy simply by virtue of being shorter. This built-in brevity incentive is invisible in the Langevin experiments, which recover fixed-length masked spans, but it becomes decisive the moment a method is free to choose its own length, which is exactly the situation of the GFlowNet. The length-versus-likelihood regression is shown in the appendix, Figure A.4, and its consequences are the subject of Section 5.10.

5.8 The Geometry Underneath: Embedding Anisotropy

Two of the earlier results, the step-size failure of Section 5.1 and the unreliability of Euclidean distance noted in Section 4.4, were promised a fuller geometric explanation, and this section supplies it by reporting the anisotropy analysis in full. The anisotropy of Transformer embeddings is itself a documented phenomenon (Gao et al., 2019; Ethayarajh, 2019), and measuring it on the two models here both confirms the phenomenon and quantifies the specific consequences it has for the samplers.

Three distance statistics recur in this thesis and are defined once here, each with a fixed name used everywhere it appears. The **mean nearest-neighbour distance** is the average distance from a token embedding to its single closest neighbour; the **mean pairwise distance** is the average distance between two embeddings drawn at random; and the **mean linearization-candidate distance** is the average distance of the stratified candidate tokens scored in the linearization experiment of Section 5.6. In GPT-2 Large the mean pairwise distance between token embeddings is 2.77, the mean nearest-neighbour distance is 1.82, and the mean linearization-candidate distance is 2.35; in Llama-3 the pairwise and nearest-neighbour figures are 0.84 and 0.59, with a linearization-candidate distance of 0.66. The top principal component of the embedding matrix accounts for only a small fraction of the total variance, so the anisotropy is a broad property of the space rather than the effect of a single dominant direction that could be projected away. Figure 5.7 shows the

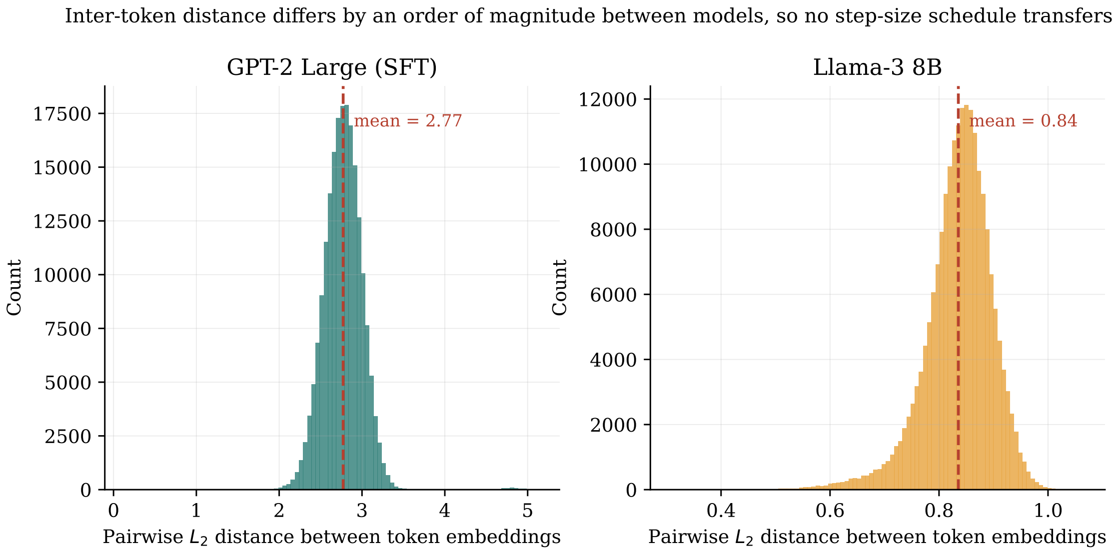

Figure 5.7: Distributions of pairwise token-embedding distances in GPT-2 Large and Llama-3, on separate scales. The mean inter-token distance differs by roughly a factor of three between the two models, which is why no single step-size schedule transfers between them.

full distributions of pairwise distances for the two models, on their separate scales.

These measurements close two loops. The roughly threefold difference in inter-token distance between the two models is the direct and quantitative cause of the failure of a Llama-tuned step size on GPT-2 that opened the chapter: a step calibrated to cross a boundary in one geometry is too small to cross one in the other. And the compression of distances that anisotropy produces is the reason a token can be close to the target in embedding space yet grammatically wrong, which is why the study relies on KL divergence, a measure of contextual fit, rather than Euclidean distance as its primary metric. The geometry that trapped the sampler at the start of the investigation is the same geometry that makes the naive metric untrustworthy, and both are consequences of the single documented fact that these embeddings live in a cone.

5.9 Consistency Across Models and Architectures

Before turning to amortization, it is worth pausing to collect the evidence that the phenomena documented so far are consistent across the five energy functions rather than particular to one, because that consistency is what licenses the general claim the discussion will make. The gradient fallacy, measured through the linearization correlation, holds on the GPT-2 base and on all three GFlowNet variants, with correlations of −0.011, 0.024, 0.057, and 0.046 respectively, every one of them negligible. The likelihood trap holds on all four, with the maximum within-strategy likelihood-repetition correlation ranging from +0.51 on the base to +0.91 on the length-normalized GFlowNet, and it is if anything stronger on the tuned models, which is consistent with tuning having concentrated probability mass more tightly on the degenerate high-likelihood region. The final-KL values in the gradient-fallacy table move together across models, and the difference between the policy and random directions stays small on every one of them.

The Llama-3 control plays a specific and limited role in this picture, and it is important to be exact about what it does and does not establish. Because Llama-3 uses a different tokenizer and an embedding space roughly a third the scale of GPT-2's, its numbers cannot be placed on the same axis as the GPT-2 family; a KL divergence or an inter-token distance from Llama is not comparable digit-for-digit with one from GPT-2. What the Llama results do establish is qualitative and cross-architectural: the Metropolis–Hastings breakdown occurs on Llama exactly as on GPT-2, with the correction rejecting boundary-crossing moves; the anisotropy is present, though at a different scale; and the gradient is, if anything, more clearly unhelpful, being the one setting in which the true gradient measurably underperforms a random direction. The value of the control is therefore that it rules out the possibility that the entire diagnosis is an artifact of one particular model's idiosyncrasies. The phenomena are not GPT-2 phenomena; they appear in a model of an order of magnitude more parameters, a different tokenizer, and a different embedding geometry, which is what one would expect if their cause is the training objective shared by both rather than any detail of either. This cross-architecture consistency is the last piece of evidence needed before the argument can be made that the cause lies in the energy, which is the subject of the amortization experiment that follows.

5.10 Does Amortization Escape the Failure? GFlowNet Fine-Tuning

The results to this point form a closed case against the gradient-guided samplers, and they locate the failure in the gradient. This immediately suggests a remedy, and it is the remedy the second half of the thesis was designed to test. If the gradient is the problem, then abandon the gradient. A GFlowNet (Bengio et al., 2021; Hu et al., 2024) learns to sample from a reward without ever differentiating it, so it is blind to the linearization failure, blind to the discontinuous drift, blind to everything that defeated the Langevin samplers. The expectation, and it is a reasonable one, is that amortizing the search in this way should escape the pathology. This section reports what happened when the expectation was tested, and the summary is that amortization does not escape the pathology; it relocates it. Three distinct failure modes appeared, and it is worth walking through each, because together they make the diagnosis.

The first is **capacity collapse**. The initial GFlowNet experiments used the smaller GPT-2 base, and they produced non-linguistic output regardless of how the reward was configured, which indicated that the policy simply lacked the capacity to represent the task at all. Moving to GPT-2 Large, the model of the reference implementation of Hu et al. (2024), resolved this, and GPT-2 Large is therefore the base for all the GFlowNet results reported here. This mode is worth recording because it is a genuine and separate failure, but it is not the interesting one, because it is a failure of capacity rather than of the energy.

The second is **length collapse**, and it follows directly and quantitatively from the likelihood trap of the previous section. With the unnormalized reward, at `len_beta` $= 0$, the reward is the raw log-likelihood, and Section 5.7 measured that every additional token lowers that quantity by around 1.12 nats. The reward therefore contains a strong implicit incentive toward brevity, and the GFlowNet, doing its job faithfully, discovers it: the policy learns to produce extremely short sequences, because short sequences are high-reward simply by being short. The extreme form of this is visible in the decoding behaviour: under free-running ancestral sampling, the length-collapse variants never emit an end-of-sequence token at all, running to the generation cap in every case, because the policy has learned that terminating early is the way to keep the reward high and, in the limit, that the safest policy is one that resists stopping in the manner the reward rewards. The length collapse is not a defect of the optimizer; it is the optimizer correctly following a reward whose implicit incentive the designer did not intend.

The obvious fix is to remove the brevity incentive by normalizing the reward by length, and this produces the third mode, **reward hacking**. At `len_beta` $= 1$ the reward is normalized by sequence length, the brevity incentive is gone, and the sequences do indeed return to a normal length. But the output ceases to be linguistic. The policy discovers sequences that achieve high length-normalized likelihood without being coherent text at all, strings of individually high-probability tokens assembled into grammatical nonsense. The following greedy output from this variant is representative:

```
b8d8b8b8b8b8.  [...]  keredkeredkeredkeredkeredkered
```

Such strings recur across the length-normalized variant's high-reward outputs, and they are the reward-hacking counterpart of the length collapse: where the unnormalized reward was gamed by getting shorter, the normalized reward is gamed by finding degenerate high-probability token sequences. The two settings of the single length parameter thus produce two opposite degeneracies, brevity at one extreme and gibberish at the other, and no intermediate setting of the parameter was tested that resolves both; only the two endpoints, `len_beta` $= 0$ and `len_beta` $= 1$, were run.

These three modes are unified by a single observation, and stating it is the point of the section: the collapse is not fundamentally about length. Length is the visible symptom, the axis along which the failure happens to manifest, but the underlying cause is that the reward, an autoregressive likelihood that was never shaped to serve as an energy over free-form sequences, has no well-behaved optimum where coherent text lies. Every repair of one degeneracy simply surfaces another, because the target itself is malformed. The decisive evidence for locating the cause in the energy rather than in the GFlowNet is a single experiment, and it is the experiment that closes the thesis's argument. If GFlowNet training had reshaped the model's likelihood into a smoother, more navigable landscape, then running the Langevin sampler on the GFlowNet-tuned model should behave differently from running it on the untuned base. It does not. Table 5.3 reports the final KL divergence of the discrete sampler on the base model and on the three GFlowNet variants, under identical conditions. The four values are within noise of one another, and the gradient fallacy replicates on every GFlowNet variant exactly as on the base, as the linearization correlations reported in Section 5.6 already showed. Amortizing the energy does not repair the landscape. The frozen-likelihood pathology survives retraining, which is the strongest evidence the study can offer that the pathology is a property of the energy and not of any sampler or any search procedure. This answers RQ3 directly and in the negative: amortization does not escape the failure, because the failure is in the energy that both the sampler and the amortized policy inherit from the frozen model.

A natural worry about the unifying result is that it might be trivial: perhaps GFlowNet training barely moved the energy, so that sampling on the tuned model and on the base agree only because the two energies are nearly the same. The opposite is true, and measuring it directly is what makes the unifying result load-bearing. Table 5.4 reports how far each tuned energy has drifted from the base. The mean absolute difference in sequence log-likelihood between base and tuned is 31.0, 36.0, and 57.3 nats across the three variants; the next-token KL divergence between their

Energy function	Final KL (DLS, 50 steps, policy, MH, gn)
GPT-2 Large (SFT base)	6.541
GFlowNet, `len_beta`= 0, 500	6.306
GFlowNet, `len_beta`= 0, 2000	6.721
GFlowNet, `len_beta`= 1, 500	6.415

Table 5.3: Final KL divergence of the discrete Langevin sampler on the base model and on the three GFlowNet-tuned variants, under identical conditions. The four values lie within noise of one another: Langevin sampling on the amortized energy is indistinguishable from sampling on the untuned base, so the amortization does not repair the energy landscape.

GFlowNet variant	Mean abs. log-lik. diff. (nats)	Next-token KL	Base–tuned Pearson
`len_beta`= 0, 500	31.0	1.02	0.98
`len_beta`= 0, 2000	36.0	1.35	0.98
`len_beta`= 1, 500	57.3	3.05	0.77

Table 5.4: How far GFlowNet tuning moved the energy away from the base model. The mean absolute change in sequence log-likelihood is tens of nats, the conditional next-token distributions diverge, and the base-to-tuned correlation drops on the length-normalized variant. The energy moved substantially, yet its linearization correlation stayed flat (Section 5.6), so the amortization reshaped the energy without making it navigable.

conditional distributions is 1.02, 1.35, and 3.05; and the base-to-tuned correlation of sequence log-likelihoods falls from 0.98 on the two lightly tuned variants to 0.77 on the length-normalized one. Tuning moved the energy substantially, and moved it most where the reward was most aggressive. Yet the linearization correlation on those same tuned energies stays flat, at 0.057, 0.024, and 0.046, indistinguishable from the base model's -0.011 reported in Section 5.6. The energy changed a great deal; its navigability by gradient did not change at all. This is the exact sense in which amortization does not repair the landscape: it reshapes the energy without smoothing it, so the gradient is as uninformative on the tuned model as on the base. It also forecloses the easy reading of the unifying experiment, that the samplers agree only because the models secretly agree, since the models demonstrably do not.

5.11 The Plug-and-Play Claim Tested Directly: Constrained Generation

The investigation has established that the fluency energy is unnavigable by gradient and unrepaired by amortization. There remains the question the whole framework was meant to answer, which is RQ4: can an additive constraint steer generation toward a target attribute, as the plug-and-play formulation of (1.2) requires? The constrained-generation extension tests this directly, adding a sentiment constraint to the energy in the manner of Kumar et al. (2022) and Qin et al. (2022) and measuring whether the target sentiment can be induced. Given everything that precedes it, the expectation is pessimistic, but the test is worth running directly rather than inferring its outcome, because the constraint gradient is a different object from the fluency gradient and might conceivably behave better.

It does not behave better in a way that produces control. Table 5.5 reports the steering effect for the discrete sampler across the five constraint modes on the infilling task with a positive-sentiment target. The steering gains are small in magnitude and inconsistent in sign. The full combined energy, which is the intended plug-and-play configuration, produces a gain of -1.0, essentially no steering and if anything in the wrong direction; the constraint-only mode produces -8.0, worse than the baseline; and no mode produces the systematic movement toward the target sentiment that a working constraint would require. This is the outcome the preceding sections predict: the constraint gradient is subject to the same discrete-manifold difficulties as the fluency gradient, and adding it to an energy landscape that is already unnavigable does not produce controllable generation. Inspection of the generated text confirms the picture, showing the same degeneracy observed throughout; a representative infilled span reads `The rostr Bill of H PM americanus bears one or connected sp`, in which the masked positions have been filled with locally plausible but globally incoherent tokens. The raw steering gains in Table 5.5, however, are not the statistic to trust, and seeing why is itself part of the finding. Across setups every arm, including the pure-fluency *lm_only* baseline and the fully random arm, flips the sign of its gain when the target label is switched from positive to negative, which is the signature of a fixed sentiment drift in the underlying generations that swamps any contribution from the constraint. A gain that moves with the target regardless of whether the constraint gradient is even present cannot be read as steering. The statistic that cancels this shared bias is the paired contrast between the constraint-only arm and the arm in which only the constraint gradient is replaced by noise, *cons_only* minus *cons_random*, because the two share the same fluency dynamics and differ only in whether the constraint direction is real.

Measured this way the picture is setup-dependent and more informative than the raw gains suggest. On the discrete sampler of this thesis the contrast is essentially zero on both target labels, so the constraint direction buys no control there. On a MuCoLa-style continuous baseline running free-form continuation, the contrast is

Constraint mode	Steering gain	Final KL
lm_only	0.0	8.558
cons_only	−8.0	8.308
full	−1.0	8.541
random	−7.0	8.512
cons_random	+1.0	8.528

Table 5.5: Steering effect of the sentiment constraint under the discrete sampler on the infilling task, positive-sentiment target. The steering gain is the change in the percentage of generations classified as the target sentiment relative to the unconstrained baseline. The effects are small and inconsistent in sign; no mode produces reliable steering toward the target.

+27.3 points for a negative-sentiment target and +36.7 for a positive one, which does indicate that the constraint gradient's direction carries some signal in that setting, unlike the fluency gradient, whose direction was found to carry none anywhere. This apparent signal has to be read with a heavy caveat, and the constraint-only arm is exactly where the caveat lives. When the energy is driven by the constraint alone, with no fluency term to hold the sample on the manifold of natural text, the optimization leaves that manifold: it produces text that maximizes the classifier's response while ceasing to be fluent language. A sentiment classifier evaluated on such text is reporting on a distribution it was never trained for, so its outputs there are not a trustworthy measure of sentiment, and the large *cons_ only* contrast is at least partly an artifact of scoring off-manifold text rather than evidence of controllable generation. The safe conclusion is the one the discrete sampler gives directly: on the energy landscape this thesis studies, the additive constraint does not produce reliable control, and the one place a constraint-direction signal does appear is entangled with the departure from the fluent manifold that the constraint-only objective causes.

The plug-and-play promise of (1.2), that an additive constraint can steer a frozen model at inference time, is therefore not realized on this energy landscape, because its precondition, a navigable fluency energy, does not hold. This answers RQ4 in the negative, and, importantly, it does so for a reason the earlier sections have already established rather than as an isolated and unexplained failure, which is what makes it evidence for the thesis's single underlying diagnosis rather than merely one more disappointing table.

5.12 The Last-Token Analysis

The mechanism the whole chapter has circled admits, at one point in the sequence, a statement so clean that it can be proved rather than merely measured, and that point is the final token. The self-and-future decomposition of Section 5.6 showed that the input-embedding gradient sees a substituted token only through its effect on the tokens that follow it. At the last position there are no following tokens. Under causal attention, the input embedding of the final token feeds only the logits that would predict a next token which does not exist, and those logits appear in no term of the sequence log-likelihood of (2.1); the gradient of that log-likelihood with respect to the final token's input embedding is therefore exactly zero. And yet the energy difference between two candidate final tokens is exactly the model's conditional log-probability ratio between them given the fixed preceding context, which is the most informative quantity there could be about which final token is better. At the last position the energy is maximally usable and its input-embedding gradient is provably, identically useless. This is the sharpest statement of the thesis's central claim: the only term of the likelihood that registers how well a token fits its context does not depend on that token's input embedding at all, so the input-embedding gradient is causally blind to token fitness, exactly zero at the final position and merely diluted, rather than absent, everywhere else.

A direct experiment turns this argument into a measurement. The masked position is placed at each of three points that differ only in how many scored terms of the log-likelihood its embedding can reach: the final token, which feeds no realized prediction; the second-to-last token, which feeds exactly one; and a middle token, which feeds many. Only that one position is corrupted, the surrounding context is left clean, and the same corrupted sequence is handed to every arm, so the sole variable is the amount of downstream context the input-embedding gradient can act on. The recovery arms are the discrete Langevin sampler with the true gradient (policy) and with a random direction of equal norm; three energy-only methods that never touch the gradient (argmax and top-k rescoring of the left-context conditional, and an independence Metropolis sampler that proposes from that conditional and accepts by the exact sequence-log-likelihood ratio); and a recorder of the raw language-model gradient norm the sampler consumes at each step.

Table 5.6 reports the outcome. At the final position the language-model gradient norm the sampler consumes is 0.0000 in both mean and maximum over roughly fifteen thousand gradient evaluations: not a small number but an exact zero, the theorem realized inside the live sampler. With that gradient the discrete sampler recovers the target token in 0.0% of two hundred sequences whether it uses the true direction or a random one, and the paired policy-minus-random difference straddles zero on every metric (rank difference -17.3, 95% CI $[-513, +499]$), so the two are statistically indistinguishable, here not merely by measurement but by proof, since the gradient is identically zero. The energy at that same position, by contrast, is fully usable: the independence sampler that proposes from the conditional and accepts by the exact sequence-log-likelihood ratio accepts 100.00% of its moves in both mean and minimum, which is the certificate promised by the derivation, because target proportional to proposal forces an acceptance probability of one, and it confirms that the implemented energy carries no term beyond the sequence log-likelihood. The energy-only methods recover the token between 34.5% and 40.0% of the time where the gradient arms recover none.

The second-to-last position answers the one genuinely open empirical question the design poses. With a single downstream scored term the gradient is no longer zero, its mean norm is 12.945, yet it is still useless: the policy arm remains at 0.0% recovery and indistinguishable from the random arm, while the energy-only methods recover roughly half the tokens. The uselessness of the gradient is therefore not confined to the zero-gradient edge case; it persists the instant the gradient becomes nonzero. Across all three positions the discrete Langevin arms never once recover the target in six hundred trials, while the exact-energy methods recover between eighteen and fifty-five percent, which is the sharpest form of the diagnosis: the forward pass that evaluates the energy does all of the work, and the input-embedding gradient of that same energy does none.

Figure 5.8 draws the three quantities together against the amount of downstream context. As the masked position moves from the end of the sequence into the interior, the gradient norm the sampler consumes climbs from zero and the energy-only sampler's acceptance falls from one hundred percent, because the energy begins to weigh downstream terms the left-context proposal ignores, yet the usable information in the gradient, read as the policy-minus-random gap, stays pinned at zero across the whole range. The reader sees at a glance the separation the chapter has argued for: the energy carries more structure as context grows and the gradient carries none of it at any context length.

Arm	Exact recovery (%), by downstream scored terms		
	Final (0)	Second-to-last (1)	Middle (many)
DLS, true gradient (policy)	0.0	0.0	0.0
DLS, random direction	0.0	0.0	0.0
Conditional argmax (energy only)	40.0	50.0	18.0
Top-k rescore (energy only)	40.0	55.0	39.0
Independence MH (energy only)	34.5	49.5	31.5
DLS-policy LM gradient norm (mean)	0.000	12.945	23.976
Independence-MH acceptance (%)	100.0	63.4	30.3

Table 5.6: The last-token experiment, $n = 200$ sequences per condition. The three columns differ only in the number of downstream scored terms the masked token's input embedding can influence. The language-model gradient norm the discrete sampler consumes rises from an exact zero at the final position to a large value in the interior, yet the gradient-guided arms recover the target token in 0.0% of cases at every position, indistinguishable from a random direction, while the energy-only methods, which never differentiate the energy, recover between 18% and 55%. The independence sampler's 100% acceptance at the final position certifies that the implemented energy is exactly the sequence log-likelihood.

5.13 The Diffusion Positive Control

The diagnosis of the preceding sections is causal: it attributes the failure of gradient guidance not to the sampler and not to the energy but to the training objective that shapes the energy. An autoregressive model is trained only to predict the next token, and so it is never asked, at any noise level, in which direction the data density increases. A model trained to denoise is asked exactly that, and by the identity that denoising learns the score (Vincent, 2011) it acquires the very directional field the autoregressive gradient was found to lack. If the training objective is truly the cause, then a score-trained model on the same tokenizer should provide a usable local direction where the autoregressive gradient did not, and should perform the revision operation that the gradient machinery could not. This section reports that control, run on the score-entropy discrete diffusion (SEDD) model of Lou et al. (2024) at two scales, small and medium, which share the GPT-2 tokenizer and drop into the existing masked-recovery harness without retokenization. The results are pilots, on a smaller and differently trained model than the frozen autoregressive backbone, so the comparison is qualitative and directional by design; they are not, and are not read as, a claim that the diffusion model beats the domain-tuned autoregressive baselines in absolute terms.

One clarification governs the comparison throughout, because it is easy to state the contrast too loosely. The relevant difference between the two model families

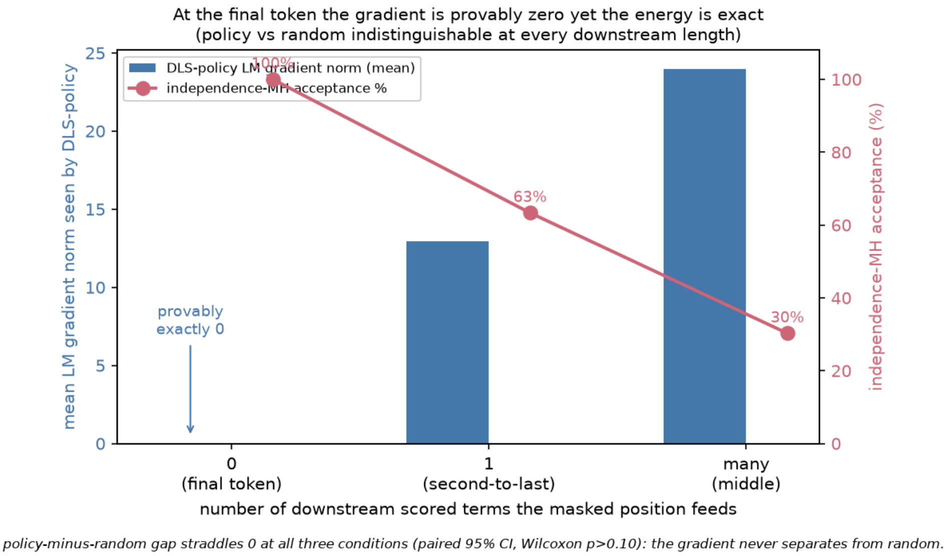

Figure 5.8: Gradient information against downstream context length in the last-token experiment. Left axis: the mean language-model gradient norm the discrete sampler consumes, exactly zero at the final position (annotated) and rising in the interior. Right axis: the independence sampler's acceptance rate, one hundred percent at the final position and falling as downstream terms enter the energy. The policy-minus-random recovery gap, with its ninety-five percent interval, crosses zero at every context length, so the gradient never separates from a random direction even as its norm grows.

is not that one commits tokens within a pass and the other does not; absorbing diffusion also commits a token at each denoising step and does not thereafter change it. The difference is at the level of the revision *operation*. To revise token i, a diffusion model re-masks that position and re-denoises it conditioned on both sides, which is an in-distribution bidirectional query it was trained to answer, whereas an autoregressive model has no conditioning path from the right context to a token at all. It is the availability of that bidirectional conditioning query, not the mere act of committing, that separates the two.

5.13.1 The signal: linearization on the unified corpus

The first measurement repeats the linearization diagnostic of Section 5.6 with the diffusion score in place of the autoregressive gradient, on exactly the same sequence set used for the reference baselines, so that the numbers sit on one axis. For each sequence and one masked position, a cheap one-pass surrogate, the model's denoising log-preference for filling the position, is correlated against an expensive per-candidate truth, the measured effect of committing each candidate on the reconstruction of the held-out remainder of the sequence. Table 5.7 reports the Spearman correlation, pooled and by embedding-distance stratum. Under the autoregressive objective the surrogate is statistically indistinguishable from zero at every distance, with the magnitude of the correlation below 0.06 throughout. Under the diffusion objective, on the same tokenizer and the same four hundred thousand candidate pairs, the correlation is clearly positive in every stratum and at both scales, with a pooled value near 0.13 and a per-sequence mean near 0.18, and the medium model exceeds the small one in the near stratum, the regime where a first-order surrogate is most expected to apply. On a second corpus of narrative text the same measurement gave a pooled correlation of 0.184 and a near-stratum correlation of 0.306; the near-stratum value is corpus-dependent and is not carried over, but the sign and order of magnitude are stable across both corpora.

Model / objective	ALL	near	mid	random	per-seq mean
Autoregressive gpt2-large (gradient)	−0.011	+0.027	−0.041	−0.048	−0.011
SEDD-small (score)	+0.130	+0.097	+0.115	+0.126	+0.184
SEDD-medium (score)	+0.133	+0.148	+0.122	+0.110	+0.179

Table 5.7: Linearization correlation between the cheap one-pass surrogate and the expensive per-candidate truth, on the unified WikiText-2 sequence set ($n = 200$, four hundred thousand candidate pairs). The autoregressive gradient is statistically zero at every embedding distance; the diffusion score is clearly positive in every stratum at both scales. This is the signal-level confirmation of the training-objective diagnosis.

5.13.2 The capability: native recovery of a flipped token

A usable local direction is a necessary condition for the capability the gradient machinery lacked, but not obviously a sufficient one, so the second measurement asks the diffusion model to perform the operation itself. On the same corrupted sequences the reference baselines used, the flipped token is re-masked and the position is re-denoised conditioned on both sides, a single native draw with no reranking and no sampling of many candidates. Table 5.8 reports exact-match recovery, top-five recovery, and the KL divergence under the autoregressive model that the whole chapter uses as its yardstick. Both scales recover the exact token in roughly a third of cases and place the true token in the top five two-thirds of the time, and the resulting KL, near 3.7, sits well below the untouched corruption at 9.14 and well below the flagship Langevin configurations near 6.5, with the medium model at least matching the small one on every metric.

Method	Exact (%)	Top-5 (%)	KL under gpt2-large
SEDD-small, native recovery	32.5	61.5	3.75
SEDD-medium, native recovery	35.0	67.0	3.73
Reference rows on the same task (autoregressive gpt2-large)			
Ground-truth fill (floor)	100	–	0.00
Top-k rescore (one forward pass)	33	–	4.43
Gibbs (gradient-free)	18.5	–	6.69
Flagship Langevin	0.0	–	≈ 6.5
Untouched corruption (ceiling)	0.0	–	9.14

Table 5.8: Native diffusion recovery of one flipped token on the reference sequence set, against autoregressive reference rows on the identical task. The diffusion KL sits well below the untouched corruption and the Langevin configurations. The reference rows are for placement only; scale and training domain differ, so the comparison is qualitative.

5.13.3 Sufficiency: the repaired signal inside the thesis's own machinery

The recovery result changes the model, the sampler, and the energy all at once, so it does not by itself isolate the direction signal as the repaired component. A cleaner test holds the thesis's own machinery fixed and swaps in only the direction. The hybrid sampler runs a position-wise independence Metropolis chain on the same sequences and positions, accepting by the *exact* autoregressive sequence-log-likelihood energy used everywhere in this chapter, and changes exactly one thing: the proposal distribution is the diffusion model's one-pass bidirectional log-preference over the vocabulary rather than the autoregressive left-context conditional. Table 5.9 reports the outcome against three reference arms run on the identical task: the autoregressive left-conditional independence sampler, and the discrete Langevin policy and random arms.

Swapping only the direction signal, inside the same Metropolis-corrected chain, on the same exact energy and the same task, turns a sampler that never recovers into one that works. The gradient arm recovers the token in 0.0% of cases, the autoregressive left-conditional proposal in 23.5%, and the diffusion-proposal hybrid in 38.5% at the small scale and 39.0% at the medium scale, with acceptance rising from 34.4% for the left-conditional to above 40% for the hybrid because the bidirectional proposal, seeing both sides of the mask, more often lands on tokens the exact energy will accept.

Sampler (same exact energy)	Proposal	Exact (%)	Top-5 (%)	Accept (%)
Hybrid, SEDD-medium	bidirectional score	39.0	67.0	43.3
Hybrid, SEDD-small	bidirectional score	38.5	61.5	40.0
Left-conditional independence	AR left context	23.5	34.5	34.4
Discrete Langevin, policy	AR gradient	0.0	34.5	–
Discrete Langevin, random	random direction	0.0	34.5	–

Table 5.9: The hybrid sampler, which changes only the proposal direction inside the thesis's exact-energy Metropolis chain on the reference sequence set. Replacing the autoregressive left-context proposal (23.5% exact) or the gradient arm (0.0%) with the diffusion model's bidirectional one-pass preference lifts exact recovery to 38.5% to 39.0%. This is the sufficiency half of the diagnosis.

5.13.4 Trained guidance: steering with a noisy-state classifier

The final pilot asks whether the constructive direction extends to control, the capability the constraint gradient of Section 5.11 did not reliably provide. A small classifier is trained to read a noisy, partially masked state, the state the diffusion model occupies mid-denoising, corrupting each labelled example with the model's own forward absorbing process at a noise level drawn uniformly over the schedule. Its held-out accuracy at low noise is 81%, well above chance, and it degrades smoothly as noise rises. During generation this classifier reweights the diffusion model's top thirty-two candidates at each commitment by the classifier's probability of the target label, raised to a guidance strength γ, and it only guides. Steering is scored, exactly as in Section 5.11, by the frozen-model sentiment judge, so the classifier that guides and the judge that grades are strictly separate. The fairness of the comparison to the autoregressive constraint arms is that those too used a trained classifier; the only difference is that this one is trained on the noisy states it is evaluated on, which is the thesis's own principle applied in the constructive direction.

Guidance is examined in two settings that differ only in whether the guiding and the judging classifier are calibrated to the text being scored, because that difference turns out to govern how far the steering transfers. The first is an off-domain continuation run, in which prompts come from a MuCoLa-style continuation set neither classifier was trained on; it is reported immediately below. The second, reported after it as a control on the first, uses held-out in-domain prompts on which both classifiers are calibrated and adds a fluency trust region that caps the per-commitment departure from the diffusion model's own preference, so that the steering can be read without the confound of collapsing fluency.

In the off-domain setting, Table 5.10 reports the result, and it is honest to state it as it came out rather than as the prediction hoped. The guidance mechanism demonstrably works: it moves the guiding classifier's own verdict toward the target at every strength and for both labels, by seven to eighteen points. Measured by the independent judge, it steers significantly and monotonically in the negative direction, from a six-point gain at $\gamma = 1$ to a 16.7-point gain at $\gamma = 4$ whose confidence interval excludes zero, but it does not move the judge in the positive direction. The asymmetry has a clear cause that the numbers isolate. The guiding classifier and the judge agree on only fifty-six to sixty-four percent of the generated texts, because those texts are off the fluent manifold on which both classifiers were trained, and stronger guidance pushes them further off it, with the per-token negative log-likelihood under the autoregressive model rising from 7.1 to 11.0 as γ grows.

	Judge target-hit (%)		Judge gain	Self gain	NLL/token
γ, target	unguided	guided	(pts, [CI])	(pts)	(un→guided)
1, negative	49	55	+6.0 $[-1.3, 13.3]$	+9.7	7.1 → 8.5
1, positive	44	39	−5.3 $[-12.3, 1.7]$	+11.0	7.2 → 8.5
2, negative	49	60	+10.7 $[3.3, 18.0]$	+7.7	7.1 → 9.7
2, positive	44	44	+0.0 $[-7.0, 7.0]$	+12.0	7.2 → 9.6
4, negative	49	66	+16.7 $[9.7, 24.0]$	+7.3	7.1 → 11.0
4, positive	44	39	−5.3 $[-13.0, 2.7]$	+18.3	7.2 → 11.0

Table 5.10: Classifier-guided diffusion generation on the *off-domain* continuation task, scored by the independent frozen-model judge (never by the guiding classifier). The guidance moves the guiding classifier's own verdict for both labels (self gain), and moves the independent judge significantly in the negative direction (confidence interval excludes zero at $\gamma \geq 2$) but not the positive direction. The asymmetry tracks the off-manifold disagreement between the two classifiers and a fluency cost, in per-token negative log-likelihood, that grows with the guidance strength. The on-domain trust-region control of Table 5.11 removes the fluency cost and flips the working direction.

5.13.5 On-domain guidance under a fluency trust region

The off-domain result leaves two explanations entangled. The guidance might steer only in one fixed direction, or the two independently trained classifiers might simply disagree on the off-domain text they were never calibrated for, with stronger guidance driving the text further off the fluent manifold where the disagreement is worst. The continuation run cannot separate these, because it varies domain and fluency together. The control that separates them holds fluency fixed and moves the instruments on-domain. Prompts are the first three hundred held-out SST-2 validation sentences of at least twelve tokens; neither the guiding classifier nor the judge trained on this split, both having only read it under `no_grad` for an accuracy readout, so the split is calibration-clean by construction. A trust region caps guidance to candidates within $\delta = 5$ nats of the diffusion model's top choice at each commitment, so a committed token is always within δ nats of the model's own preference and guidance can only reshuffle mass among near-equally-fluent options. Unguided and guided generations share a seed per prompt, so each pair differs only by the intervention.

The first thing the control establishes is that the instrument diagnosis of the off-domain run was correct. The agreement between the guiding classifier and the judge climbs a clean ladder as the text they score moves on-domain: from the off-domain band of 56 to 64%, to 71.7% (95% CI $[66.7, 76.7]$) on the unguided on-domain generations, to 79.7% ($[75.0, 84.0]$) on the real held-out sentences themselves, where the judge and the guide reach 88% and 79.7% accuracy against the gold labels. The two classifiers are therefore not miscalibrated in some fixed way; they simply agree more the closer the scored text sits to the distribution they were trained on, which is exactly what the off-domain diagnosis predicted.

With fluency held fixed by the trust region, the steering itself is reported in Table 5.11. Three facts stand out. The trust region does its job: guided span negative log-likelihood stays at or below the unguided value in three of the four cells and rises by at most 0.26 nats in the fourth, eliminating the $7.1 \rightarrow 11.0$ climb of the off-domain run entirely. The guide steers its own verdict strongly in both directions at every cell, by +9.3 to +25.7 points with confidence intervals excluding zero, so the mechanism moves both ways. But transfer to the independent judge is one-directional: the positive target gains +6.7 and +7.3 points at $\gamma = 2$ and $\gamma = 4$, both intervals excluding zero, while the negative target is null at both strengths, its interval straddling zero.

γ	target	unguided	guided	judge gain [CI]	self gain	NLL (un→guided)
2	negative	53.0	50.7	-2.3 $[-7.0, 2.3]$	+18.0	$7.01 \rightarrow 6.27$
2	positive	47.0	53.7	$+6.7$ $[1.7, 11.7]$	+9.3	$7.01 \rightarrow 5.30$
4	negative	53.0	52.3	-0.7 $[-5.3, 4.0]$	+25.7	$7.01 \rightarrow 7.28$
4	positive	47.0	54.3	$+7.3$ $[2.7, 12.0]$	+17.7	$7.01 \rightarrow 6.01$

Table 5.11: On-domain, trust-region guided diffusion generation on held-out SST-2 prompts ($n = 300$ pairs per cell), scored by the independent judge. Guided target-hit percentage against the paired unguided baseline, with a paired bootstrap 95% confidence interval; the self gain is the guiding classifier's own verdict; the last column is the span negative log-likelihood before and after guidance. The trust region keeps fluency at or below the unguided level, and the judge gain is positive with an interval excluding zero for the positive target at both strengths but null for the negative target. Compare the off-domain run of Table 5.10, where the working direction was the opposite.

The working direction has therefore flipped relative to the off-domain run: there the judge moved on the negative target and not the positive, here on the positive and not the negative, with fluency no longer a confound. This flip is the decisive evidence about what the residual limit is and is not. It is not a fluency artifact, because the trust region removed the fluency cost and the asymmetry remained. And it is not a fixed unreachable sentiment, because the reachable direction moved when the setting changed. What remains is an instrument-alignment effect: the two independently trained classifiers still disagree on about 28% of the on-domain generations, and steering toward one classifier's notion of a label registers on the other only where the two happen to align on the generated text. The claim the data licenses is therefore setting-contingent and deliberately narrow: with a single noisy guide at this scale, one-directional transfer to an independent judge is reachable under bounded fluency, but symmetric transfer at both labels is not, and which direction transfers tracks instrument alignment rather than a general "positive steers, negative does not" rule.

It is tempting to explain the flip by supposing the residual disagreement falls asymmetrically by class, so that the guide's positive verdict transfers to the judge more reliably than its negative verdict. Table 5.12 tests that supposition directly on the neutral calibration surface, the unguided on-domain generations, and does not support it. The per-class accuracies rule out a silent label-map flip: the guide is 79.4% accurate on negative and 79.9% on positive held-out sentences, the judge 85.8% and 89.9%, both balanced. On the unguided generations the judge confirms the guide's positive verdict 67.9% of the time and its negative verdict 75.7%, a paired difference of -7.7 points whose 95% confidence interval $[-18.0, +2.6]$ includes zero and in fact leans, if at all, the other way. The two conditional agreement rates are statistically indistinguishable, so the confusion table confirms the overall alignment gap that the ladder measures but does not identify a by-class asymmetry that would by itself explain the transfer direction. The honest reading is that the transfer asymmetry is real and observed but its class-level cause is not resolved by this data, and no further mechanism is asserted beyond the alignment gap. Representative guided pairs from both settings, drawn by a fixed seed and shown unfiltered, are collected in Appendix A.5, including the off-domain high-γ pairs whose fluency the trust region repairs.

Taken together, these four pilots move the diffusion control from a proposed experiment to a completed one. The signal the autoregressive gradient lacked is present in the diffusion score; the recovery the gradient machinery could not perform is performed natively; the repaired direction works inside the thesis's own exact-

		Judge verdict	
		negative	positive
Guide verdict	negative	109	35
	positive	50	106

Table 5.12: Per-class agreement of the guide (noisy classifier) and the independent judge (frozen GPT-2 sentiment head) on the 300 unguided on-domain generations (the neutral calibration surface; guided text is excluded to avoid circularity). The judge confirms the guide's positive verdict 67.9% of the time (95% CI $[60.4, 75.3]$) and its negative verdict 75.7% ($[68.5, 82.4]$); the paired difference (positive minus negative) is -7.7 points ($[-18.0, 2.6]$). The two conditional agreement rates are statistically indistinguishable (their paired-difference 95% CI includes zero), and the point estimates do not favour the positive channel. So the residual disagreement confirms that the two independently trained instruments are only partially aligned (71.7% agreement, up from 56–64% off-domain), but this table does not identify a by-class asymmetry that would by itself explain why positive steering transferred to the judge and negative steering did not; that transfer asymmetry is reported as observed and setting-contingent, and no class-level mechanism is claimed beyond the overall alignment gap. Overall agreement 71.7% $[66.3, 76.7]$.

energy chain; and trained guidance steers, with a reach limited by the same off-manifold fragility the thesis documents elsewhere. The training objective is thereby isolated as the causal variable, at both the signal level and the capability level.

6 Discussion

The results of the previous chapter are, on their surface, a collection of specific findings: a step size that does not transfer between models, an annealing artifact that mimics convergence, a Metropolis–Hastings correction that rescues one sampler and destroys another, a gradient that carries no information, a likelihood trap in which the best-scoring text is the worst text, three distinct modes of GFlowNet collapse, and a constraint that does not steer. Taken one at a time, each is a result about a method. Taken together, they are evidence for a single claim about the object all of these methods share. This chapter draws them into that single account. It first states explicitly which research question each part of the results answers and in which section, so that the mapping from questions to evidence is unambiguous. It then presents the unified explanation and argues for its strength on the grounds that it is reached by two independent routes. It connects each finding back to the specific prior work of Chapter 3, so that the contribution can be located precisely against the literature. It states carefully what the results do and do not license one to conclude. And it closes with the experiment that would falsify the central explanation, which is, in the author's view, the most important thing the thesis leaves for future work.

6.1 Answers to the Research Questions

It is useful to state plainly which research question each result answers and in which section of Chapter 5 the answer is established, both because the reader deserves an explicit map and because the discipline of writing the map down is a check that every question posed in the introduction has in fact been answered by evidence rather than by assertion.

RQ1 asked whether the frozen likelihood can be sampled effectively with faithful Langevin dynamics on the discrete manifold, and if not, why. It is answered in the negative across three sections. Section 5.1 shows that the sampler cannot even be made to move without model-specific calibration, which already denies the strong plug-and-play reading. Section 5.5 shows, through the direction-versus-magnitude ablation, that once the sampler does move, following the true gradient is statistically indistinguishable from following a random direction of equal norm, so the gradient is not a usable search direction. And Section 5.6 supplies the mechanism behind that fact, namely that the linearization radius of the surrogate the sampler uses is smaller than the smallest admissible discrete step, so the gradient is inapplicable rather than merely imprecise, and is compounded by the structural blindness of the input-embedding gradient to the self-term of the energy change. The three sections together answer both the "whether" and the "why" of RQ1. Section 5.12 then sharpens the "why" to a theorem at the final token, where the input-embedding gradient of the sequence log-likelihood is exactly zero while the energy difference between candidate tokens is exactly the model's conditional log-ratio, and Section 5.13 confirms by positive control that the defect lies in the training objective, since a score-trained model on the same tokenizer supplies the usable local direction the autoregressive gradient lacks and performs the recovery the gradient-guided sampler could not.

RQ2 asked about the role of the Metropolis–Hastings correction in each sampler. It is answered in Sections 5.2 and 5.3, and the answer is that the correction plays opposite roles in the two samplers, which is itself informative. For the discrete sampler the correction is beneficial: Section 5.2 shows that it removes the quenching artifact and produces smooth early convergence, so here theoretical correctness and empirical performance align. For the continuous sampler the same correction is catastrophic: Section 5.3 shows that it rejects almost every useful move, and the decomposition of the acceptance ratio attributes the collapse specifically to the proposal term, connecting it to the non-Lipschitz drift induced by nearest-neighbour projection. The correction does not make the continuous sampler work; it reveals that the continuous sampler was only ever functioning as a biased optimizer, and that its apparent successes were successes of optimization under early stopping rather than of sampling.

RQ3 asked whether amortization with a GFlowNet escapes the failure. It is answered in Section 5.10, and the answer is no. Three distinct failure modes appear, capacity collapse, length collapse, and reward hacking, and the unifying experiment shows that Langevin sampling on the amortized energy is indistinguishable from sampling on the untuned base, which locates the cause in the energy rather than in the search procedure.

RQ4 asked whether an additive constraint can steer generation on this landscape. It is answered in Section 5.11, and the answer is that it does not produce reliable control: on the discrete sampler the trustworthy paired contrast between a real and a randomized constraint direction is essentially zero, so the additive constraint does not steer, and the failure follows from the same defect established for the fluency energy rather than being a separate phenomenon. The finding is nonetheless more precise than a blanket null, and the precision matters: unlike the fluency gradient, the constraint gradient does carry a directional signal, but that signal cannot rescue generation once a constraint-only objective drives the sample off the fluent manifold. The constructive counterpart of this appears in the guidance pilots of Section 5.13.4 and Section 5.13.5: a classifier trained on the noisy states it scores does steer an independent judge, but only in a direction that tracks how far the guiding and judging instruments agree, and only once fluency is held fixed by a trust region, which is what it costs to buy the control the additive constraint could not.

6.2 A Single Defect, Reached by Two Routes

The organizing claim of this thesis is that every failure catalogued above is a surface manifestation of one underlying defect: the frozen likelihood of an autoregressive language model is not a usable energy function for sampling, because the training objective that produces it never shapes it to be one. The argument for this claim was made incrementally through the results; here it is stated as a whole.

Recall the distinction drawn in Section 2.1. Teacher forcing optimizes each local conditional $p_{\mathrm{LM}}(x_t \mid x_{<t})$ to predict the next token given a correct prefix drawn from the data. That is the entire content of the objective, and modern models satisfy it superbly. What the objective never does is constrain the behaviour of the joint log-likelihood $\log p_{\mathrm{LM}}(\boldsymbol{x})$ as a function of $\boldsymbol{x}$ when $\boldsymbol{x}$ is varied off the data manifold, which is exactly the regime in which every energy-based sampler operates, because sampling is precisely the business of moving through sequence space, including through its off-distribution regions, in search of good sequences. The joint likelihood is therefore never asked to be smooth, never asked to be a calibrated measure of quality, and never asked to have a gradient that points toward better sequences. Energy-based decoding borrows this quantity and treats it as though it had all three properties. The results show, in three complementary ways, that it has none of them: it is not smooth, as the discontinuous Voronoi landscape of Section 5.3 demonstrates; it is not a good measure of quality, as the likelihood trap of Section 5.7 demonstrates; and its gradient does not point toward better sequences, as the gradient fallacy of Section 5.5 and the linearization failure of Section 5.6 demonstrate. These are not three separate deficiencies but three faces of one absence, the absence of any training pressure toward a well-behaved global energy.

The same account, read constructively, predicts what would restore each missing property, and the diffusion control of Section 5.13 bears the prediction out end to end. The energy itself was never the problem: a gradient-free Gibbs sampler and a single forward pass with top-k rescoring both work on the frozen likelihood, so the information needed to recover good text is present in the energy and only the attempt to extract a search direction from its input-embedding gradient fails. That gradient is causally blind to how well a token fits its own context, exactly zero at the final position and merely diluted elsewhere. A model trained by score matching rather than by teacher forcing supplies precisely the direction the autoregressive gradient lacks, the linearization correlation moving from statistically zero to clearly positive; that repaired direction works inside the thesis's own exact-energy Metropolis chain, where swapping only the proposal lifts recovery from zero to thirty-nine percent; and a classifier trained on the noisy states it is evaluated on steers generation, with a reach bounded by the off-manifold fragility of classifiers that the constraint experiment already exposed. The defect is thus not only diagnosed but localized by its repair: what teacher forcing withholds, a score objective provides, and the capability the plug-and-play promise assumed for free is available only at the price of training a model to have it.

There is a cleaner and more mechanical way to see why the *gradient* in particular is the defective part, and it follows directly from the decomposition of Section 5.6. The only term in the likelihood that registers how well a token fits its own left context enters through a discrete index into the output softmax, not through the continuous input embedding that the sampler differentiates. The input-embedding gradient can therefore see a substituted token only through that token's effect on the *later* tokens it feeds forward, and it is structurally blind to the token's own fit. This blindness is partial in the interior of a sequence, where a token still shapes its successors, and it becomes total at the final position, where the token's embedding feeds no realized prediction at all: the gradient of the sequence log-likelihood with respect to the last token's input embedding is exactly zero, even though the energy difference between candidate final tokens is precisely the model's conditional log-ratio and is therefore maximally informative. At that position the energy knows exactly which token is best and the gradient provably cannot, which is the sharpest possible statement of the whole diagnosis: what fails is not the information in the energy but the attempt to extract a search direction from its input-embedding gradient. Making this last-position argument exact, and pairing it with an energy-only sampler that succeeds on the same task where the gradient arm fails, is the natural culmination of the diagnosis and the immediate next analysis the work points to.

The strength of the evidence for this claim rests on a feature of the study's design that is worth making explicit, because it is what elevates the conclusion above a collection of negative results. The claim is reached by two routes that share no machinery. The gradient-guided route differentiates the likelihood, and it fails: the gradient is uninformative because its surrogate is uncorrelated with the true energy change, and the correction, applied honestly, rejects every useful move. The amortized route never touches the gradient at all; it treats the likelihood only as a scalar reward, and it fails too, because the reward has no well-behaved optimum where coherent text lies, so the learned policy collapses in whichever direction the reward's implicit incentives push it. Two methods with nothing in common beyond the energy they use should not fail in ways that trace back to the same property of that energy, unless the energy is indeed the common cause. And the unifying experiment removes even the last doubt: running the gradient-guided sampler on the amortized energy reproduces exactly the behaviour of running it on the base, so passing the energy through the amortization changes nothing about the sampler's fate. The energy, and not the sampler, and not the search procedure, is the problem.

The likelihood trap ties the two routes together at the level of intuition, and it explains an otherwise puzzling feature of the whole picture. Because low energy is degenerate text, the goal that both methods pursue is the wrong goal, and this casts the failure of the gradient-guided samplers in an unexpected light. Their inability to reach low energy, which the earlier sections established as a failure, is in a sense a reprieve, because reaching low energy, as both the quenching phase of the discrete sampler and the reward hacking of the GFlowNet show, produces bad text. The methods are protected from their own objective by their inability to optimize it, and the moments when they do optimize it effectively, beam search on the length-normalized GFlowNet being the clearest, are exactly the moments when their output degenerates into strings like the ones displayed in Section 5.7. A method that worked, in the narrow sense of reaching the energy's minimum, would produce gibberish; a method that produces fluent text does so by failing to reach the minimum. This is the deepest sense in which the frozen likelihood is not an energy worth sampling.

6.3 Connections to the Related Work

The findings of this thesis stand in a definite relationship to the literature reviewed in Chapter 3, and making that relationship explicit locates the contribution precisely rather than leaving the reader to infer it.

The likelihood trap measured in Section 5.7 is a direct, quantitative reproduction on the study's own models of the neural-text-degeneration phenomenon that Holtzman et al. (2020) identified: that the most probable sequences under a neural language model are degenerate, and that maximization-based decoding produces worse text than sampling. Chapter 3 presented that as a known hazard of decoding. The contribution here is to re-derive it as a structural obstacle for energy-based sampling specifically, because in the energy-based framing the degeneracy of high-probability text is not merely a decoding pitfall to be avoided but a statement that the *target* of the sampling procedure is undesirable, which is a more damaging thing to say about a method that is defined by its pursuit of that target. The monotone likelihood-repetition relationship measured here is the quantitative form of that statement.

The anisotropy results of Section 5.8 confirm, on GPT-2 Large and Llama-3, the representation-degeneration and embedding-anisotropy phenomena documented by Gao et al. (2019) and Ethayarajh (2019). Chapter 3 raised anisotropy as a reason to distrust Euclidean distance in embedding space, and the measurements here quantify that distrust. But they also draw a consequence those works did not consider: because the anisotropy scale differs by roughly a factor of three between the two models, a single step-size schedule cannot transfer between them, which is the concrete and previously unremarked cause of the calibration failure that opened the results chapter. The anisotropy literature explains why a token can be close in embedding space yet wrong; this thesis adds that the same geometry makes the samplers model-specific in a way that undercuts their claim to be plug-and-play.

The Metropolis–Hastings breakdown of Section 5.3 is the empirical counterpart to a gap noted in Section 3.2: that the energy-based decoding literature frequently omits the correction, and that where it is included, the difficulty of computing the reverse proposal on a projected landscape is seldom examined. This thesis takes the correction seriously, implements it faithfully following the convergence theory of Roberts and Tweedie (1996), and measures exactly what the common omission was concealing. Seen through this lens, methods such as COLD (Qin et al., 2022) and MuCoLa (Kumar et al., 2022), which the continuous sampler of this thesis follows, are best understood not as samplers from the stated energy but as biased optimizers, and their reported successes are successes of optimization under early stopping rather than of sampling. This is not a criticism of those works, which are careful about what they claim, but a clarification of what the methods actually do, and it is a clarification the correction makes unavoidable. The same lens sharpens the relationship to the annealed stochastic-gradient Langevin dynamics of Welling and Teh (2011), whose guarantee is that annealing the step size toward zero makes the chain converge to the target distribution. The quenching effect of Section 5.2 is what that recipe degenerates into once the correction is dropped: the annealed noise does not deliver a fair draw from the distribution but freezes the chain into whichever mode it last occupied, so the very schedule that the theory of Welling and Teh (2011) relies on becomes, without the correction, the mechanism of a biased optimization.

Finally, the GFlowNet results of Section 5.10 qualify, without contradicting, the picture presented by Hu et al. (2024), whose formulation and codebase this thesis builds on. Their work showed that amortized sampling from a frozen-model posterior can yield diverse, transferable samples on tasks where the reward is well-behaved, and nothing in this thesis disputes that. The narrower and more critical finding here is that when the reward is the raw frozen likelihood used as an energy over free-form sequences, the amortization inherits the pathology of that energy and the learned policy collapses. The two findings are compatible, and the axis that reconciles them is precisely the one this thesis foregrounds: whether the reward defines a landscape with a coherent-text optimum. Where it does, amortization is useful; where it is the raw autoregressive likelihood, it is not, and the contribution here is to show, by the unifying experiment, exactly which case the frozen likelihood falls into.

Finally, the gradient-free samplers isolated in Section 3.2, Mix-and-Match (Mireshghallah et al., 2022) and the twisted sequential Monte Carlo of Lew et al. (2023) and Zhao et al. (2024), stand in a confirming rather than a contradicting relationship to the central result. Each samples from a frozen-model energy without differentiating it, and each works. Chapter 3 raised this as a hint; the Gibbs baseline of Section 5.5.2 converts the hint into direct evidence by running a gradient-free sampler on the identical energy the Langevin methods use and recovering coherent text where they cannot. Taken together, the external methods and the in-platform Gibbs result draw the same line: the frozen likelihood is a serviceable object to *evaluate*, and to search without gradients, and an unusable object to *differentiate*. This is the empirical warrant for the narrowed claim of the conclusion, and it is why the thesis does not overreach into asserting that the energy is useless, only that its gradient is.

6.4 What the Results Do and Do Not Show

It is important to state the limits of these conclusions precisely, both out of scientific honesty and because a diagnosis is only useful if its scope is clear. The thesis shows that the frozen likelihood of a *standard autoregressive* language model is a poor energy for gradient-guided and amortized sampling on discrete text, and it explains why, in terms of the training objective and the geometry of the token embedding space. It does not show that energy-based controllable generation is impossible in general. It shows that a particular and widely used instantiation of it fails, and it identifies the specific property responsible for the failure, which is a more useful result than a blanket impossibility claim precisely because it points to what would have to change.

Several caveats bound the empirical claims, and they are stated here rather than buried. The KL-divergence metric depends on the position of the corrupted token within the sequence, which is why, as Section 4.4 explained, the central results are argued from the absence of a gap between conditions rather than from small measured differences; a null gap is robust to a positionally noisy metric in a way that a narrow measured win would not be, and the reader should read the gradient-fallacy result as "no reliable difference" rather than as a precise numerical equality. The Llama-3 results support only qualitative cross-architecture claims, because the tokenizer and embedding scale differ from the GPT-2 family, so the Llama measurements establish that the phenomena occur across architectures without licensing a numerical comparison between the two. The linearization result, as Section 5.6 was careful to note, is that the surrogate is uniformly uninformative across the admissible range of distances rather than informative up to a sharp radius and useless beyond it; the language of a "linearization radius" is a convenient summary of a decay, not a claim of a clean threshold, and the honest reading of Figure 5.5 is that the correlation is low everywhere the sampler operates. And the constrained-generation extension establishes the absence of reliable steering on this energy, not the impossibility of steering on a better one.

The thesis is also candid about the relationship between its title and its content, because that candour is part of the diagnostic honesty the work aims for. The original objective was controllable generation via faithful Langevin dynamics, conceived as a two-stage plan in which the first stage would establish that energy-guided sampling on a frozen model is viable and the second would add the constraint term and perform sentiment and toxicity control. The first stage did not hold. Faced with that, there were two paths: to proceed to the second stage regardless, building control on a foundation the study had shown to be unsound and thereby producing numbers whose meaning could not be interpreted, or to turn to the more valuable question of why the foundation fails. The work took the second path. What it delivers is therefore a systematic diagnosis of the failure of a prerequisite, with the mechanisms identified, measured, and cross-checked, and with the constrained-generation extension included as direct evidence of the downstream consequence, rather than a demonstration of successful control. A diagnosis of why a widely pursued approach does not work, when it is supported by measurement and reaches a single explanation by independent routes, is in the author's view a more useful contribution to the field than a marginal and uninterpretable positive result would have been.

6.5 Implications and the Falsifying Experiment

The most valuable consequence of locating the defect in the training objective, rather than in the sampler or the search procedure, is that it makes a sharp and testable prediction, and a diagnosis that makes a falsifiable prediction is worth more than one that does not. If the reason the frozen autoregressive gradient is useless is that teacher forcing never shapes a score function over sequences, then a model whose training objective *does* shape such a function should not suffer the same fate, and running the identical protocol of this thesis on such a model would either confirm the diagnosis or refute it.

Diffusion language models are exactly such models (Li et al., 2022; Lou et al., 2024). A diffusion model is trained, by construction, to denoise corrupted inputs, and denoising is equivalent to learning the gradient of the log-density of the data, the score function, by the identity of Vincent (2011) that also underpins score-based generative modeling (Song and Ermon, 2019; Ho et al., 2020). Where the autoregressive model is trained only to predict the next token and is never asked about the shape of its energy off the data manifold, the diffusion model is trained precisely to know, at every noise level, in which direction the data density increases, which is exactly the information the Langevin sampler needs and exactly the information the autoregressive gradient was found to lack. Running the diagnostic protocol of this thesis on a diffusion language model therefore constitutes a positive control in the strict experimental sense. If the linearization correlation is high where it was zero, if the gradient outperforms a random direction where it did not, and if the sampler recovers coherent text where it produced degeneracy, then the training objective is isolated as the causal variable and the diagnosis of this thesis is confirmed. If, on the other hand, the diffusion model fails in the same ways, then the diagnosis is wrong, the cause lies deeper than the training objective, and a different explanation must be sought. Either outcome is informative, which is the mark of a well-posed control, and the experiment is inexpensive because score-based discrete diffusion models that share the GPT-2 tokenizer are available off the shelf. This was, in the author's assessment, the single most important test the thesis could run, and it was run; the design of this thesis, in particular its reliance on a masked-recovery task that a diffusion model can perform without modification, was shaped throughout to make it clean and immediate, and Section 5.13 reports its outcome.

The concrete instrument for the control is the score-entropy discrete diffusion model of Lou et al. (2024), which shares the GPT-2 tokenizer and drops into the existing masked-recovery harness without retokenization. This control has now been run, at two model scales, and its outcome is reported in full in Section 5.13. Every arm of the prediction is borne out. The linearization correlation that was statistically indistinguishable from zero for the autoregressive gradient is clearly positive for the diffusion score in every embedding-distance stratum and at both scales; the model natively recovers the flipped token that the gradient-guided sampler never once recovered; and, in the sharpest form of the test, replacing only the proposal direction inside the thesis's own exact-energy Metropolis chain, and changing nothing else, lifts exact recovery from zero for the gradient arm to thirty-nine percent. A trained noisy-state classifier steers generation on an independent judge as well, though its reach is bounded and setting-contingent: off-domain and without a trust region it moved the judge in the negative direction at a growing fluency cost, while on held-out in-domain prompts with a trust region that holds fluency fixed the working direction flipped to positive, so the reachable direction tracks where the guiding and judging classifiers happen to agree rather than a fixed sentiment, exactly the off-manifold fragility of classifiers documented for the constraint-only arm in Section 5.11. The correlations are modest and the diffusion models are smaller and differently trained than the frozen backbone, so these are pilots and are read as such, but they point one way: the training objective is the causal variable, at the signal level and at the capability level. The result also reframes the promise the thesis set out to examine. The appeal of energy-based control was that a fresh constraint could be imposed on a frozen model with no training at all; the diagnosis is that this promise fails for a structural reason, because teacher forcing never shapes a navigable score, and the pilot shows what the capability actually costs, which is a model trained for it, by a score-matching objective for the direction signal and by a noisy-state classifier for the guidance.

A second and complementary direction is to attack the defect at its source rather than route around it. If the problem is that the likelihood is not a score function, then one option is to make it one, by fine-tuning the model so that its sequence log-density is aligned directly with a target energy using a score-matching objective. This is a lighter intervention than the reinforcement-learning-style training of a GFlowNet, and it is directed precisely at the identified deficiency rather than at a downstream symptom of it. Only once an energy landscape is known to be navigable does the additive constraint term of the plug-and-play framework become meaningful, because only then is there a fluency energy for the constraint to be added to that a sampler can actually follow, and at that point the original goal of controllable generation becomes reachable by the route this thesis set out to take. The contribution of the present work to that longer programme is a map of where the ground gives way, drawn carefully enough that the next traveller knows both which step is unsafe and why, together with a falsifiable hypothesis about what firmer ground would look like.

7 Conclusion

This thesis set out to realize the promise of energy-based controllable text generation, in which a frozen autoregressive language model is treated as an energy function and steered at inference time by sampling from it with gradient-guided Markov chain Monte Carlo. The appeal of that promise is real: a single frozen model, augmented with an additive constraint at generation time, could in principle be steered toward any objective without the cost of retraining. Across a systematic study of one hundred and forty-five sampling configurations, five energy functions, and two samplers, complemented by four diagnostic experiments, the thesis found that the central premise on which the promise rests does not hold. The frozen likelihood of an autoregressive language model does not admit local gradient-based navigation on discrete text: its gradient is not a usable search direction. The narrower phrasing is deliberate and is earned by the evidence, because the energy itself proves searchable by gradient-free means, a single top-k rescoring pass and a Gibbs sampler both recovering coherent text on the same benchmark; what fails is not the evaluation of the energy but the attempt to follow its gradient.

The contribution of the thesis is not this negative observation alone, which the difficulty the field has had with these methods already hints at, but the mechanistic account that accompanies it, established by independent means and supported at every step by direct measurement. Following the true gradient was shown to be statistically indistinguishable from following a random direction of equal magnitude, on all five energy functions, once a normalization confound was removed by an ablation designed for that purpose. This gradient fallacy was explained by a linearization failure: the first-order surrogate that the discrete sampler uses to rank candidate tokens is uncorrelated with the true change in energy, because the smallest admissible discrete step is far larger than the radius over which the local linear model is valid, and the input-embedding gradient is in addition structurally blind to a substantial part of the energy change. The Metropolis–Hastings correction, applied faithfully rather than omitted, was shown to paralyze the continuous sampler, and the trace analysis located the cause precisely in the proposal term of the acceptance ratio, which collapses because the nearest-neighbour projection makes the drift discontinuous at every cell boundary. A likelihood trap was measured on the study's own models, showing that the lowest-energy text is the most degenerate, so that success at the optimization objective is failure at the underlying goal. Turning to amortized inference, the GFlowNet experiments exhibited three distinct failure modes, and a unifying experiment showed that Langevin sampling on the GFlowNet-tuned energy is indistinguishable from sampling on the untuned base, demonstrating that the pathology survives amortization and is therefore a property of the energy rather than of any sampler. A constrained-generation extension refined rather than merely repeated this picture: unlike the fluency gradient, the constraint gradient does carry a directional signal, but it does not produce reliable control on the discrete sampler, and where a signal appears it is entangled with the departure from the fluent manifold that a constraint-only objective causes.

Taken together, these findings locate the cause of the failure in the training objective of autoregressive language models, which shapes accurate local conditional distributions but never a smooth global energy over sequences. This diagnosis is not an endpoint but a signpost, because it makes a falsifiable prediction: a diffusion language model, whose objective does shape a score function over the data, should not fail in the same way. The thesis runs that positive control rather than only proposing it. A score-entropy discrete diffusion model on the same tokenizer, tested at two scales, supplies a clearly positive local direction where the autoregressive gradient was statistically zero, natively recovers the flipped token the gradient-guided sampler never recovered, and, in the sharpest form of the test, lifts recovery from zero to thirty-nine percent when it replaces only the proposal inside the thesis's own exact-energy chain; a trained noisy-state classifier steers an independent judge, with a reach bounded and setting-contingent for the same off-manifold reasons the constraint experiment exposed. These are pilots on smaller, differently trained models, and are read as such, but every arm of the prediction is borne out, which isolates the training objective as the causal variable and points to the complementary route of aligning a model's sequence density with a target energy by score matching. What the thesis provides to the larger effort of building inference-time controllable generation on frozen models is therefore a precise characterization of where and why the most common approach breaks down, drawn carefully enough to guide what comes next, together with a positive control that confirms the explanation. A negative result about a method, when it is accompanied by a mechanism, a measurement, and a falsifiable hypothesis, is a positive result about understanding, and it is in that spirit that this work is offered.

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A Appendix

A.1 Additional Diagnostic Figures

This section collects supporting figures referenced in Chapter 5. Each figure is discussed at the point of reference in the results chapter, as indicated in the captions below.

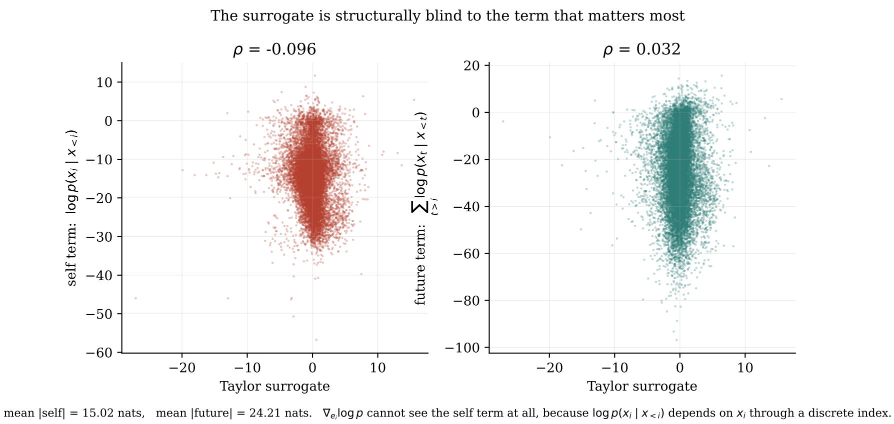

Figure A.1: Decomposition of the true energy change into the self term and the future term, plotted against the gradient surrogate, for GPT-2 Large. The surrogate correlates weakly with the future term and is anti-correlated with the self term, to which it is structurally blind. This figure supports the argument of Section 5.6.

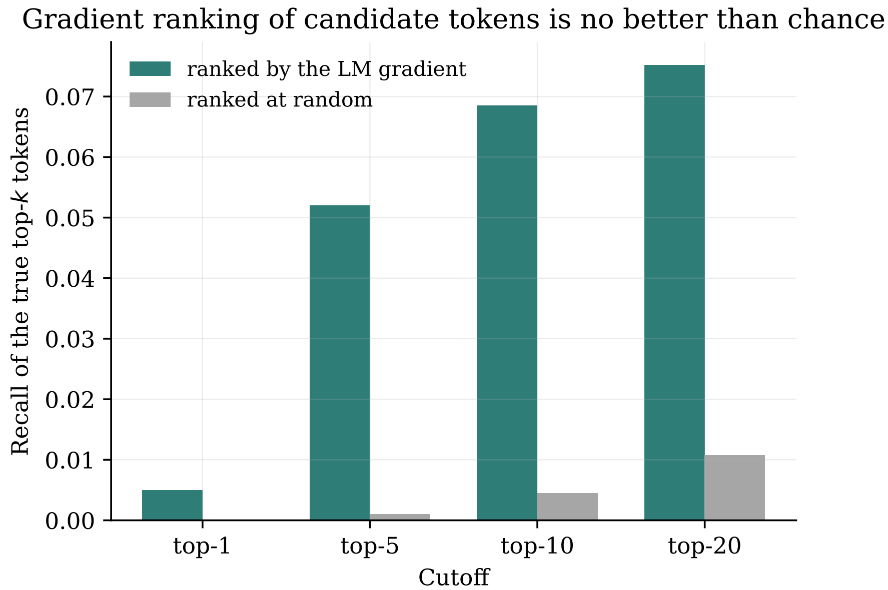

Figure A.2: Recall of the true top-k token substitutions by the gradient-ranked top-k, against a random-ranking baseline, for GPT-2 Large. Gradient ranking does not exceed chance, corroborating the null correlation reported in Section 5.6.

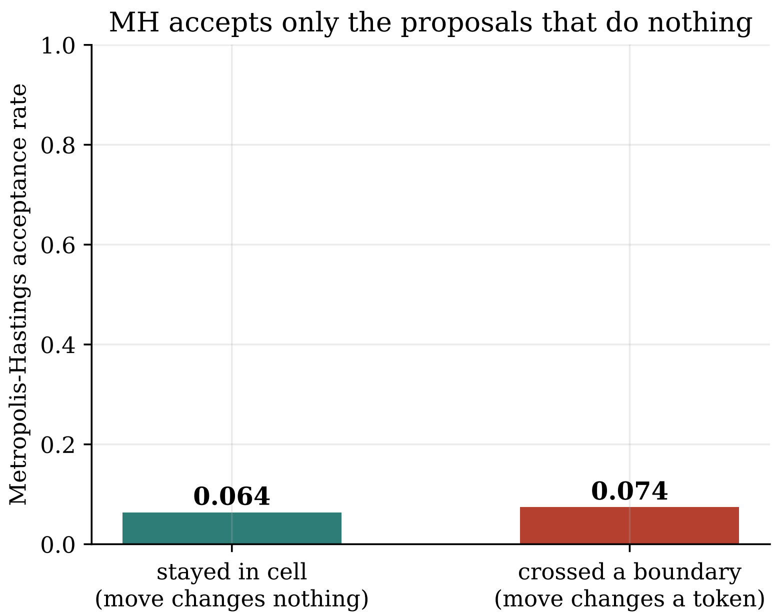

Figure A.3: Metropolis–Hastings acceptance rate in the continuous sampler, conditioned on whether the proposal crossed a Voronoi cell boundary. Moves that change a token are almost never accepted. This figure supports Section 5.3.

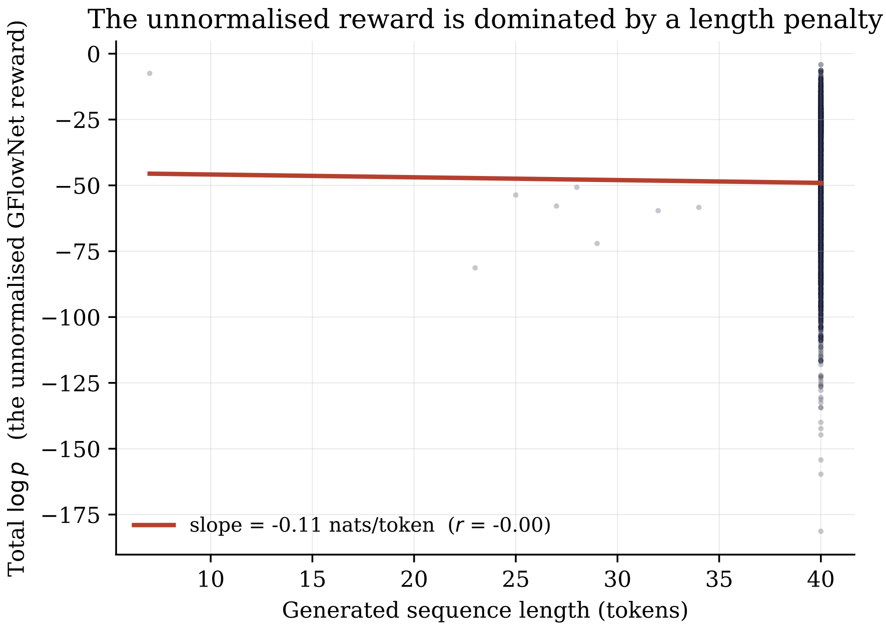

Figure A.4: Total log-likelihood against generated length for GPT-2 Large. The negative slope, approximately -1.12 nats per token, is the implicit brevity incentive in an unnormalized-likelihood reward, discussed in Section 5.7 and Section 5.10. The slope is estimable only where the forty-token length cap does not bind; for GPT-2 Large nearly all generations reach the cap, so the uncensored estimate shown here is the informative one, while on the length-normalized GFlowNet, where length varies, the censored slope steepens to -2.36.

A.2 Full Configuration Grid

Table A.1 reports the final KL divergence for a representative subset of the full configuration grid across the four GPT-2-family energy functions, drawn from the aggregated results file `final_kl_by_model.csv`. The complete grid comprises the factorial combination of sampler (DLS, CLS), proposal method (policy, grad-norm-preserved, random), Metropolis–Hastings correction (on, off), gradient normalization (on, off), and schedule length (50, 100), evaluated on each of the five energy functions, for a total of 145 configurations.

The count arises as $145 = 5 \times 29$, five energy functions times twenty-nine configurations each, and the twenty-nine is worth setting out because the naive factorial overcounts it. A fully crossed design over the five binary and ternary axes above would give $2 \times 3 \times 2 \times 2 \times 2 = 48$ cells per energy function, or 240 in total, but two prunings reduce this. The continuous sampler is run for only the policy and random proposal methods, not the grad-norm-preserved method, and across four of the gradient-normalization and schedule variants, giving 8 CLS configurations per energy function; the discrete sampler is run for all three proposal methods across seven of the correction, normalization, and schedule variants, giving 21 DLS configurations. The sum is $8 + 21 = 29$ per energy function. The count was verified directly against the result folders, which hold 29 configurations for GPT-2 Large (`results_gpt2_v2`), 29 for Llama-3 (`results_llama`), and 87 across the three GFlowNet variants (`results_gfn`), and $29 + 29 + 87 = 145$.

Configuration	GPT-2 Large	GFN lb0-500	GFN lb0-2000	GFN lb1-500
DLS policy, MH, gn, s50	6.541	6.306	6.721	6.415
DLS policy, no-MH, gn, s50	9.499	9.195	10.085	8.669
DLS random, MH, gn, s50	6.370	6.105	6.548	5.525
DLS gnp-random, MH, gn, s50	6.370	6.105	6.548	5.525
DLS policy, MH, no-gn, s50	6.474	6.009	6.698	6.008
DLS gnp-random, MH, no-gn, s50	6.430	6.001	6.552	5.338
CLS policy, MH, no-gn, s50	8.083	7.799	8.198	13.732
CLS policy, no-MH, no-gn, s50	8.083	7.799	8.198	13.732

Table A.1: Representative subset of the final-KL configuration grid, from `final_kl_by_model.csv`. Note that the DLS policy and grad-norm-preserved random rows are near-identical under both normalization settings, which is the gradient fallacy of Section 5.5. Note also that the CLS rows with and without the Metropolis–Hastings correction are identical to three decimals, because under the correction the sampler rejects essentially all proposals and does not move, as established in Section 5.3.

A.3 Computational Cost

Because one conclusion of the thesis is that gradient-free methods match or beat the gradient-guided samplers on the same energy, it is worth setting out the computational cost of each, which differs by more than a constant factor. Table A.2 gives the number of network evaluations each method spends to recover a masked span, counted as forward and backward passes, where S denotes the schedule length (either 50 or 100), M the number of masked positions, T the number of Gibbs sweeps, and k the number of candidates a rescoring pass considers.

Method	Forward passes	Backward passes	Note
DLS / CLS, no correction	S	S	one gradient per step
DLS / CLS, with correction	$2S$	$2S$	extra gradient at the proposed point
Conditional argmax/sample	1	0	a single forward pass
Top-k rescore	$1 + k$	0	rescore k candidates
Gibbs	$M\,T$	0	M positions, T sweeps, no gradient

Table A.2: Network evaluations per recovered span. The gradient-guided samplers spend a backward pass at every step, doubled when the Metropolis–Hastings correction is enabled, whereas the gradient-free baselines spend none. The single-pass rescoring method, which was the strongest baseline of Section 5.5.2, costs a small constant number of forward passes and no backward pass at all.

The measured wall-clock times are consistent with this accounting and give a sense of the absolute scale on the hardware of Section 4.8. The discrete sampler on the free-form continuation task took 2157 seconds for one hundred sequences, about 21.6 seconds per sequence over a fifty-step schedule; the full suite of gradient-free baselines, including the Gibbs sampler, took 215 seconds for two hundred sequences, roughly one second per sequence; and rescoring the six hundred recovered sequences with the external Llama-3 judge, a single forward pass each, took 24.8 seconds in total. The comparison is stark: the gradient-free baselines reach comparable or better recovery quality at a fraction of the per-sequence cost, because they never pay for the backward pass whose product this thesis finds to be uninformative.

A.4 Geometry of Sampler Trajectories in Embedding Space

This section examines the spatial paths the samplers trace through the embedding space, which gives a concrete counterpart to the acceptance statistics of Section 5.3 and the trajectory summaries of Section 5.4. The primary figure is deliberately *not* a projection. A two-dimensional projection of a 1280-dimensional space cannot carry the argument, as the explained-variance figure below makes plain, so the load-bearing figure reports exact $L2$ distances in the full space and the projection is demoted to a single illustrative panel. All distances and all nearest-token decoding here use the full 50,257 $\times$ 1280 embedding matrix of the model, not any subsample.

Figure A.5 is the primary view. For one seeded trace (index 5, chosen once by `np.random.default_rng(0)`), it plots, at every step and on a symmetric-log axis so that exact zeros remain visible, the $L2$ distance from the sampler's state to the ground-truth token embedding, and for the continuous sampler also the distance to the nearest token of any kind, the off-manifold measure. Because `collect_traces` advances the random-number state across configurations, trace index 5 is the same source sentence in every panel but its randomly masked position, and hence its ground-truth token, differs by configuration; each panel therefore uses its own exact ground-truth embedding, and the cross-panel claim is about the on-manifold-versus-off-manifold dynamics, which does not require a shared token. The token strip under each axis shows the decoded token at each change, and for the continuous sampler the steps at which the proposal was rejected, inferred exactly from identical consecutive states, are marked as ticks.

The distances make the geometry exact. The discrete sampler, policy and random alike, sits at distance zero from the token manifold at every step, because its state is always a genuine token embedding; it visits about five distinct tokens over the run and ends about 2.2 units from the ground truth, a token or so away. The continuous sampler with the correction on is quenched: its state sits between 118 and 151 units from the nearest token across the run and it changes its projected token about twice in fifty steps, the spatial form of the near-total rejection measured in Section 5.3. With the correction off it visits about 45 of 50 cells and drifts as far as roughly 980 units off the manifold before collapsing back to about 17 units at the end. Against a mean nearest-neighbour token spacing of 1.82 and a mean pairwise spacing of 2.77, an off-manifold distance of 120 to 150 is a factor of roughly 65 to 83 times the spacing between neighbouring tokens, so the continuous state spends its run in essentially empty space where the projected energy is a poor guide, which is why the boundary-crossing correction almost always rejects.

Figure A.6 is the secondary, illustrative view: the same trace projected onto the first two principal components of the full vocabulary embedding matrix. It is shown for intuition only, and its own caption records why it cannot be the evidence: the two components capture just 3.3% of the embedding variance, so the projection discards almost all of the geometry the distances above measure exactly. Read for what it can show, it agrees with the distances: the discrete trajectories stay inside the vocabulary cone while the continuous ones escape far outside it.

Read together, the two figures make the central geometric claim visible: the samplers either fail to move in any directed way, the discrete sampler and the free continuous sampler both wandering without bending toward the ground-truth token, or fail to move at all, the corrected continuous sampler frozen far off the manifold, and in neither case does the path converge on the target, which is the spatial signature of a gradient that carries no directional signal.

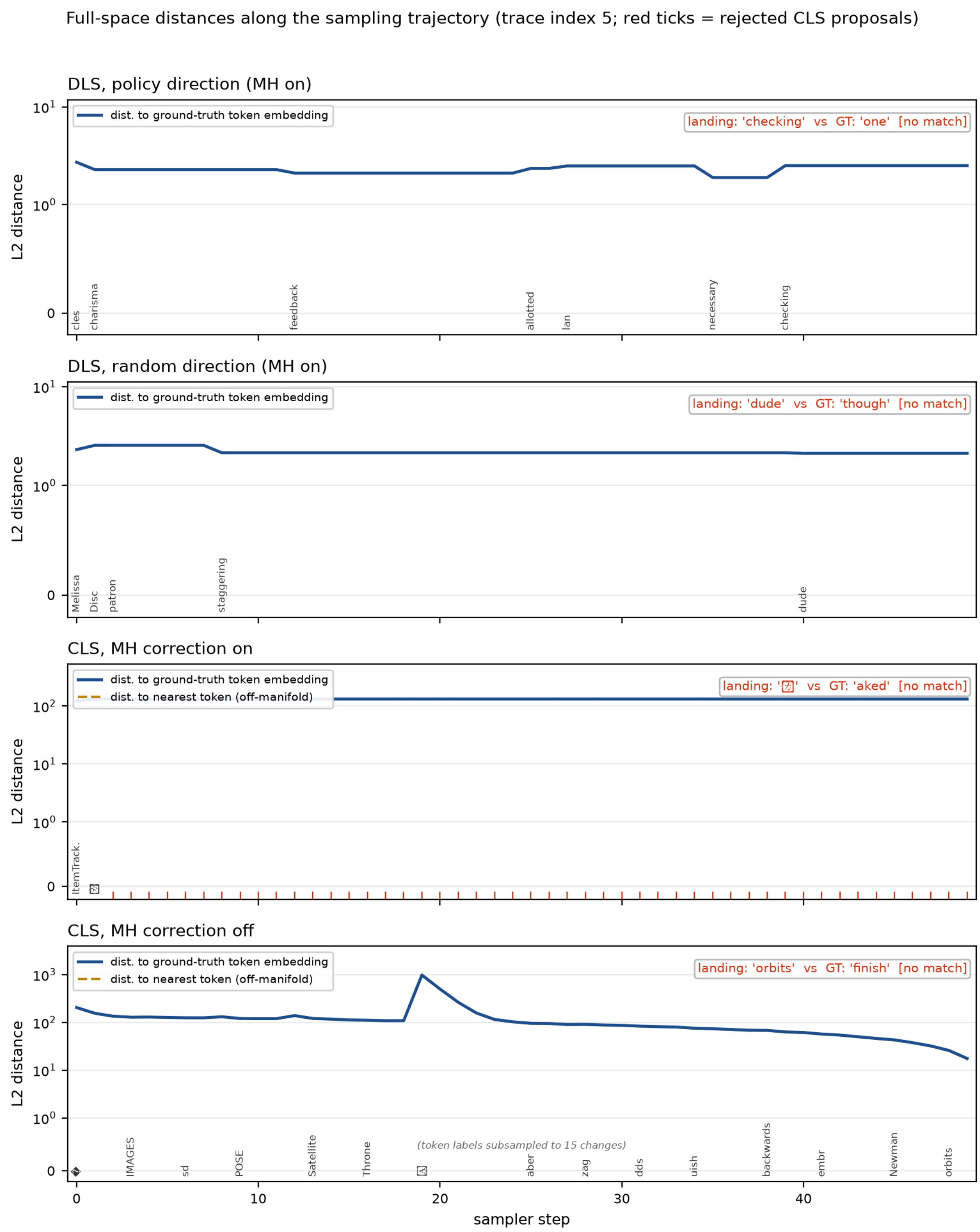

Figure A.5: Full-space distances along the sampling trajectory for one seeded trace (index 5), on a symmetric-log axis. Solid line: $L2$ distance to the ground-truth token embedding; dashed line (continuous sampler only): $L2$ distance to the nearest token of any kind. The discrete sampler sits exactly on the token manifold at every step (nearest-token distance identically zero, so it is not drawn) and moves from token to token; the continuous sampler with the correction on is quenched roughly 120 to 150 units off the nearest token and its proposals are rejected at almost every step (red ticks); with the correction off it wanders as far as about 980 units before collapsing back toward a token. Neither continuous variant lands on the ground truth.

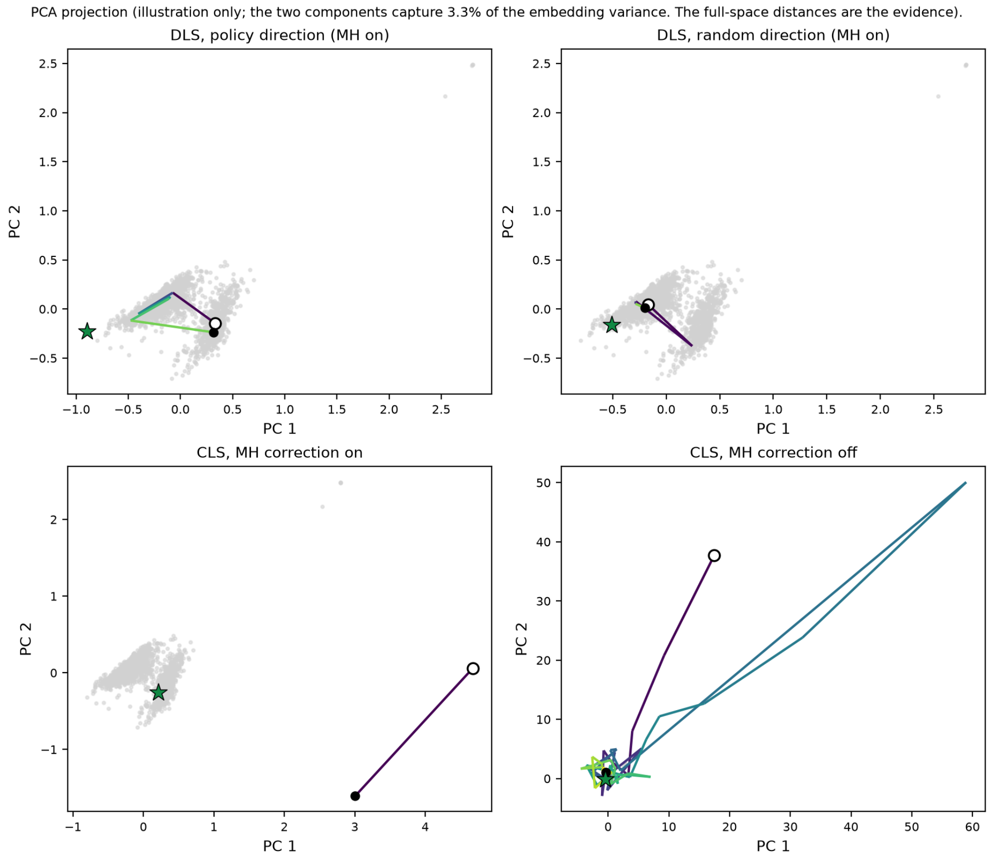

Figure A.6: PCA projection of the same seeded trace, one panel per configuration, with the vocabulary embeddings as a grey cloud, the start state as an open circle, the end as a filled dot, and the ground-truth token as a star. This projection is illustration only: its two components capture 3.3% of the embedding variance, and the full-space distances of Figure A.5 are the evidence. The discrete trajectories stay inside the cone; the continuous trajectories, especially with the correction off, leave it entirely.

A.5 Guided-Generation Examples

The following pairs make the classifier-guided steering of Section 5.13.5 visible in text. The on-domain draws are seeded and unfiltered (`np.random.default_rng(0)`, three pairs per cell), so dull or crude draws are kept; the statistics are in Table 5.11 and Table 5.12, and these are illustration. For each pair, g is the guiding classifier's verdict and j the independent judge's, so a pair where they disagree shows the partial instrument alignment directly rather than a failure. The second block places the highest-fluency-cost pairs from the off-domain run beside them, where guidance without a trust region collapses into word salad at a negative log-likelihood near 16; the on-domain trust-region pairs stay bounded. Garbled bytes the diffusion model emits are shown as `[?]`.

Table A.3: On-domain trust-region guided generations (G-prime). Draws are seeded (`default_rng(0)`, three pairs per cell) and unfiltered; the statistics are in Table 5.12 and the G-prime results table. g and j are the guide's and the independent judge's verdict. Pairs where they disagree show the partial instrument alignment directly.

Cell: $\gamma = 2$, target negative	
Prompt	bleakly funny, its characters all the more
Unguided	bleakly funny, its characters all the more often point to negation of value: will you have a sense of humor? or will you have [g:pos, j:neg; NLL 6.57]
Guided	bleakly funny, its characters all the more so, their than the the-a and b are the moreso. characters than- the the [g:pos, j:pos; NLL 7.92]
Prompt	all that's missing is the spontaneity
Unguided	all that's missing is the spontaneity, and the courage to pick up undercooked stock,and also the fierce muscle of expectation that will [g:pos, j:neg; NLL 7.10]
Guided	all that's missing is the spontaneity. When they leave the missing is the spontaneity. When they leave doing what they are supposed [g:neg, j:neg; NLL 4.44]
Prompt	the quality of the art combined with the humor and

Unguided	the quality of the art combined with the humor and character of the characters proves that the game is good.In my opinion, I want to answer you [g:pos, j:pos; NLL 4.84]
Guided	the quality of the art combined with the humor and simplicity of the art create a very that is is out of its own right, the arethe and [g:pos, j:pos; NLL 6.52]
Cell: $\gamma = 2$, target positive	
Prompt	you don't have to know about music to
Unguided	you don't have to know about music to be musical– even if you don't know about music Audio 1 11 1 [g:neg, j:pos; NLL 5.36]
Guided	you don't have to know about music to play music you t if you don't listen to music about music music y y y n s y [g:pos, j:neg; NLL 7.10]
Prompt	the emotions are raw and will strike a nerve with
Unguided	the emotions are raw and will strike a nerve with a few steps across the dark for the firstthe bell goes ringing and this quote will be pure in [g:pos, j:pos; NLL 9.96]
Guided	the emotions are raw and will strike a nerve withthe-number-of-people-who-want-to-make-their-better- [g:pos, j:pos; NLL 5.12]
Prompt	it's a cookie-cutter movie,
Unguided	it's a cookie-cutter movie, same that's where the movies come out, the same majority of them come out of the same industries [g:neg, j:neg; NLL 6.38]
Guided	it's a cookie-cutter movie, hitit,s a cookie- cut -ter movie, hitit,s [g:neg, j:neg; NLL 6.22]
Cell: $\gamma = 4$, target negative	
Prompt	determined to be fun, and bouncy,
Unguided	determined to be fun, and bouncy, and quirky, and exciting, and colourful, and play-determined to be fun-determined [g:pos, j:pos; NLL 4.57]

Guided	determined to be fun, and bouncy, didn[?] wasd that to to to and havedd andd, wasd of to, [g:pos, j:pos; NLL 10.83]
Prompt	a psychological thriller with a genuinely spooky premise and
Unguided	a psychological thriller with a genuinely spooky premise and it's only since afterhigh" that he's in jail for full house SH NOT FOR SUPIT [g:pos, j:pos; NLL 9.64]
Guided	a psychological thriller with a genuinely spooky premise and a overall sort of the to the the[?] [?]s n's done" with [g:pos, j:pos; NLL 10.71]
Prompt	the best film about baseball to hit theaters since field
Unguided	the best film about baseball to hit theaters since fieldy Chavez. This was a game fieldy Chavez. Not bad fieldy Chavez. Made the cut [g:neg, j:pos; NLL 5.92]
Guided	the best film about baseball to hit theaters since fielders are athletes were his for his's Baseball baseball to baseball, be baseball baseball, the the [g:neg, j:pos; NLL 8.59]
Cell: $\gamma = 4$, target positive	
Prompt	pacino is brilliant as the sleep-deprived
Unguided	pacino is brilliant as the sleep-deprived - is adding already of a sleep-deprived (cowned) bristol is brilliant as is [g:neg, j:pos; NLL 8.40]
Guided	pacino is brilliant as the sleep-deprived-pacino is the sleep-deprivedpacpacpacpacino is brilliant as the - [g:pos, j:neg; NLL 8.92]
Prompt	once (kim) begins to overplay the
Unguided	once (kim) begins to overplay the BILLJOUSE and, others, plays it entirely. is it every (kim) [g:neg, j:neg; NLL 7.05]
Guided	once (kim) begins to overplay the significance of In that regard, it's important away see the first the[?] the interested in [g:pos, j:pos; NLL 10.23]

Prompt	stealing harvard is evidence that the farrell
Unguided	stealing harvard is evidence that the farrell men have not been subject to pressure and blackmail for decades.watch this official and loyalists deceitful [g:neg, j:pos; NLL 7.85]
Guided	stealing harvard is evidence that the farrells are from. of.rell far. has stolen from from that the is the farrell far [g:neg, j:neg; NLL 8.08]

Table A.4: Trust-region before/after. The three highest-NLL guided continuations from the Phase 4 off-domain run (no trust region), where guidance pushed text off the fluent manifold (guided NLL near 11 versus unguided near 7). The on-domain gamma-4 pairs in Table A.3 stay bounded, which is the trust region working.

Prompt	The book
Unguided	The book was sold!The book was never sold.The book will also never be sold!Something[?] [g:neg, j:neg; NLL 7.12]
Guided	The book traded tended end preview store where ’in minorities posed failing Photo minutes united Koreaorean media 4MattIn [g:neg, j:pos; NLL 16.01]
Prompt	The city
Unguided	The city of provides Toronto I walk nowhere near as fast as I can walk Hyfening speed is a bad [g:neg, j:neg; NLL 8.30]
Guided	The city proudly absorb 67 loss unemployment mixed off "bah thousand Baronety mile flat mono seaw dialectically larger [g:neg, j:neg; NLL 15.97]
Prompt	The potato
Unguided	The potato located in The House of Commons’s foodbox.Smart, well-organized, food-arrant [g:pos, j:neg; NLL 7.27]

Guided	The potato problem replied strongly ugly likeThe Tomckswicking meant liter commenting his next early Manhattan critic Angelagi [g:neg, j:neg; NLL 15.92]

A.6 Qualitative Recovery and Steering Showcase

A.6.1 Qualitative infill recovery showcase

The ten sequences were drawn once with a fixed rule and never filtered: `np.random.default_rng(0).choice(200,10,replace=False)` over the kl-baselines set (WikiText-2 validation, one random interior mask, `data_seed` 0), sorted. Failures are kept in. The corrupted and recovered token at the masked position is shown in bold; a check mark means the recovered token equals the ground truth. Numbers cross-reference Sections 5.5.2 and 5.13.

Table A.5: Infill recovery on ten seeded sequences.

Method	Recovered token (in context)	Hit
Sequence idx 3 ground truth: 'fish' corrupted: 'moments'		
original: The diet of H. gammarus mostly consists of other benthic invertebrates. These include crabs, molluscs, sea urchins, starfish and polychaete worms.		
*corrupted: The diet of H. gammarus mostly consists of other benthic invertebrates. These include crabs, molluscs, sea urchins, star**moments** and polychaete worms.*		
DLS policy (MH, GN)	The diet of H. gammarus mostly consists of other benthic invertebrates. These include crabs, molluscs, sea urchins, star**passages** and polychaete worms.	
DLS random dir	The diet of H. gammarus mostly consists of other benthic invertebrates. These include crabs, molluscs, sea urchins, star**fossils** and polychaete worms.	
CLS flagship (policy, MH, GN, free, s50)	The diet of H. gammarus mostly consists of other benthic invertebrates. These include crabs, molluscs, sea urchins, star**specialization** and polychaete worms.	
Metropolized Gibbs	The diet of H. gammarus mostly consists of other benthic invertebrates. These include crabs, molluscs, sea urchins, star**fish** and polychaete worms.	✓
Conditional argmax	The diet of H. gammarus mostly consists of other benthic invertebrates. These include crabs, molluscs, sea urchins, star**fish** and polychaete worms.	✓
Conditional top-k rescore	The diet of H. gammarus mostly consists of other benthic invertebrates. These include crabs, molluscs, sea urchins, star**fish** and polychaete worms.	✓
Left-conditional (independence MH)	The diet of H. gammarus mostly consists of other benthic invertebrates. These include crabs, molluscs, sea urchins, star**fish** and polychaete worms.	✓

Method	Recovered token (in context)	Hit
SEDD-small recovery	The diet of H. gammarus mostly consists of other benthic invertebrates. These include crabs, molluscs, sea urchins, star**lings** and polychaete worms.	
SEDD-medium recovery	The diet of H. gammarus mostly consists of other benthic invertebrates. These include crabs, molluscs, sea urchins, star**lings** and polychaete worms.	
Hybrid (SEDD proposal, exact-energy MH)	The diet of H. gammarus mostly consists of other benthic invertebrates. These include crabs, molluscs, sea urchins, star**fish** and polychaete worms.	✓
Sequence idx 8 ground truth: 'by' corrupted: 'Info' *original: 2 Highlands Historic District — roughly bounded by 15th St, 34th Ave, 19th St, and 36th Ave.* *corrupted: 2 Highlands Historic District — roughly bounded**Info** 15th St, 34th Ave, 19th St, and 36th Ave.*		
DLS policy (MH, GN)	2 Highlands Historic District — roughly bounded**lay** 15th St, 34th Ave, 19th St, and 36th Ave.	
DLS random dir	2 Highlands Historic District — roughly bounded**AHL** 15th St, 34th Ave, 19th St, and 36th Ave.	
CLS flagship (policy, MH, GN, free, s50)	2 Highlands Historic District — roughly bounded**Registered** 15th St, 34th Ave, 19th St, and 36th Ave.	
Metropolized Gibbs	2 Highlands Historic District — roughly bounded**by** 15th St, 34th Ave, 19th St, and 36th Ave.	✓
Conditional argmax	2 Highlands Historic District — roughly bounded**by** 15th St, 34th Ave, 19th St, and 36th Ave.	✓
Conditional top-k rescore	2 Highlands Historic District — roughly bounded**by** 15th St, 34th Ave, 19th St, and 36th Ave.	✓
Left-conditional (independence MH)	2 Highlands Historic District — roughly bounded**by** 15th St, 34th Ave, 19th St, and 36th Ave.	✓
SEDD-small recovery	2 Highlands Historic District — roughly bounded**by** 15th St, 34th Ave, 19th St, and 36th Ave.	✓
SEDD-medium recovery	2 Highlands Historic District — roughly bounded**by** 15th St, 34th Ave, 19th St, and 36th Ave.	✓
Hybrid (SEDD proposal, exact-energy MH)	2 Highlands Historic District — roughly bounded**by** 15th St, 34th Ave, 19th St, and 36th Ave.	✓
Sequence idx 14 ground truth: 'Road' corrupted: 'bolstered' *original: 8 Poplar Springs Road Historic District — roughly bounded by 29th St, 23rd Ave, 22nd St, and 29th Ave.* *corrupted: 8 Poplar Springs**bolstered** Historic District — roughly bounded by 29th St, 23rd Ave, 22nd St, and 29th Ave.*		
DLS policy (MH, GN)	8 Poplar Springs**Gathering** Historic District — roughly bounded by 29th St, 23rd Ave, 22nd St, and 29th Ave.	

Method	Recovered token (in context)	Hit
DLS random dir	8 Poplar Springs**Cove** Historic District — roughly bounded by 29th St, 23rd Ave, 22nd St, and 29th Ave.	
CLS flagship (policy, MH, GN, free, s50)	8 Poplar Springs**ded** Historic District — roughly bounded by 29th St, 23rd Ave, 22nd St, and 29th Ave.	
Metropolized Gibbs	8 Poplar Springs**Road** Historic District — roughly bounded by 29th St, 23rd Ave, 22nd St, and 29th Ave.	✓
Conditional argmax	8 Poplar Springs**Road** Historic District — roughly bounded by 29th St, 23rd Ave, 22nd St, and 29th Ave.	✓
Conditional top-k rescore	8 Poplar Springs**Village** Historic District — roughly bounded by 29th St, 23rd Ave, 22nd St, and 29th Ave.	
Left-conditional (independence MH)	8 Poplar Springs**Lane** Historic District — roughly bounded by 29th St, 23rd Ave, 22nd St, and 29th Ave.	
SEDD-small recovery	8 Poplar Springs**Springs** Historic District — roughly bounded by 29th St, 23rd Ave, 22nd St, and 29th Ave.	
SEDD-medium recovery	8 Poplar Springs**National** Historic District — roughly bounded by 29th St, 23rd Ave, 22nd St, and 29th Ave.	
Hybrid (SEDD proposal, exact-energy MH)	8 Poplar Springs**Area** Historic District — roughly bounded by 29th St, 23rd Ave, 22nd St, and 29th Ave.	

Sequence idx 34 ground truth: 'resided' corrupted: 'By'

original: Lewis McAllister, a businessman in Tuscaloosa, Alabama, was the first Republican to serve in the Mississippi House of Representatives since Reconstruction, 1962 @-@ 1968 ; he resided in Meridian prior to 1971.

*corrupted: Lewis McAllister, a businessman in Tuscaloosa, Alabama, was the first Republican to serve in the Mississippi House of Representatives since Reconstruction, 1962 @-@ 1968 ; he**By** in Meridian prior to 1971.*

Method	Recovered token (in context)	Hit
DLS policy (MH, GN)	Lewis McAllister, a businessman in Tuscaloosa, Alabama, was the first Republican to serve in the Mississippi House of Representatives since Reconstruction, 1962 @-@ 1968 ; he**Client** in Meridian prior to 1971.	
DLS random dir	Lewis McAllister, a businessman in Tuscaloosa, Alabama, was the first Republican to serve in the Mississippi House of Representatives since Reconstruction, 1962 @-@ 1968 ; he**Client** in Meridian prior to 1971.	
CLS flagship (policy, MH, GN, free, s50)	Lewis McAllister, a businessman in Tuscaloosa, Alabama, was the first Republican to serve in the Mississippi House of Representatives since Reconstruction, 1962 @-@ 1968 ; he**Although** in Meridian prior to 1971.	
Metropolized Gibbs	Lewis McAllister, a businessman in Tuscaloosa, Alabama, was the first Republican to serve in the Mississippi House of Representatives since Reconstruction, 1962 @-@ 1968 ; he**died** in Meridian prior to 1971.	

Method	Recovered token (in context)	Hit
Conditional argmax	Lewis McAllister, a businessman in Tuscaloosa, Alabama, was the first Republican to serve in the Mississippi House of Representatives since Reconstruction, 1962 @-@ 1968 ; he**was** in Meridian prior to 1971.	
Conditional top-k rescore	Lewis McAllister, a businessman in Tuscaloosa, Alabama, was the first Republican to serve in the Mississippi House of Representatives since Reconstruction, 1962 @-@ 1968 ; he**served** in Meridian prior to 1971.	
Left-conditional (independence MH)	Lewis McAllister, a businessman in Tuscaloosa, Alabama, was the first Republican to serve in the Mississippi House of Representatives since Reconstruction, 1962 @-@ 1968 ; he**sponsored** in Meridian prior to 1971.	
SEDD-small recovery	Lewis McAllister, a businessman in Tuscaloosa, Alabama, was the first Republican to serve in the Mississippi House of Representatives since Reconstruction, 1962 @-@ 1968 ; he**lived** in Meridian prior to 1971.	
SEDD-medium recovery	Lewis McAllister, a businessman in Tuscaloosa, Alabama, was the first Republican to serve in the Mississippi House of Representatives since Reconstruction, 1962 @-@ 1968 ; he**lived** in Meridian prior to 1971.	
Hybrid (SEDD proposal, exact-energy MH)	Lewis McAllister, a businessman in Tuscaloosa, Alabama, was the first Republican to serve in the Mississippi House of Representatives since Reconstruction, 1962 @-@ 1968 ; he**resided** in Meridian prior to 1971.	✓
Sequence idx 52 ground truth: 'like' corrupted: 'Spanish' *original: ... in fairyland, so exquisite is the colouring of the roof and sides and so pellucid is the water... [with] alcoves or recesses like stalls in a church.* *corrupted: ... in fairyland, so exquisite is the colouring of the roof and sides and so pellucid is the water... [with] alcoves or recesses**Spanish** stalls in a church.*		
DLS policy (MH, GN)	... in fairyland, so exquisite is the colouring of the roof and sides and so pellucid is the water... [with] alcoves or recesses**right** stalls in a church.	
DLS random dir	... in fairyland, so exquisite is the colouring of the roof and sides and so pellucid is the water... [with] alcoves or recesses**Democratic** stalls in a church.	
CLS flagship (policy, MH, GN, free, s50)	... in fairyland, so exquisite is the colouring of the roof and sides and so pellucid is the water... [with] alcoves or recesses**Spanish** stalls in a church.	
Metropolized Gibbs	... in fairyland, so exquisite is the colouring of the roof and sides and so pellucid is the water... [with] alcoves or recesses**in** stalls in a church.	

Method	Recovered token (in context)	Hit
Conditional argmax	... in fairyland, so exquisite is the colouring of the roof and sides and so pellucid is the water... [with] alcoves or recesses**for** stalls in a church.	
Conditional top-k rescore	... in fairyland, so exquisite is the colouring of the roof and sides and so pellucid is the water... [with] alcoves or recesses**or** stalls in a church.	
Left-conditional (independence MH)	... in fairyland, so exquisite is the colouring of the roof and sides and so pellucid is the water... [with] alcoves or recesses**of** stalls in a church.	
SEDD-small recovery	... in fairyland, so exquisite is the colouring of the roof and sides and so pellucid is the water... [with] alcoves or recesses**or** stalls in a church.	
SEDD-medium recovery	... in fairyland, so exquisite is the colouring of the roof and sides and so pellucid is the water... [with] alcoves or recesses**of** stalls in a church.	
Hybrid (SEDD proposal, exact-energy MH)	... in fairyland, so exquisite is the colouring of the roof and sides and so pellucid is the water... [with] alcoves or recesses**between** stalls in a church.	

Sequence idx 60 ground truth: 'NT' corrupted: 'abase'
original: The $ 261 million project was started in 2002 and completed in December 2005. It was designed by the HNTB Corporation and built by Zachry Construction Corporation.
*corrupted: The $ 261 million project was started in 2002 and completed in December 2005. It was designed by the H**abase**B Corporation and built by Zachry Construction Corporation.*

DLS policy (MH, GN)	The $ 261 million project was started in 2002 and completed in December 2005. It was designed by the H**v**B Corporation and built by Zachry Construction Corporation.	
DLS random dir	The $ 261 million project was started in 2002 and completed in December 2005. It was designed by the H**BUT**B Corporation and built by Zachry Construction Corporation.	
CLS flagship (policy, MH, GN, free, s50)	The $ 261 million project was started in 2002 and completed in December 2005. It was designed by the H**abase**B Corporation and built by Zachry Construction Corporation.	
Metropolized Gibbs	The $ 261 million project was started in 2002 and completed in December 2005. It was designed by the H**&**B Corporation and built by Zachry Construction Corporation.	
Conditional argmax	The $ 261 million project was started in 2002 and completed in December 2005. It was designed by the H**&**B Corporation and built by Zachry Construction Corporation.	
Conditional top-k rescore	The $ 261 million project was started in 2002 and completed in December 2005. It was designed by the H**NT**B Corporation and built by Zachry Construction Corporation.	✓

Method	Recovered token (in context)	Hit
Left-conditional (independence MH)	The $ 261 million project was started in 2002 and completed in December 2005. It was designed by the H**Z**B Corporation and built by Zachry Construction Corporation.	
SEDD-small recovery	The $ 261 million project was started in 2002 and completed in December 2005. It was designed by the H**LS**B Corporation and built by Zachry Construction Corporation.	
SEDD-medium recovery	The $ 261 million project was started in 2002 and completed in December 2005. It was designed by the H**CL**B Corporation and built by Zachry Construction Corporation.	
Hybrid (SEDD proposal, exact-energy MH)	The $ 261 million project was started in 2002 and completed in December 2005. It was designed by the H**NT**B Corporation and built by Zachry Construction Corporation.	✓
Sequence idx 98 ground truth: 'a' corrupted: 'it' *original: Late in the afternoon, Edson stepped onto a grenade box and addressed his exhausted troops, saying,* *corrupted: Late in the afternoon, Edson stepped onto**it** grenade box and addressed his exhausted troops, saying,*		
DLS policy (MH, GN)	Late in the afternoon, Edson stepped onto**1** grenade box and addressed his exhausted troops, saying,	
DLS random dir	Late in the afternoon, Edson stepped onto**it** grenade box and addressed his exhausted troops, saying,	
CLS flagship (policy, MH, GN, free, s50)	Late in the afternoon, Edson stepped onto**.** grenade box and addressed his exhausted troops, saying,	
Metropolized Gibbs	Late in the afternoon, Edson stepped onto**the** grenade box and addressed his exhausted troops, saying,	
Conditional argmax	Late in the afternoon, Edson stepped onto**the** grenade box and addressed his exhausted troops, saying,	
Conditional top-k rescore	Late in the afternoon, Edson stepped onto**the** grenade box and addressed his exhausted troops, saying,	
Left-conditional (independence MH)	Late in the afternoon, Edson stepped onto**the** grenade box and addressed his exhausted troops, saying,	
SEDD-small recovery	Late in the afternoon, Edson stepped onto**the** grenade box and addressed his exhausted troops, saying,	
SEDD-medium recovery	Late in the afternoon, Edson stepped onto**the** grenade box and addressed his exhausted troops, saying,	
Hybrid (SEDD proposal, exact-energy MH)	Late in the afternoon, Edson stepped onto**the** grenade box and addressed his exhausted troops, saying,	
Sequence idx 122 ground truth: 'close' corrupted: 'atop' *original: Group R is another twin @-@ pyramid complex, dated to 790. It is close to the Maler Causeway.* *corrupted: Group R is another twin @-@ pyramid complex, dated to 790. It is**atop** to the Maler Causeway.*		

Method	Recovered token (in context)	Hit
DLS policy (MH, GN)	Group R is another twin @-@ pyramid complex, dated to 790. It is**guardian** to the Maler Causeway.	
DLS random dir	Group R is another twin @-@ pyramid complex, dated to 790. It is**guardian** to the Maler Causeway.	
CLS flagship (policy, MH, GN, free, s50)	Group R is another twin @-@ pyramid complex, dated to 790. It is**transparent** to the Maler Causeway.	
Metropolized Gibbs	Group R is another twin @-@ pyramid complex, dated to 790. It is**atop** to the Maler Causeway.	
Conditional argmax	Group R is another twin @-@ pyramid complex, dated to 790. It is**the** to the Maler Causeway.	
Conditional top-k rescore	Group R is another twin @-@ pyramid complex, dated to 790. It is**home** to the Maler Causeway.	
Left-conditional (independence MH)	Group R is another twin @-@ pyramid complex, dated to 790. It is**home** to the Maler Causeway.	
SEDD-small recovery	Group R is another twin @-@ pyramid complex, dated to 790. It is**linked** to the Maler Causeway.	
SEDD-medium recovery	Group R is another twin @-@ pyramid complex, dated to 790. It is**adjacent** to the Maler Causeway.	
Hybrid (SEDD proposal, exact-energy MH)	Group R is another twin @-@ pyramid complex, dated to 790. It is**anchored** to the Maler Causeway.	
Sequence idx 162 ground truth: '=' corrupted: 'deliveries' *original: = = = Day 6 of the trial = = =* *corrupted: =**deliveries** = Day 6 of the trial = = =*		
DLS policy (MH, GN)	=**result** = Day 6 of the trial = = =	
DLS random dir	=**iatus** = Day 6 of the trial = = =	
CLS flagship (policy, MH, GN, free, s50)	=**logger** = Day 6 of the trial = = =	
Metropolized Gibbs	=**a** = Day 6 of the trial = = =	
Conditional argmax	=**a** = Day 6 of the trial = = =	
Conditional top-k rescore	=**en** = Day 6 of the trial = = =	
Left-conditional (independence MH)	=**en** = Day 6 of the trial = = =	
SEDD-small recovery	=**=** = Day 6 of the trial = = =	✓
SEDD-medium recovery	=**=** = Day 6 of the trial = = =	✓
Hybrid (SEDD proposal, exact-energy MH)	=**=** = Day 6 of the trial = = =	
Sequence idx 199 ground truth: '1990' corrupted: 'ede' *original: = = = Global security, late @-@ 1990s = = =* *corrupted: = = = Global security, late @-@**ede**s = = =*		
DLS policy (MH, GN)	= = = Global security, late @-@**OR**s = = =	
DLS random dir	= = = Global security, late @-@**o**s = = =	

Method	Recovered token (in context)	Hit
CLS flagship (policy, MH, GN, free, s50)	= = = Global security, late @-@**[?]**s = = =	
Metropolized Gibbs	= = = Global security, late @-@**;**s = = =	
Conditional argmax	= = = Global security, late @-@**date**s = = =	
Conditional top-k rescore	= = = Global security, late @-@**'**s = = =	
Left-conditional (independence MH)	= = = Global security, late @-@**tech**s = = =	
SEDD-small recovery	= = = Global security, late @-@**@**s = = =	
SEDD-medium recovery	= = = Global security, late @-@**@**s = = =	
Hybrid (SEDD proposal, exact-energy MH)	= = = Global security, late @-@**'**s = = =	

A.7 Use of AI-Tools

The author of this thesis has used AI tools, specifically Google AI Studio (Gemini 3.1 Pro) and Claude (Anthropic), in the preparation of this work, and documents their use here as required.

The tools were used first for code support. When errors were encountered in the implementation of the sampling and evaluation code, they assisted in diagnosing those errors and in checking the code for compliance with the methods described in the accompanying papers. They were used in particular to help write and debug the job-scheduling and GPU-utilization scripts that distribute the experiments across the available hardware, and the diagnostic and plotting scripts. They were also used in the analysis, again as support in implementing the analysis code and resolving the errors encountered in it, and in some cases in assembling the result tables.

In the revision of the manuscript, Claude was additionally used, through an agentic coding interface, to apply a structured and author-specified set of edits to the LaTeX sources of this thesis. That use comprised matching the department template, transcribing verified numerical results from the experimental output files into the tables and the text, adding cross-references and formatting, and copy-editing prose for clarity. Every edit was made against an explicit list of changes prepared by the author, every quantitative claim so transcribed was traced to the result file that produced it and annotated with its source in the document, and the author reviewed the output of each step.

All other parts of the thesis, including the design of the experiments, the synthesis of the background and related work, the analysis and interpretation of the results, and the arguments and conclusions drawn from them, were carried out solely by the author, who has verified all quantitative claims against the underlying experimental output and assumes full responsibility for the entire content of this work.